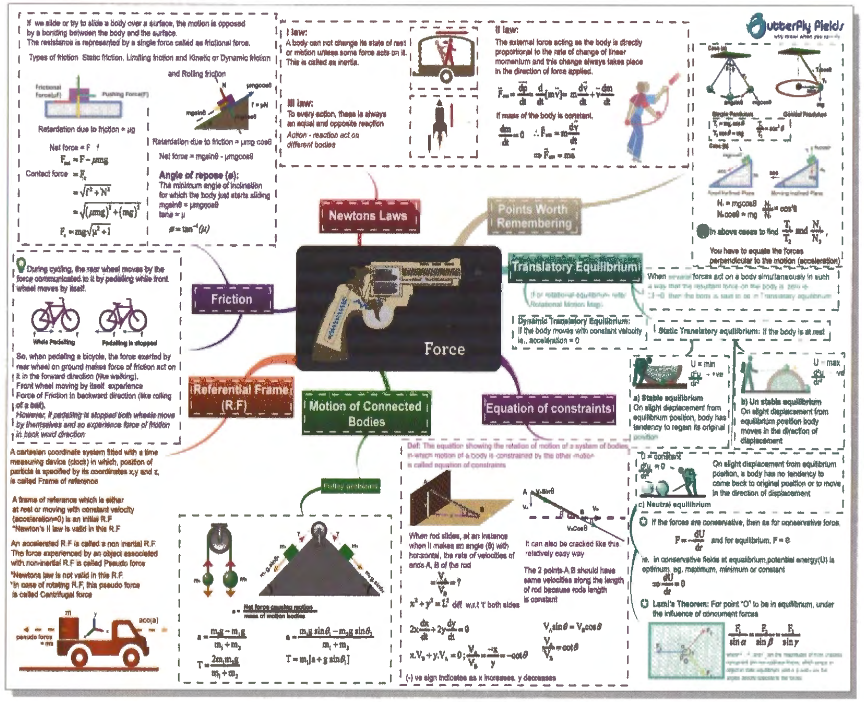
What is a Concept Map?

A Concept map is a colourful, visual form of representing key concepts covered in a chapter. It can be used as an effective overview, learning and revision tool.

At its heart is a central topic of the chapter. This is then explored by means of branches representing main ideas, which all connect to this central idea.

Concept maps are metacognitive tools that give students awareness towards the process of learning and allow the representation of knowledge structures in thematic fields.



Why use a Concept Map?

Students:

Brain is the key to effective learning for a student and more effectively the student uses it, the more successful he/she will be. If the brain of the student is stimulated with correct thinking and effective learning tools, excellent results can be obtained. **Understanding lists, lines, sequences, letters and numbers** is the work of the **LEFT BRAIN.** While **interpreting colour, images, rhythms and spatial awareness** is the work of the **RIGHT BRAIN.** Through a Concept Map both Right & Left parts of the brain would be put into use.

As a learning strategy, concept mapping stimulate learners' **commitment and interest towards the concepts,** which is very important for better learning. Students can use it as tool for effective revision of not just the current class chapters but also the topics covered previously.

Teachers:

As a teaching support, concept maps are also helpful, as they permit the establishment of networks of concepts that originate from a central concept in a coherent, structured and integrated manner, **guiding the sequence of teaching.** Concept maps can also be used by teachers for **curriculum planning.** Using this instrument means giving a new meaning to education, as well as a new meaning to the concepts of teaching, learning, and assessment of learning.

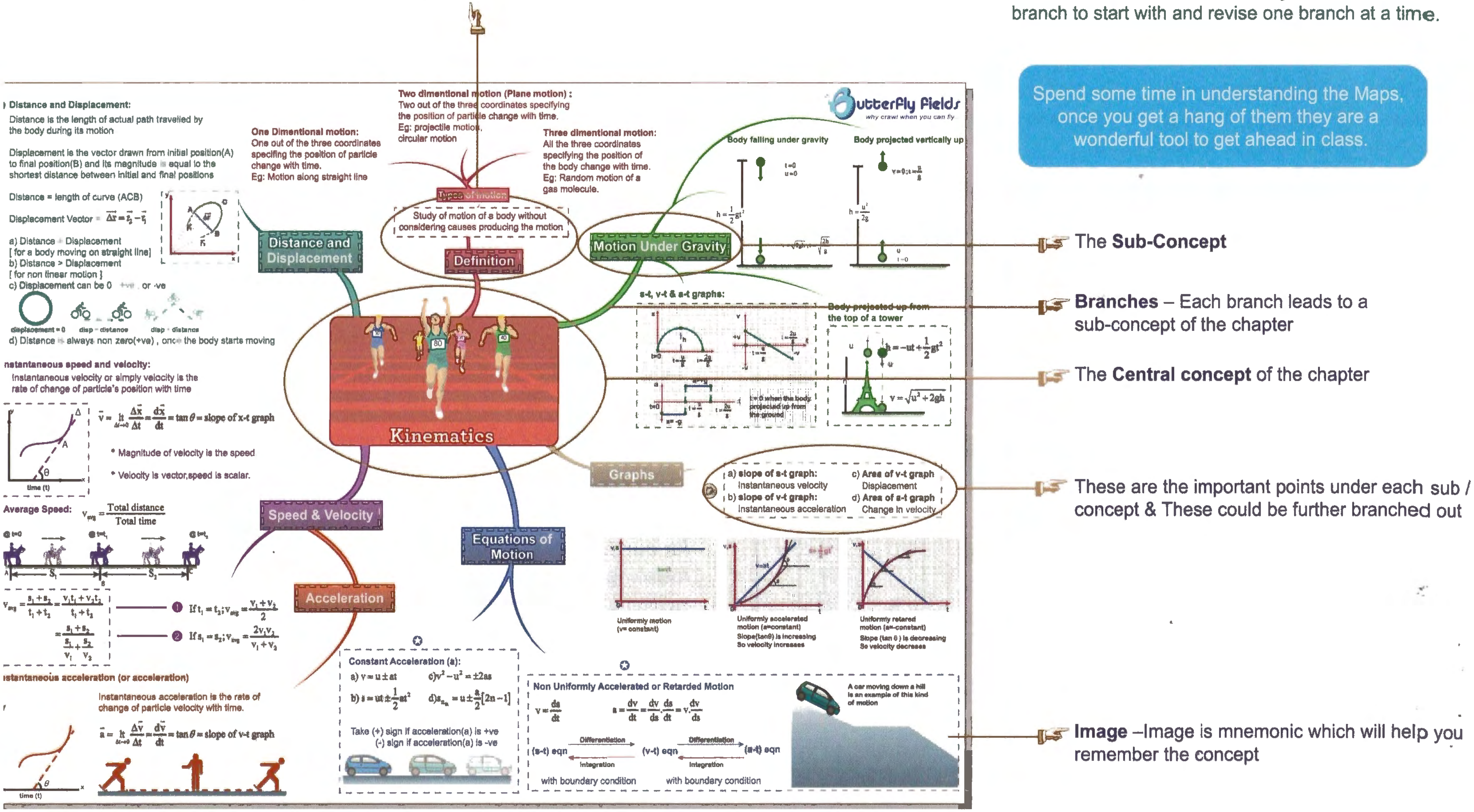
Each main topic has been transformed into a Map. For example, given below is the map for KINEMATICS which includes an introduction to the map.

If you take the above branch as an example, we can understand that KINEMATICS has a Definition - Study of motion of a body without considering causes producing the motion(correspondingly the image).

Important Points

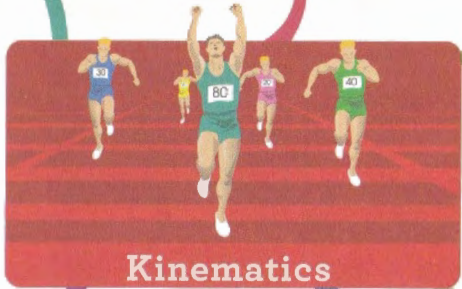
- The sub-concepts and the points under them are colour coded .i.e. each branch is identified with its colour.
- Based on the colour you can identify the sub-concept under which it will come.
- Refer to the text book and identify the first branch to start with and revise one branch at a time.

Spend some time in understanding the Maps, once you get a hang of them they are a wonderful tool to get ahead in class.



A to Z of PHYSICS

Kinematics



Definition

Study of motion of a body without considering causes producing the motion

Types of motion

One Dimentional motion:
One out of the three coordinates specifing the position of particle change with time.
Eg: Motion along straight line

Two dimentional motion (Plane motion) :
Two out of the three coordinates specifying the position of particle change with time.
Eg: projectile motion, circular motion

Three dimentional motion:
All the three coordinates specifying the position of the body change with time.
Eg: Random motion of a gas molecule.

Distance and Displacement

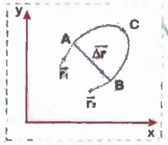
Distance and Displacement:

Distance is the length of actual path travelled by the body during its motion.

Displacement is the vector drawn from initial position(A) to final position(B) and its magnitude is equal to the shortest distance between initial and final positions.

Distance = length of curve (ACB)

Displacement Vector = $\vec{\Delta r} = \vec{r_2} - \vec{r_1}$

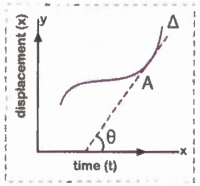


a) Distance = Displacement
[for a body moving on straight line]
b) Distance > Displacement
[for non linear motion]
c) Displacement can be 0 , +ve , or -ve

d) Distance is always non zero(+ve) , once the body starts moving

Speed & Velocity

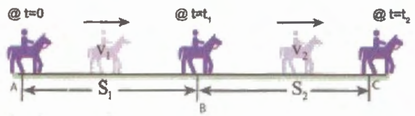
Instantaneous speed and velocity:
Instantaneous velocity or simply velocity is the rate of change of particle's position with time



$$\vec{v} = \lim_{\Delta t \to 0} \frac{\Delta \vec{x}}{\Delta t} = \frac{d\vec{x}}{dt} = \tan\theta = \text{slope of x-t graph}$$

- Magnitude of velocity is the speed.
- Velocity is vector,speed is scalar.

Average Speed: $v_{avg} = \dfrac{\text{Total distance}}{\text{Total time}}$



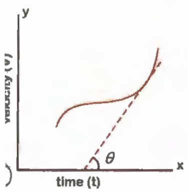
$$v_{avg} = \frac{s_1+s_2}{t_1+t_2} = \frac{v_1t_1+v_2t_2}{t_1+t_2} = \frac{s_1+s_2}{\frac{s_1}{v_1}+\frac{s_2}{v_2}}$$

1. If $t_1 = t_2; v_{avg} = \dfrac{v_1+v_2}{2}$
2. If $s_1 = s_2; v_{avg} = \dfrac{2v_1v_2}{v_1+v_2}$

Acceleration

Instantaneous acceleration (or acceleration)

Instantaneous acceleration is the rate of change of particle velocity with time.



$$\vec{a} = \lim_{\Delta t \to 0} \frac{\Delta \vec{v}}{\Delta t} = \frac{d\vec{v}}{dt} = \tan\theta = \text{slope of v-t graph}$$

Equations of Motion

Constant Acceleration (a):

a) $v = u \pm at$

b) $s = ut \pm \frac{1}{2}at^2$

c) $v^2 - u^2 = \pm 2as$

d) $s_{n_{th}} = u \pm \frac{a}{2}[2n-1]$

Take (+) sign if acceleration(a) is +ve
(-) sign if acceleration(a) is -ve

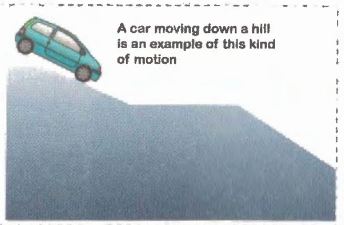
Non Uniformly Accelerated or Retarded Motion

$$v = \frac{ds}{dt} \qquad a = \frac{dv}{dt} = \frac{dv}{ds}\cdot\frac{ds}{dt} = v.\frac{dv}{ds}$$

(s-t) eqn $\xrightarrow{\text{Differentiation}}$ (v-t) eqn $\xrightarrow{\text{Differentiation}}$ (a-t) eqn

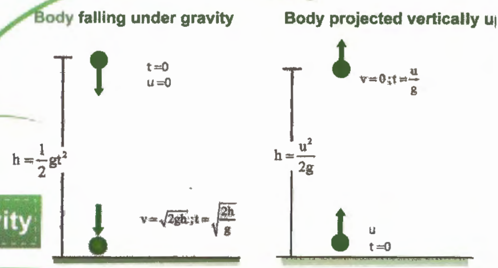
(s-t) eqn $\xleftarrow{\text{Integration}}$ (v-t) eqn $\xleftarrow{\text{Integration}}$ (a-t) eqn

with boundary condition — with boundary condition

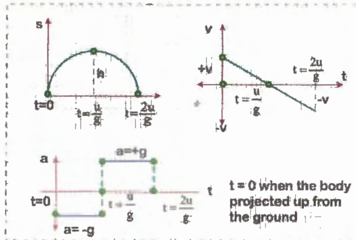


A car moving down a hill is an example of this kind of motion

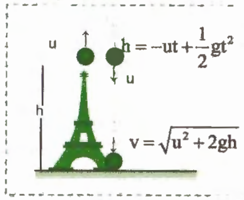
Motion Under Gravity



s-t, v-t & a-t graphs:

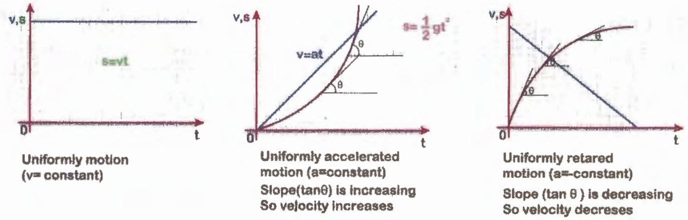


Body projected up from the top of a tower



Graphs

a) **slope of s-t graph:** Instantaneous velocity
b) **slope of v-t graph:** Instantaneous acceleration
c) **Area of v-t graph** Displacement
d) **Area of a-t graph** Change in velocity

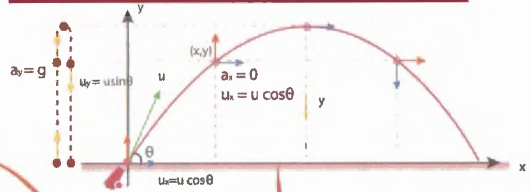


Projectile Motion

Definition

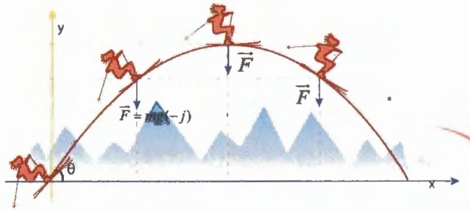
Projectile is a body projected at an angle other than 90° with the horizontal, travelling along a **parabolic path**.
It's motion is **simultaneous superposition of two motions** - one horizontal and the other vertical.

Projectile motion ⇒ Horizontal motion + Vertical motion



Force

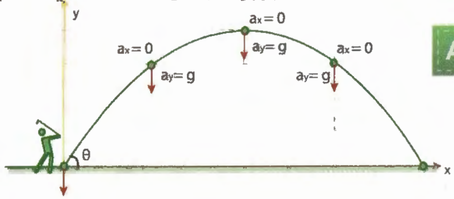
Force (F) on the projectile:
Air resistance force on projectile is not considered. The one and only one force on the projectile is its weight (mg)



Acceleration

Acceleration(a) of projectile:
As no forces are considered to be acting on projectile other than its weight, which acts along - y axis.

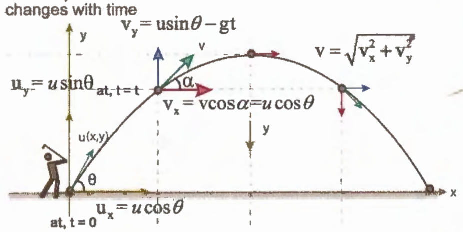
$F_x = 0;\ a_x = 0;\ \vec{F}_y = m\vec{g};\ \vec{a}_y = g(-\hat{j})$



Velocity

Velocity (v) of a projectile:
as, $a_x = 0$, Horizontal component of its velocity, $u_x = \cos\theta$ (constant)
but as $a_y \neq 0$, vertial component of its velocity (v_y) changes with time

$v_y = u\sin\theta - gt$



Displacement

Displacement of a projectile at time = t

$x = u_x t + \frac{1}{2}a_x t^2 \quad (u_x = u\cos\theta, a_x = 0)$

$x = u\cos\theta . t\ ;\ t = \dfrac{x}{u\cos\theta}$

$y = u_y t + \frac{1}{2}a_y t^2 \quad (u_y = u\sin\theta, a_y = -g)$

$y = u\sin\theta . t - \frac{1}{2}gt^2 = x\tan\theta - x^2\dfrac{g}{2u^2\cos^2\theta}$ (parabola)

$y = Ax - Bx^2 \quad ; A = \tan\theta\ ; B = g/2u^2\cos^2\theta$

Displacement $= OP = \sqrt{x^2 + y^2}$

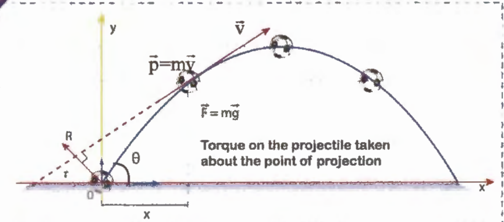
Angular Momentum

Angular momentum (L), Torque(τ) on the projectile

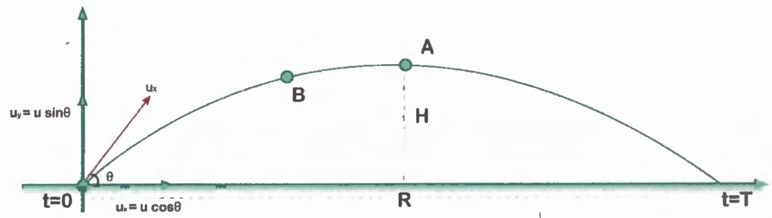
$\tau_0 = F \times \perp$ ar distance(x)

$\tau_0 = mg \times x\cos\theta \times t$

Angular momentum of projectile
$L_0 = mvr \quad (r = OR)$



R,H,T & Energy



Range (R), Maximum Height (H) and Time of flight (T) of projectile

$H = \dfrac{u^2\sin^2\theta}{2g} = \dfrac{u_y^2}{2g}$

$R = \dfrac{u^2\sin 2\theta}{g} = \dfrac{2.u_x.u_y}{g}$

$T = \dfrac{2u\sin\theta}{g} = \dfrac{2.u_y}{g}$

- When R → maximum, θ = 45°
  then $H = \dfrac{R_{max}}{4}$

For complimentary angles (θ, 90-θ), Ranges will be same for same u

$\dfrac{H_1}{H_2} = \tan^2\theta$

Energy of a projectile:

TE at (o) = P.E + K.E = $0 + \frac{1}{2}mu^2$

TE at (A) = P.E + K.E = $mgH + \frac{1}{2}mv$

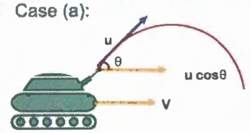
$= mg\left[\dfrac{u^2\sin^2\theta}{2g}\right] + \frac{1}{2}mu^2\cos^2\theta = \frac{1}{2}mu^2$

At 'A': $\dfrac{K.E}{P.E} = \cot^2\theta$

As $TE_O = TE_A$, Mechanical energy is conserv

Expressions for R, H, T if the point of prjection (O) is moving

Case (a):



$u_x = v + u\cos\theta$

$u_y = u\sin\theta$

Case (b):

$u_x = u\cos\theta$

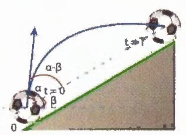
$u_y = v + u\sin\theta$

Substitute the values of u_x, u_y in H, R, T formulae

Projectile Special Cases

Case(a): Projectile thrown up

$T = \dfrac{2u\sin(\alpha - \beta)}{g\cos\beta}$

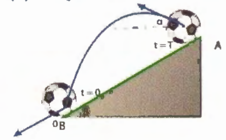


$R = AB = \dfrac{2u^2}{g\cos^2\beta}[\cos\alpha.\sin(\alpha - \beta)]$

For maximum range, $\alpha = \dfrac{\pi}{4} + \dfrac{\beta}{2}$

$R_{max} = \dfrac{u^2}{g(1 + \sin\beta)}$

Case(b): Projectile thrown down



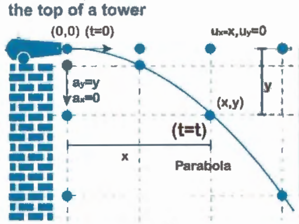
$T = \dfrac{2u\sin(\alpha + \beta)}{g\cos\beta}$

$R = AB = \dfrac{2u^2}{g\cos^2\beta}[\cos\alpha.\sin(\alpha + \beta)]$

For maximum range, $\alpha = \dfrac{\pi}{4} - \dfrac{\beta}{2}$

$R_{max} = \dfrac{u^2}{g(1 - \sin\beta)}$

Projecting a body horizontal from the top of a tower



h

$x = u_x t + \frac{1}{2}a_x t$

$x = u.t;\ t = \dfrac{x}{u}$

$y = u_y t + \frac{1}{2}a_y t^2$

$y = \frac{1}{2}gt^2 = \dfrac{g}{2u^2}x^2$

(parabola $y = Ax^2$)

Time of flight: $T = \sqrt{\dfrac{2h}{g}}$

Range(R) = u× T

Circular Motion

Motion on the Circumference of Circle (A Two Dimensional Motion)

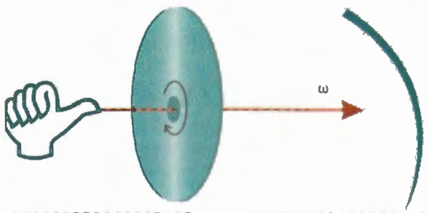
Velocity

Angular velocity (or) Angular frequency (ω):

Average angular velocity (ω) $= \dfrac{\Delta\theta}{\Delta t}$

Instantaneous angular velocity $= \dfrac{d\theta}{dt}$

ω is a vector quantity. When right hand fingers are rotated in anti-clock wise direction (ie., direction of rotation) thumb is along $\vec{\omega}$.



Time Period (T): Time for one rotation

Frequency (f): No of rotations in 1 sec

$\therefore T = \dfrac{1}{f}; \omega = \dfrac{2\pi}{T} = 2\pi f$

Relation between linear velocity (v) and angular velocity (ω) is:

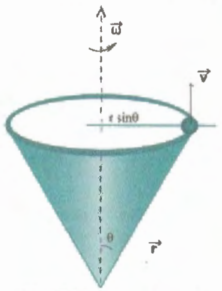
$\vec{v} = \vec{\omega} \times \vec{r}$

$\vec{v}$ is in the plane of circular path;

$\vec{\omega}$ is perpendicular to the circular path;

$v = \omega .\ r.\ \sin\theta$

θ - The angle between $\vec{\omega}$ & $\vec{r}$



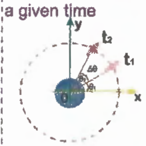
Displacement

Kinematics equations in circular motion with constant α

$$\begin{cases} \omega_f = \omega_i + \alpha t \\ \theta = \omega_i t + \dfrac{1}{2}\alpha t^2 \\ \omega_f^2 = \omega_i^2 + 2\alpha\theta \end{cases} \quad \frac{S}{\theta} = \frac{V}{\omega} = \frac{a}{\alpha} = r$$

Angular Displacement:

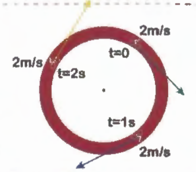
It is the angle traced by the radius vector at the axis of the circle in a given time



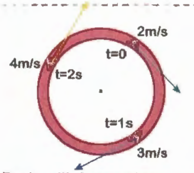
$\Delta\theta = \theta_2 - \theta_1$

$\Delta t = t_2 - t_1$

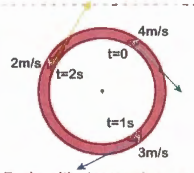
Types



Body with constant speed (Uniform circular motion)



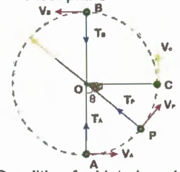
Body with increasing speed (Non uniform accelerating circular motion)



Body with decreasing speed (Non uniform decelerating circular motion)

Dynamics of Circular Motion

Motion of a body connected to a string in a vertical plane



Condition for V_A to be minimum, for which the body just completes the circular path

$v_A = \sqrt{5gR}; T_A = 6mg$

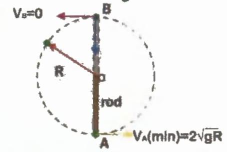
$v_B = \sqrt{gR}; T_B = 0$

$v_C = \sqrt{2gR}; T_C = 3mg$

$v_P = \sqrt{gR(3+2\cos\theta)}$

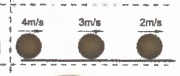
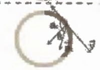
$T_P = \sqrt{3mg(1+\cos\theta)}$

Note: Instead of string if the body is connected to a light rod

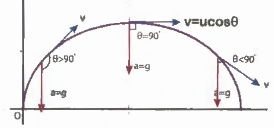


Classification of Motion

Classification of motion of a body based on a_c, a_t
[θ - The angle between velocity and acceleration vectors]

θ = 0	$a_c = 0; a_t > 0$	Linear accelerated	
θ = 180°	$a_c = 0; a_t < 0$	Linear decelerated	
θ = 90°	$a_c \neq 0; a_t = 0$ $a = a_c$	Uniform circular	
θ Acute	$a_c \neq 0; a_t > 0$	Non uniform accelerated circular	
θ Obtuse	$a_c \neq 0; a_t < 0$	Non uniform decelerated circular	

Simulation of Projectile motion with circular motion



Projectile motion resembler

a) Decelerating circular motion up to maximum height (**θ > 90°**)

b) Uniform circular motion at (**θ = 90°**) maximum height

c) Accelerating circular motion for balance path (**θ < 90°**).

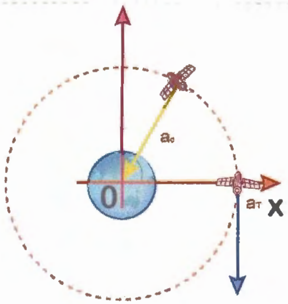
Radius of Curvature

Any curved path can be assumed to be made of infinite number of circular arcs. Radius of curvature of a curve is the radius of the circle of which this curve is a part.

To find the radius of curvature at a point take accelerations component perpendicular to velocity. Then this is the case of uniform circular motion. Then apply

$a_c = \dfrac{v^2}{R}; R = \dfrac{v^2}{a_c}$

Acceleration



From $\vec{v} = \vec{\omega} \times \vec{r}$

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d}{dt}\left[\vec{\omega} \times \vec{r}\right] = \vec{\omega} \times \frac{d\vec{r}}{dt} + \frac{d\vec{\omega}}{dt} \times \vec{r}$$

$$= \vec{\omega} \times \vec{v} + \vec{\alpha} \times \vec{r} = \vec{a}_c + \vec{a}_t = \left|\vec{a}\right| = a = \sqrt{a_c^2 + a_t^2}$$

The term ($\vec{\alpha} \times \vec{r}$) indicates a vector directed along tangent (ie., along velocity) and is called Tangential acceleration (a_t).

a_t is due to the change in magnitude of velocity (speed).

$= \dfrac{d}{dt}[r.\omega] = r.\dfrac{d\omega}{dt} = r.\alpha$

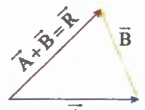
$\vec{\alpha}$ – The angular acceleration, $\vec{\alpha} = \dfrac{d\vec{\omega}}{dt}$

The term ($\vec{\omega} \times \vec{v}$) indicates a vector directed towards the centre of circular orbit and is called Centripetal acceleration (a_c) and is due to the change in direction of velocity

$a_c = v.\omega = r.\omega^2 = \dfrac{v^2}{2}$

Angular Acceleration has the same sign as the angular velocity when I speed up and the opposite sign when I slow down

Vectors

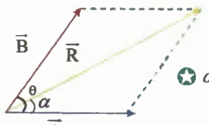


Addition of Vectors $(\vec{A}+\vec{B})$

a) Addition of vector obeys **cummutative law:** $\vec{A}+\vec{B}=\vec{B}+\vec{A}$

b) It also obeys **Associate law:** $\vec{A}+(\vec{B}+\vec{C})=(\vec{A}+\vec{B})+\vec{C}$

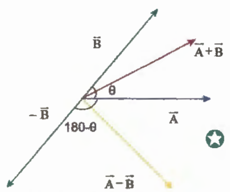
Parallelogram law of vector addition



$$R=\sqrt{A^2+B^2+2AB\cos\theta}$$

$$\alpha=\tan^{-1}\left[\frac{B\sin\theta}{A+B\cos\theta}\right]$$

Substraction of Vectors $(\vec{A}-\vec{B})$



Angle between $\vec{A},-\vec{B}$ is (180-θ)

$$|\vec{A}-\vec{B}|=\sqrt{A^2+B^2+2AB\cos(180-\theta)}$$

$$|\vec{A}-\vec{B}|=\sqrt{A^2+B^2-2AB\cos\theta}$$

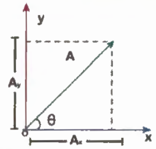
Resolution of a vector into rectangular components

A_x = x-component of $\vec{A}$ = A$\cos\theta$

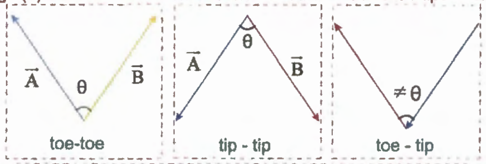
A_y = y-component of $\vec{A}$ = A$\sin\theta$

$$\vec{A}=A_x\hat{i}+A_y\hat{j}$$

$$|\vec{A}|=\sqrt{A_x^2+A_y^2}$$



Angle(θ) between two vectors is measured between their tips or toes



Is there any physical quantity which is neither scalar or vector: Ya!! It is a **Tensor Quantity.** eg., **Moment of Inertia**, because it is not unique for a given body and does not depend on direction of rotation.

Definition

A physical quantity which has **only magnitude** without any direction is called a **scalar quantity.**

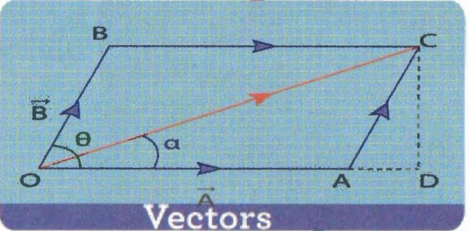
A physical quantity is called as a **vector** if it is

(i) Has both magnitude and direction

(ii) Can be resolved is rectangular components

(iii) Obeys vector law of addition

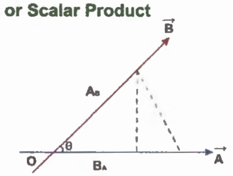
A physical quantity which has direction is called vector, but that which has direction need not be a vector. eg., Surface tension, current etc.



Operations

Products

Dot or Scalar Product



a) $\vec{A}.\vec{B}=AB\cos\theta=\vec{B}.\vec{A}$

$=\vec{A}[\vec{B}\cos\theta]$

$=A$[Component of $\vec{B}$ along $\vec{A}=\vec{B}_A$]

$=B[\vec{A}\cos\theta]$

$=B$[Component of $\vec{A}$ along $\vec{B}=A_B$]

$*\cos^{-1}\left[\frac{\vec{A}.\vec{B}}{AB}\right]=\theta$ (The angle between $\vec{A},\vec{B}$)

b) Dot Product of two perpendicular vector is zero

c) **Dot Product of unit vectors :**

$\hat{i}.\hat{i}=\hat{j}.\hat{j}=\hat{k}.\hat{k}=1$

$\hat{i}.\hat{j}=\hat{j}.\hat{k}=\hat{k}.\hat{i}=0$

d) If $\vec{A}=A_x\hat{i}+A_y\hat{j}+A_z\hat{k}$

$\vec{B}=B_x\hat{i}+B_y\hat{j}+B_z\hat{k}$

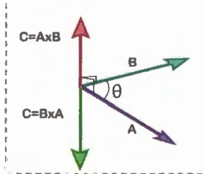
$\vec{A}.\vec{B}=A_x.B_x+A_y.B_y+A_z.B_z$

Vector or cross product:

$\vec{A}\times\vec{B}=\vec{C}=AB\sin\theta.\hat{n}$

[$\hat{n}$ – The limit vector perpendicular to the plane containing $\vec{A},\vec{B}$]

a) $|\vec{A}\times\vec{B}|=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ A_x & A_y & A_z\\ B_x & B_y & B_z\end{vmatrix}$



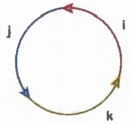
b) If 2 vectors, $\vec{A},\vec{B}$ are parallel

$\vec{A}=A_x\hat{i}+A_y\hat{j}+A_z\hat{k}$

$\vec{B}=B_x\hat{i}+B_y\hat{j}+B_z\hat{k}$

then $\frac{A_x}{B_x}=\frac{A_y}{B_y}=\frac{A_z}{B_z}$

c) Cross Product of unit vectors:



$\hat{i}\times\hat{j}=\hat{k},\hat{j}\times\hat{k}=\hat{i},\hat{k}\times\hat{i}=\hat{j}$

$\hat{i}\times\hat{i}=\hat{j}\times\hat{j}=\hat{k}\times\hat{k}=0$

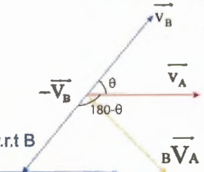
d) Cross product of 2 collinear vectors is zero

Relative Velocity

Relative velocity of a particle A with respect to particle B is defined as the velocity with which A appears to move if B is considered to be rest.

For making B to be rest, apply $-\vec{V}_B$

The resultant of $\vec{V}_A$ and $-\vec{V}_B$ is the relative velocity of A w.r.t B



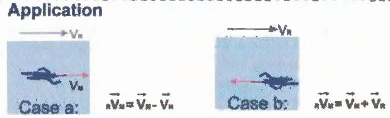
$${}_BV_A=\sqrt{V_A^2+V_B^2+2V_AV_B\cos(180-\theta)}$$

$${}_BV_A=\sqrt{V_A^2+V_B^2-2V_AV_B\cos\theta}$$

Application

Relative motion in river flow:

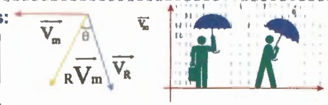
$\vec{V}_m$ & $\vec{V}_R$ are the velocities of man and river w.r.t ground is the velocity of man w.r.t river water flow



Rain problems:

$\theta=\tan^{-1}\left[\frac{V_m}{V_R}\right]$

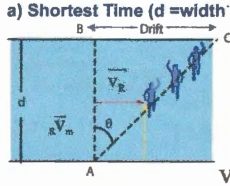
$_mV_R=\vec{V}_R+\vec{V}_m$



angle with which he should hold umbrella with vertical for not to get wet

Crossing the River

a) Shortest Time (d =width of river)

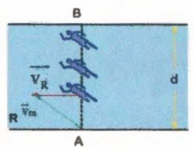


$t_{shortest}=\frac{d}{_RV_m}$

Drift(λ) $=V_R.t_{shortest}$

$V_M^2=V_R+{}_RV^2{}_M$

b) Shortest Path



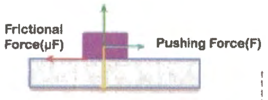
$t_{shortest\ path}=\frac{d}{V_M}=\frac{d}{\sqrt{_RV_M^2-V_R^2}}$

$\theta=\sin^{-1}\frac{V_R}{_RV_M}$

$_RV_M^2=V_M^2+V_R^2$

If we slide or try to slide a body over a surface, the motion is opposed by a bonding between the body and the surface.
The resistance is represented by a single force called as frictional force.

Types of friction: Static friction, Limiting friction and Kinetic or Dynamic friction and Rolling friction



Retardation due to friction = μg

Net force = F - f

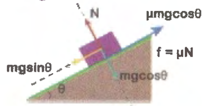
$F_{net} = F - \mu mg$

Contact force $= F_c$

$= \sqrt{f^2 + N^2}$

$= \sqrt{(\mu mg)^2 + (mg)^2}$

$F_c = mg\sqrt{\mu^2 + 1}$



Retardation due to friction = μmg cosθ

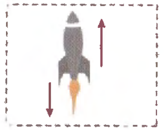
Net force = mgsinθ - μmgcosθ

Angle of repose (ø):
The minimum angle of inclination for which the body just starts sliding
mgsinθ = μmgcosθ
tanø = μ

$\varnothing = \tan^{-1}(\mu)$

I law:
A body can not change its state of rest or motion unless some force acts on it.
This is called as inertia.

III law:
To every action, there is always an equal and opposite reaction.
Action - reaction act on different bodies



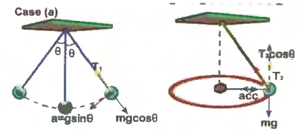
II law:
The external force acting as the body is directly proportional to the rate of change of linear momentum and this change always takes place in the direction of force applied.

$\vec{F}_{ext} = \frac{d\vec{p}}{dt} = \frac{d}{dt}[m\vec{v}] = m\frac{d\vec{v}}{dt} + \vec{v}\frac{dm}{dt}$

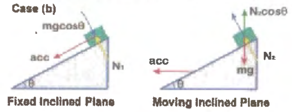
If mass of the body is constant,

$\frac{dm}{dt} = 0 \quad \therefore \vec{F}_{ext} = m\frac{d\vec{v}}{dt}$

$\Rightarrow \vec{F}_{ext} = m\vec{a}$



$T_1 = mg.\cos\theta$
$T_2\cos\theta = mg$
$\frac{T_1}{T_2} = \cos^2\theta$



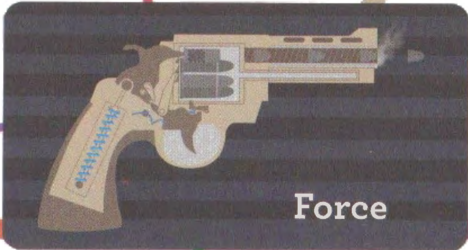
$N_1 = mg\cos\theta \quad \frac{N_1}{N_2} = \cos^2\theta$
$N_2\cos\theta = mg$

In above cases to find $\frac{T_1}{T_2}$ and $\frac{N_1}{N_2}$,

You have to equate the forces perpendicular to the motion (acceleration)

Newtons Laws

Points Worth Remembering



Force

Friction

Translatory Equilibrium

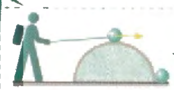
(For rotational equilibrium refer Rotational Motion Map)

When several forces act on a body simultaneously in such a way that the resultant force on the body is zero ie., ΣF=0, then the body is said to be in Translatory equilibrium

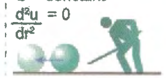
Dynamic Translatory Equilibrium:
If the body moves with constant velocity ie., acceleration = 0

Static Translatory equilibrium: If the body is at rest

a) Stable equilibrium
On slight displacement from equilibrium position, body has tendency to regain its original position



b) Un stable equilibrium
On slight displacement from equilibrium position body moves in the direction of displacement



On slight displacement from equilibrium position, a body has no tendency to come back to original position or to move in the direction of displacement

c) Neutral equilibrium

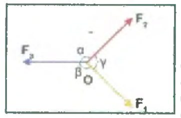
If the forces are conservative, then as for conservative force,

$F = -\frac{dU}{dr}$ and for equilibrium, F = 0

ie., in conservative fields at equalibrium, potential energy(U) is optimum. eg: maximum, minimum or constant

$\Rightarrow \frac{dU}{dr} = 0$

Lami's Theorem: For point "O" to be in equilibrium, under the influence of concurrent forces



$\frac{F_1}{\sin\alpha} = \frac{F_2}{\sin\beta} = \frac{F_3}{\sin\gamma}$

where F_1, F_2 and F_3 are the magnitudes of three coplanar concurrent and non-collinear forces, which keeps an object in static equilibrium, and α, β and γ are the angles directly opposite to the forces.

During cycling, the rear wheel moves by the force communicated to it by pedalling while front wheel moves by itself.

So, when pedaling a bicycle, the force exerted by rear wheel on ground makes force of friction act on it in the forward direction (like walking).
Front wheel moving by itself experience Force of Friction in backward direction (like rolling of a ball).
However, if pedalling is stopped both wheels move by themselves and so experience force of friction in back word direction

Referential Frame (R.F)

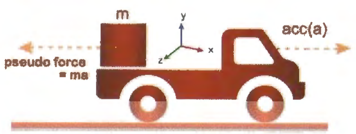
Motion of Connected Bodies

Equation of constraints

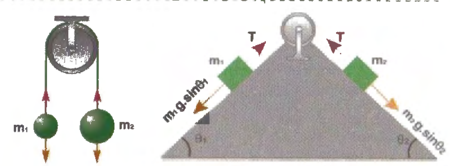
cartesian coordinate system fitted with a time easuring device (clock) in which, position of article is specified by its coordinates x,y and z, called Frame of reference

frame of referance which is either t rest or moving with constant velocity acceleration=0) is an initial R.F
Newton's II law is valid in this R.F

n accelerated R.F is called a non inertial R.F.
he force experienced by an object associated ith non-inertial R.F is called Pseudo force
Newtons law is not valid in this R.F.
n case of rotating R.F, this pseudo force called Centrifugal force



Pulley problems



$a = \frac{\text{Net force causing motion}}{\text{mass of motion bodies}}$

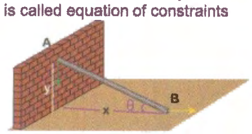
$a = \frac{m_2 g - m_1 g}{m_1 + m_2}$

$T = \frac{2m_1 m_2 g}{m_1 + m_2}$

$a = \frac{m_1 g\sin\theta_1 - m_2 g\sin\theta_2}{m_1 + m_2}$

$T = m_1[a + g\sin\theta_1]$

Def: The equation showing the relation of motion of a system of bodies in which motion of a body is constrained by the other motion is called equation of constraints



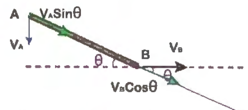
When rod slides, at an instance when it makes an angle (θ) with horizontal, the rate of velocities of ends A, B of the rod

$= \frac{V_A}{V_B} = ?$

$x^2 + y^2 = L^2$ diff. w.r.t 't' both sides

$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$

$x.V_B + y.V_A = 0; \frac{V_A}{V_B} = \frac{-x}{y} = -\cot\theta$



It can also be cracked like this, relatively easy way.

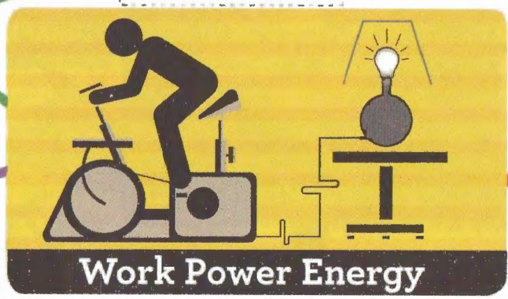
The 2 points A,B should have same velocities along the length of rod because rods length is constant

$V_A\sin\theta = V_B\cos\theta$

$\frac{V_A}{V_B} = \cot\theta$

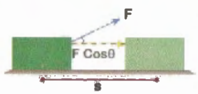
(-) ve sign indicates as x increases, y decreases

Work Power Energy

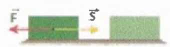


Work

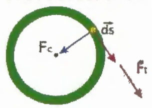
Work done by a constant force



$W = \vec{F}\cdot\vec{s} = Fs\cos\theta$



c) If θ is obtuse, work done is negative



If $\vec{F}$ and $\vec{s}$ are perpendicular, then work done will be zero, even though $F \neq 0$, $s \neq 0$.
eg., work done by centipetal force is zero, as $\vec{F}_c$ & $\vec{ds}$ are ⊥"

Work done by a variable force:
When the magnitude and direction of a force varies with position, work done by such a force (F) for an infinitesimal displacement ($\vec{ds}$) is given by, $dW = \vec{F}\cdot\vec{ds}$

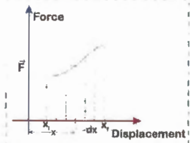
The total work done in going from A to B as shown in figure

$$W = \int_A^B \vec{F}\cdot\vec{ds} = \int_A^B (F\cos\theta).ds$$

$$dw = \vec{F}.\vec{dx}$$

$$\therefore w = \int_i^f \vec{F}.\vec{dx} = \int_{x_i}^{x_f} \text{Area of a strip of dx}$$

= Area under curve between x_i, x_f



If a force applied on a body displaces it, then work is said to be done by that force

Work done in conservative and non conservative fields

b) A force (like frictional force) is said to be non conservative force, if the work done by or against the force in moving a body from one position to another depends on path followed between those 2 points.

Work done against friction from A → B

$(w_f)_{AB} = \mu mgs$

$(w_f)_{A\to B} + (w_f)_{B\to A} \neq 0 = 2\mu mgs$

a) In gravitational field (conservative), work done is path independent.
eg. A body in taken from position A to B along different paths as shown.
Work done is same in all cases

Work depends on frame of reference.
eg., If a man lifts some weights on his head moves up a stair case, work done by the upward lifting force relative to him is zero, while relative to a person on the ground will be mgh.

Power(P)

Rock Climbers do a lot of work at small rate. The Power is less.

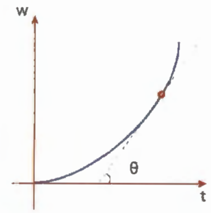
It is the rate at which work is done

Average power $= \dfrac{\Delta W}{\Delta t} = \dfrac{W}{t}$

Instantaneous power (P) $= \dfrac{dW}{dt} = \dfrac{\vec{F}.\vec{s}}{t} = \vec{F}.\dfrac{\vec{s}}{t}$

$\boxed{P = \vec{F}.\vec{V} = F.V\cos\theta}$

(θ - The angle between $\vec{F}$ & $\vec{V}$)



The slope of w-t curve gives the instantaneous power (P)

$$P = \frac{dW}{dt} = \tan\theta$$

Position and velocity (v) of a car with respect to time (t). Consider a car of mass (m) accelerating from rest, due to its engine delivering constant power (P)

Velocity: $P = F.v = ma.v = m.\dfrac{dv}{dt}.v = \text{constnat}$

$$\int_0^v v.dv = \frac{P}{m_0}.\int_0^t dt; \quad v = \left[\frac{2Pt}{m}\right]^{\frac{1}{2}}$$

Position: $v = \dfrac{ds}{dt} = \sqrt{\dfrac{2P}{m}}\sqrt{t}$

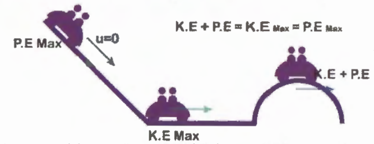
$$\int ds = \sqrt{\frac{2P}{m}}.\sqrt{t}\,dt$$

$$s = \sqrt{\frac{8P}{9m}}.t^{\frac{3}{2}}$$

Energy

It is the capacity for doing work

Law of Conservation of Energy



From work energy theorem, K2 - K1 = W = $\int \vec{F}.\vec{dr}$
But according to definition of P.E, in conservative field,

$U_2 - U_1 = -\int \vec{F}.\vec{dr}$
$K_2 - K_1 = -(U_2 - U_1)$
$K_2 + U_2 = K_1 + U_1$ ie., K + U = constant
ie., Sum of K.E & P.E is constant in presence of conservative forces

Kinetic Energy (KE):
It is the energy of a body by virtue of its motion

$$dW = \vec{F}.\vec{ds} = F.ds = m.a.ds = m.\frac{dv}{dt}.ds$$

$$= m.\frac{ds}{dt}.dv = m.v.dv$$

Work done in order to increase its velocity from zero to v

$$= W = \int_0^v mv\,dv = \frac{mv^2}{2}$$

This work done appears as K.E of body

$$W = \frac{1}{2}mv^2 = KE$$

Work-Energy theorem:
Work done by a force acting on a body = change in its K.E

$$W = \int_u^v mv\,dv = \frac{1}{2}mv^2 - \frac{1}{2}mu^2 = \Delta KE$$

Relation between KE (E) & momentum (P = mv)

$$E = \frac{P^2}{2m} = \frac{PV}{2} \Rightarrow P = \frac{2E}{v} = \sqrt{2mE}$$

Potential Energy (PE): It is the energy possessed by virtue of its position or configuration
- P.E is defined only for conservative forces
- Generally P.E is 3 types:
  a) Elastic P.E b) Electric P.E
  c) Gravitational P.E

Change in P.E: It is the work done by associated conservative force in displacing the body between the 2 points without change in K.E

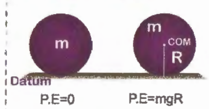
$$U_2 - U_1 = -\int_{r_1}^{r_2} \vec{F}.\vec{dr} = -dW$$

If the initial position is taken at infinity $r_1 \to \infty, r_2 = r$

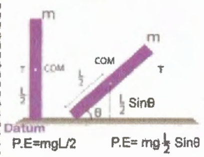
$$U = -\int_\infty^r \vec{F}.\vec{dr} = -W$$

ie., In case of conservative field (force). P.E = - work done by conservative force in shifting the body

Potential energy of different bodies:



Point mass



P.E = mgh
where 'h' is the height of COM of body above datum

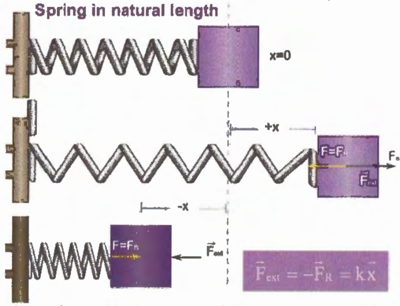
Elastic potential energy:
When spring is elongated or compressed by a displacement (x) by some external force (F_{ext}), an internal force called as Restoring force ($F = F_R$) will be developed within the spring

$\boxed{F \alpha - x \quad \text{(HOOKE'S LAW)}}$

F = -kx

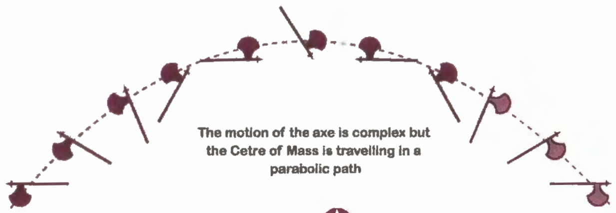
The work done by F_{ext} in stretching the spring through displacement is stored as Elastic P.E (U)

$$W = U = \frac{1}{2}kx^2 = \frac{1}{2}F.x = \frac{F^2}{2k}$$



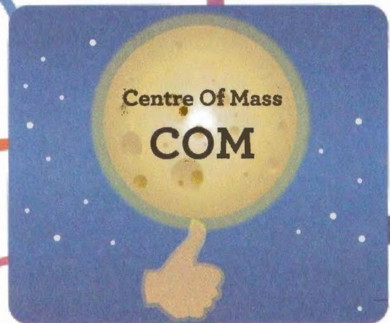
$\boxed{F_{ext} = -F_R = kx}$

COM and COG(Centre of Gravity) coincide, if the body is in **uniform gravitation field.** But as we go up, the lower portion of the body is closer to earth than the upper part (as in the case of the mountain, where the gravitational force is lesser) and so the **centre of gravity is slightly below the centre of mass.**



The point where the total mass of the body is imagined to be concentrated.

Why? because the motion of COM represents motion of entire body.



Centre of Gravity COG

Definition

Connected Bodies

$$x_{com} = \frac{L_1x_1 + L_2x_2 + \cdots}{L_1 + L_2 + \cdots}$$

One Dimensional (1d)

$$x_{com} = \frac{A_1x_1 + A_2x_2 + \cdots}{A_1 + A_2 + \cdots}$$

Two Dimensional (2d)

$$x_{com} = \frac{A_1x_1 - A_2x_2}{A_1 - A_2}$$

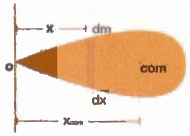
2d Distorted Bodies

$$x_{com} = \frac{V_1x_1 + V_2x_2 + \cdots}{V_1 + V_2 + \cdots}$$

Three Dimensional (3d)

Note: x_1, x_2 ....are the distances of components of bodies from origin, then x_{com} is also arrived with respect to origin

$$x_{com} = \frac{\int x\,.\,dm}{\int dm}$$



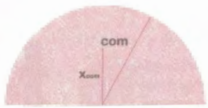
Pappus Theorem

I Law: The Volume(V) generated as a result of the revolution of a closed plane area about an axis (such that every point moves perpendicular to the plane) equals the **area (S)** of the plane, times the cirumference ($2\pi.x_{com}$) of the circle described by COM of plane, *i.e., $V = S.\ 2\pi.x_{com}$*

e.g, Consider a semi circular plate of **radius 'r'.**

It's area $S = \frac{\pi r^2}{2}$

$\frac{4}{3}\pi r^3 = \frac{\pi r^2}{2}.\ 2\pi x_{com}$ | $x_{com} = \frac{4r}{3\pi}$



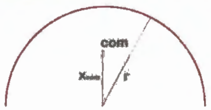
II Law: The **Surface area(S)** generated by revolution of a plane curved line about an axis (such that every point moves perpendicular to the line of curve) is equal to the product of length of the **line and the distance through which COM moves.**

eg., Consider a wire bent in the form of semi circular arc of radius 'r'.

The area swept out. i.e., Surface area of sphere generated is equal to the product of the length of plane curve and distance the COM moves.

$S = 4\pi r^2 = \pi r \times 2\pi x_{com}$

$x_{com} = \frac{2r}{\pi}$



Discrete Particles

$$x_{com} = \frac{m_1x_1 + m_2x_2 + \cdots}{m_1 + m_2 + m_3 + \cdots}$$, also similar for y_{com} and z_{com}

$$\vec{V}_{com} = \frac{m_1\frac{d\vec{r}_1}{dt} + m_2\frac{d\vec{r}_2}{dt} + \cdots}{m_1 + m_2 + \cdots} = \frac{m_1\vec{v}_1 + m_2\vec{v}_2 + \cdots}{m_1 + m_2 + \cdots}$$

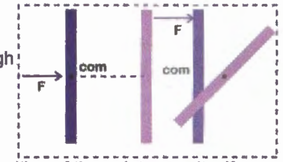
$$\vec{a}_{com} = \frac{m_1\frac{d\vec{v}_1}{dt} + m_2\frac{d\vec{v}_2}{dt} + \cdots}{m_1 + m_2 + \cdots} = \frac{m_1\vec{a}_1 + m_2\vec{a}_2 + \cdots}{m_1 + m_2 + \cdots}$$

$$\vec{a}_{com} = \frac{\vec{F}_1 + \vec{F}_2 + \cdots}{m_1 + m_2 + \cdots} = \frac{\text{net force an system}}{\text{total mass of system}}$$

Point Mass

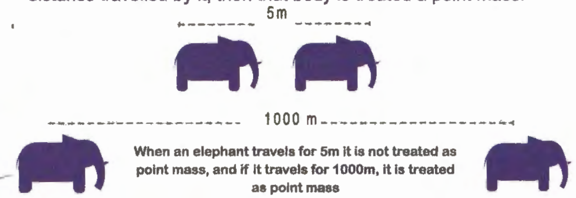
Did you Know

If the line of action of the force(F) applied on the body passes through the COM of the body, the body undergoes translatory motion, otherwise rotatory motion also



The COM of a point mass is the position of the point mass itself

If the dimensions of a body are negligible, when compared with the distance travelled by it, then that body is treated a point mass.



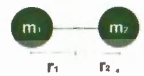
As the distance between sun and earth is veary large when compared with their sizes, they are treated as point masses

If the origin of the coordinate system is at COM.

then, $\vec{R}_{CM} = 0$, then $\frac{1}{M}.\sum m_i r_i = 0$

I.e, the sum of the moments of the masses of a system about its COM is always zero

Position of COM doesnot depend on the co-ordinate system. e.g, COM of a ring is at its centre whatever be the co-ordinate system.



COM is located towards heavier mass

If no external force acts on 2 bodies, and if they move towards each due to internal force between them (like gravitational force of attraction), then they meet at COM

If $\vec{x}_1, \vec{x}_2$ are their displacements,

then $m_1\vec{x}_1 = m_2(-\vec{x}_2)$

Net Force = Sum of external and internal forces, but vector sum of internal forces is **zero.**

Therefore **Net Force = Sum of external force**

If no external force acts, the **displacement of COM = 0**

COLLISIONS

- Can Kinetic Energy of a system be changed without changing its momentum?

A) YUP!!! In the explosion of a bomb or inelastic collision between two bodies, as force is internal, momentum is conserved while KE changes.

- Can momentum of a system be changed without changing the KE?

A) YUP again!!!! If a force acts perpendicular to motion, as in case of uniform circular motion, KE remains unchanged but as direction of velocity changes, momentum changes.

Law of Conservation of Linear Momentum (LOCOM)

WHAT IS LOCOM

From Newton's 2nd law, $\vec{F}_{ext} = \frac{d\vec{P}}{dt}$, if the external force $\vec{F}_{ext}$ is zero; the linear momentum $\vec{P}$ will remain constant.

LOCOM is independent of frame of reference though linear momentum depends on frame of reference.

LOCOM is the same for the person inside and outside the bus

NOTE: LOCOM is equivalent to Newton's 3rd law of motion. From LOCOM, $P_1 + P_2 = cons$

differentiating both sides with time

$$\frac{d\vec{P}_1}{dt} + \frac{d\vec{P}_2}{dt} = 0 \Rightarrow \vec{F}_1 + \vec{F}_2 = 0$$

i.e, for every action there is an equal and opposite reaction, which is the Newton's 3rd law. $\vec{F}_1 = -\vec{F}_2$

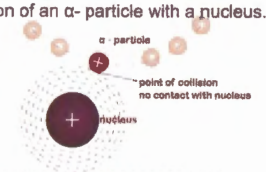
Definition

An isolated event in which a strong force acts between 2 or more bodies for a short time.

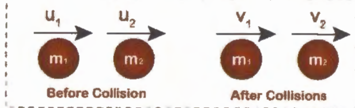
DURING COLLISION

Bodies may be in physical contact. eg: Collision between 2 billard balls

Bodies may not be in physical contact. eg: collision of an α- particle with a nucleus.



Newton's Law of Restitution



Before Collision	After Collision
For collision to occur $u_1 > u_2$	For seperation to occur $v_2 > v_1$
Relative velocity of approach = $u_1 - u_2$	Relative velocity of separation = $v_2 - v_1$

Coefficient of restitution(e) = $(v_2 - v_1)/(u_1 - u_2)$; 'e' does not depend on mass and site of bodies, but depends on body material.

System of Variable Mass

$\vec{V}_{rel}$ is relative velocity of mass entering or leaving the system w.r.t the system.

dm/dt is rate of change of mass of system. **$\vec{V}$.dm/dt** is the THRUST. Thrust is the force exerted on the system by the mass either entering or leaving the it.

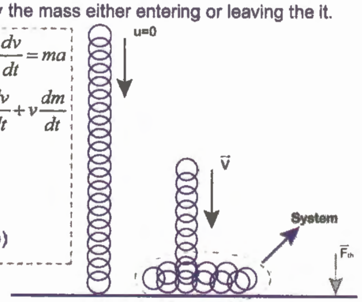
If mass (m) is constant, dm/dt = 0 $F_{ext} = m\frac{dv}{dt} = ma$

If mass (m) is variable, dm/dt ≠ 0 $F_{ext} = m\frac{dv}{dt} + v\frac{dm}{dt}$

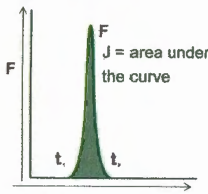
$V = \vec{v}_{rel} = \vec{v}_{mass\ entering/leaving\ system} - \vec{v}_{system}$

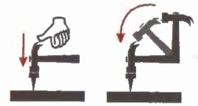
$\vec{V}_{rel} = \vec{V}_{chain} - \vec{V}_{system}$

$= V(-\hat{j}) - 0 = V(-\hat{j})$; dm/dt > 0 (+ve)



Impulse (J)





t=maximum Pressing nail in the thumb

t=minimum Hammering

J Is a very short duration force and is responsible for change in momentum.

$$J = \int F.dt = \int m.\frac{dv}{dt}.dt = \int m.dv = \Delta P$$

TYPES BASED ON

Energy

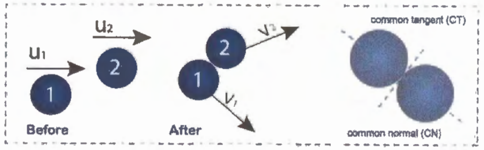
Both in elastic and inelastic collisions, momentum is conserved along the line of impact.

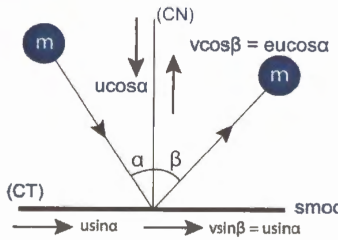
Elastic Collision:
Both KE and momentum of system remain constant before and after momentum. (e = 1)

Inelastic Collision:
Only **momentum** is conserved but there is loss of **KE**. (0 < e <1)

Line of Impact

Oblique Collisions





Momentum and velocities change along CN but not along CT.

IN CASE OF SMOOTH SURFACES

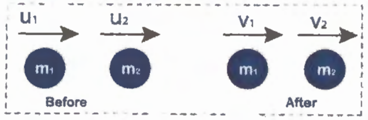
$$v = u\sqrt{\sin^2\alpha + e^2\cos^2\alpha}$$

$$\beta = \tan^{-1}[\frac{\tan\alpha}{e}]$$

1) For elastic collision (e),
v = u, β = α

2) **J_x = 0,**
J_y = mucosα(1 + e)

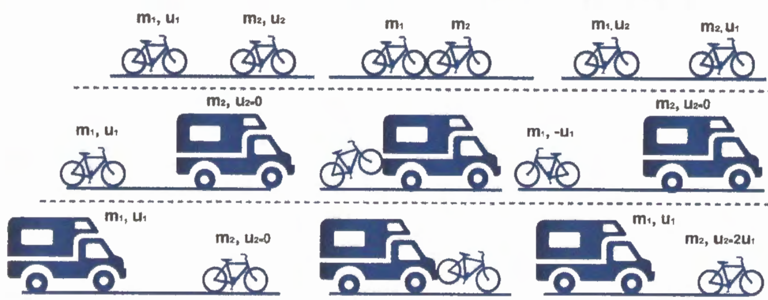
Head On Collisions



$$v_2 = [\frac{m_2 - em_1}{m_2 + m_1}]u_2 + [\frac{m_1 + em_1}{m_1 + m_2}]u_1$$

$$v_1 = [\frac{m_1 - em_2}{m_1 + m_2}]u_1 + [\frac{m_2 + em_2}{m_1 + m_2}]u_2$$

SPECIAL CASES



- If m₁ = m₂ for elastic collision velocities interchange.
- If m₂ >>> m₁ & u₂ = 0 then after collision, m₂ remains at rest and m₁ bounces back with same velocity.
- If m₁ >>> m₂ and u₂ = 0 then m₁ moves with same velicity and m₂ moves with double the velocity of m₁

For Gravitation

Definition

Gravitation is the force of attraction between any two bodies.

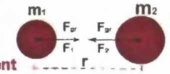
On Earth, this gravitational force is called Gravity.
Gravity is a special case of gravitation

Did you Know Gravitational Force is:

- Always attractive & does not depend on medium
- Applicable in case of point masses only
- A two body interaction. i.e., Gravitational force between two particles is independent of presence of other particles
- A conservative force. ie. work done by it is path independent

$F = -dv/dr;\ v = -\int \vec{E}.d\vec{r}$

Newton's Law of Gravitation:



Gravitation force $F_{gr} = \dfrac{G.m_1m_2}{r^2}$

G = Universal gravitational constant

$G = 6.67 \times 10^{-11}\ N\text{-}m^2/kg^2$

Gravitational Potential (V) Gravitational Field (E)

V

V at a point (P) is the work done in bringing unit mass (m=1) from infinity(∞) to that point without acceleration

Point Mass

$V_p = \dfrac{-GM}{r}$ $\quad V_p \propto \dfrac{1}{r}$

E

E at a point is the gravitational force experienced by a unit mass kept at that point

$E_p = \dfrac{-GM}{r^2}$ $\quad E_p \propto \dfrac{1}{r^2}$

Points to Remember

Gravitational potential is a scalar quantity and is always negative and is zero (ie, maximum) at a finite distance.

'E' is a vector quantity and is directed towards the point mass.

'E' at a point in the field is the accelaration of unit mass placed at that point.

Force experienced at a point mass(m) kept at a point is gravitational field, and is given by

$\vec{F} = m\vec{E}$

Relation between E and V

$\vec{E} = -\text{gradient } v$

$\vec{E} = -\dfrac{\partial v}{\partial x}\hat{i} + \dfrac{\partial v}{\partial y}\hat{j} + \dfrac{\partial v}{\partial z}\hat{k}$

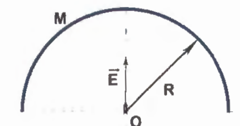
$\dfrac{\partial v}{\partial x}$ = Partial derivative of potential function 'v' w.r.t x, assuming y,z constant

$dv = -\vec{E}.\vec{dr}$

$\vec{dr} = dx(\hat{i}) + dy(\hat{j}) + dz(\hat{k})$

$\vec{E} = E_x(\hat{i}) + E_y(\hat{j}) + E_z(\hat{k})$

V, E Due to semi-circular arc



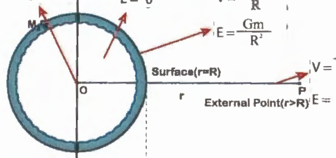
At point O; $V = -\dfrac{GM}{R}$ $\quad E = \dfrac{2GM}{\pi R^2}$

V, E Due to rod at a point on its axis

At point P;

$V = -\dfrac{GM}{L}\log_e\left[\dfrac{a+L}{a}\right]$ $\quad E = \dfrac{GM}{L(a+L)}$

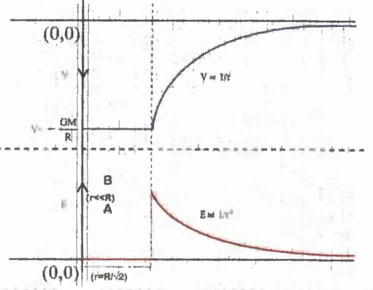
V, E Due to Spherical Shell



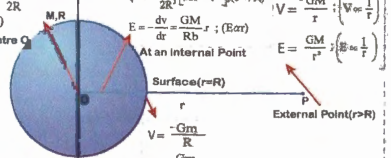
Inside: $V = \dfrac{-Gm}{R}$, $E = 0$

Surface(r=R): $V = \dfrac{-Gm}{R}$, $E = \dfrac{Gm}{R^2}$

External Point(r>R): $V = -\dfrac{GM}{r}$; $\left(V \propto \dfrac{1}{r}\right)$, $E = \dfrac{GM}{r^2}$; $\left(E \propto \dfrac{1}{r^2}\right)$



V, E Due to Solid Sphere

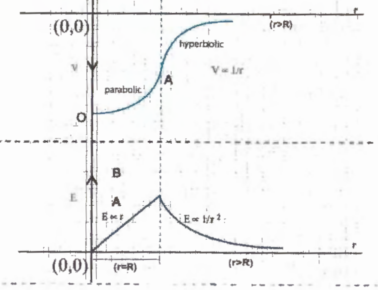


At centre O: $V = \dfrac{-3Gm}{2R}$, $E = 0$

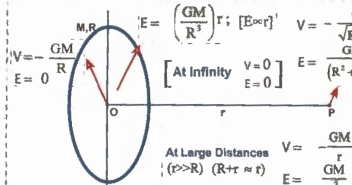
At an internal Point: $V = -\dfrac{Gm}{2R^3}[3R^2 - r^2]$; (0→A), $E = -\dfrac{dv}{dr} = \dfrac{GM}{Rb}r$; $(E \propto r)$

Surface(r=R): $V = \dfrac{-Gm}{R}$, $E = \dfrac{Gm}{R^2}$

External Point(r>R): $V = -\dfrac{GM}{r}$; $\left(V \propto \dfrac{1}{r}\right)$, $E = \dfrac{GM}{r^2}$; $\left(E \propto \dfrac{1}{r^2}\right)$



V, E Due to Circular Ring

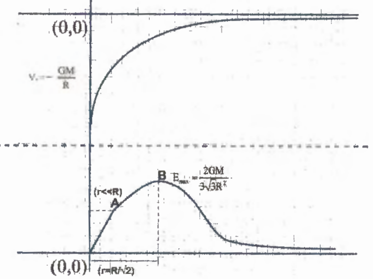


At centre: $V = -\dfrac{GM}{R}$, $E = 0$

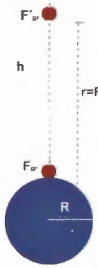
$E = \left(\dfrac{GM}{R^3}\right)r$; $[E \propto r]$ $\quad V = -\dfrac{GM}{\sqrt{R^2+r^2}}$ $\quad E = \dfrac{GMr}{(R^2+r^2)^{3/2}}$

At Infinity $\left[\begin{array}{l} V=0 \\ E=0 \end{array}\right]$

At Large Distances (r>>R) (R+r ≈ r): $V = -\dfrac{GM}{r}$ $\quad E = \dfrac{GM}{r^2}$



for E to be maximum, $\dfrac{dE}{dr} = 0$. ie., $r = R/\sqrt{2}$ $\quad E_{max} = \dfrac{2GM}{3\sqrt{3}R^2}$



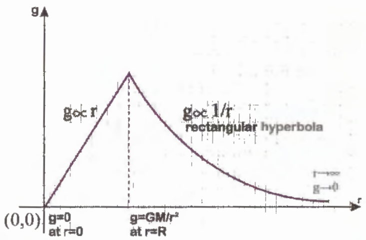
$$F'_{gr}=\frac{GM}{r^2}=mg'=\frac{GMm}{(R+h)^2}$$

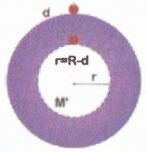
$$F_{gr}=\frac{GMm}{R^2}=mg$$

$$g'=g\left[\frac{R}{R+h}\right]^{2}$$

For small heights (h<<R)

$$g'=g\left[1-\frac{2h}{R}\right]$$





M' = Mass of (shaded) spherical particles of earth of radius r responsible for gravitational force called as EFFECTIVE MASS.

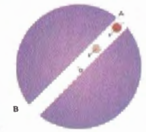
$$g=\frac{GM}{R^2}$$

$$g'=\frac{GM'}{(R-d)^2}=\frac{GM}{(R-d)^2}\left(\frac{R-d}{R}\right)^3=\frac{GM}{R^3}(R-d)$$

$$g'=\frac{GM}{R^3}r \qquad g'\propto r$$

If density (ρ) of earth is taken constant all through

$$\rho=\frac{M}{\frac{4}{3}\pi R^3}=\frac{M'}{\frac{4}{3}\pi R^3} \qquad M'=M\left(\frac{r}{R}\right)^3 \qquad M'=M\left(\frac{R-d}{R}\right)^3$$



Motion of a body dropped in to a tunnel dug along diameter or chord of earth

$$a=g'=-\frac{GM}{R^3}r$$

It resembles standard SHM eq: $a=-\omega^2x$

$$\omega=\sqrt{\frac{GM}{R^3}}=\sqrt{\frac{g}{R}}$$

$$T=\frac{2\pi}{\omega}=2\pi\sqrt{\frac{R}{g}}\approx 83.6\text{ min}$$

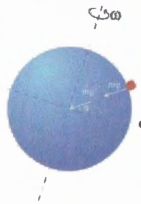
$$t_{A\to O}=t_{O\to B}=t_{B\to O}=t_{O\to A}=\frac{T}{4}$$

Above the Surface of the Earth (h)

Below the Surface of the Earth (d)

Rotation of Earth & Latitude (Φ)

Variation of Acceleration Due to Gravity



$$\omega=\frac{2\pi}{T}=\frac{2\pi}{24\times3600}\frac{rad}{sec}$$

$$mg=mg-mR\omega^2\cos^2\varnothing$$

Apparent wt = True wt − centrifugal force

$$g^1=g\left[1-\frac{R\omega^2}{g}\cos^2\varnothing\right]$$

a) at poles $(\varnothing=90^0)$ $g^1=g$

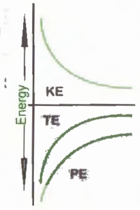
b) at equator $(\varnothing=0^0)$ $g^1=g-R\omega^2$

For body at equator to feel weightless $(mg^1=0)$

$$\omega=\sqrt{\frac{g}{r}}$$

$$T=2\pi\sqrt{\frac{R}{g}}\approx 83.6\ MIN$$

The Time Period of the earth should be 83.6 min



Energy(E)

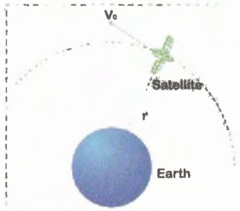
$$P.E=\frac{-GMm}{r}$$

$$K.E=\frac{1}{2}mv_0^2=\frac{GMm}{2r}$$

$$T.E=\frac{-GMm}{2r}$$

P.E : K.E : T.E = -2: +1: -1

Satellites



(V_0) Orbital Velocity

The velocity required by a satellite to put it into its orbit around the earth. Gravitational pull provides the required centripetal force

$$\frac{mV_0^2}{r}=\frac{GMm}{r^2} \qquad V_0=\sqrt{\frac{GM}{r}}\ (r=R+h)$$

Note: If satellite is very close to earth (h<<R),

$$R+h\approx R\ ;\ v_0=\sqrt{\frac{GM}{R}}=\frac{v_e}{\sqrt{2}}=7.9\text{ km/sec}$$

$$v_0=\sqrt{\frac{G.\rho\frac{4}{3}\pi R^3}{R}}=\left(\sqrt{\frac{4}{3}\rho.G.\pi}\right)R$$

V_0 does not depend on the mass of satellite

The orbit of a satellite follows a spiral path

Time Period (T)

$$\text{Time Period (T)}=\frac{2\pi r}{V_0}=\frac{2\pi r}{\sqrt{\frac{GM}{r}}}$$

$$T=\frac{2\pi}{\sqrt{GM}}r^{3/2} \qquad (T^2\propto R^3)$$

Points to remember

1) If satellite is very close to earth

$$r\approx R\ ;\ GM=gr^2$$

$$\therefore T=2\pi\sqrt{\frac{R}{g}}\approx 84\text{ min}\approx 1.4\text{ hr}$$

2) **Geostationary satellite:** Appears stationary relative to earth and its orbit is called as Parking orbit.

a) It revolves in an orbit concentric and coplanar with equitarial plane

b) It's sense of rotation is same as earth ie., from west to east with T = 24 hr.

c) Height of such geostationary satellite above earth surface = h = 6R ~ 36,000 km

Angular Momentum(L_0)

Angular momentum L_0 of satellite about the centre (O) of orbit

$$L_0=m\cdot V_0.r$$

$$=m\sqrt{\frac{GM}{r}}\cdot r$$

$$L_0=m\sqrt{GM.r}$$

To remember:When satellite is transformed to higher orbit (r2) from lower orbit (r1)

Qty	Variation	Relation with r
V_0	Decreases	$V\propto 1/\sqrt{r}$
T	Increases	$T\propto r^{3/2}$
L_0	Decreases	$P\propto 1/\sqrt{r}$
M	Increases	$L\propto\sqrt{r}$
K.E	Decreases	$k\propto 1/r$
PE	Increases	$U\propto -1/r$
T.E	Increases	$E\propto -1/r$
B.E	Decreases	$B.E\propto 1/r$

Gravitational energy of two point masses

Binding Energy (B.E)

$$TE=PE+KE=\frac{-GMm}{R}+0$$

$$B.E=|T.E|=\frac{GMm}{R}$$

B.E is the energy due to which the bodies are bound to each other or The minimum energy required to separate them to infinite distance.

Escape Velocity(Ve):

It is the minimum velocity with which a body has to be projected so as to just escape the Gravitational Field.

$$\text{T.E(total energy) on surface of earth}=\frac{-GMm}{R}+\frac{1}{2}mv_e^2$$

T.E on crossing the gravitational field $=\frac{-GMm}{r}+\frac{1}{2}mv^2=0\ \ (r\to\infty, v=0)$

$$v_e=\sqrt{\frac{2GM}{R}}=\sqrt{2GR}\approx 11.2\text{ km/sec}$$

Points to remember.

1) For a body projected with velocity Ve, T.E at any position during its journey will be zero.

2) Ve doesn't depend on mass, size and angle of projection

Maximum height reached by a body

1. T.E at maximum height $=\frac{-GMm}{R+h}$

2. T.E on surface of earth $=\frac{-GMm}{R}+\frac{1}{2}mv^2$

Solving 1, 2; $h=\frac{v^2}{2g-\frac{v^2}{R}}$

Note: If v is very small, $h=\frac{v^2}{2g}$

Work done against gravity:

T.E at height = h $=\frac{-GMm}{R+h}$

T.E on surface $=\frac{-GMm}{R}$

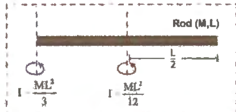
$$W=\Delta TE=\frac{-GMmh}{R^2[1+\frac{h}{R}]}=\frac{mgh}{1+\frac{h}{R}}$$

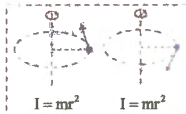
Note: If h<<R ; W = mgh

Moment of Inertia

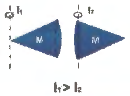
It is the property of a body due to which it opposes any change in its state of rest or of uniform rotation.

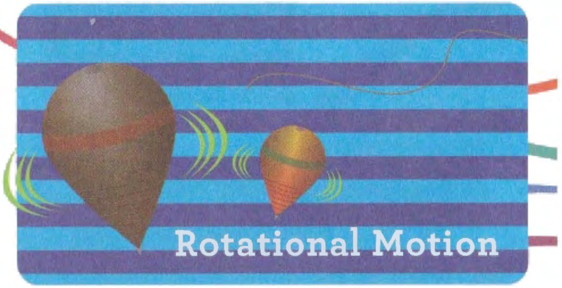
- M.I is a TENSOR, because it is not a SCALAR , as it has different values about different axis of rotation,and it is not a VECTOR, because clockwise or anticlockwise rotation does not alter the value.



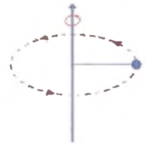


- If more mass is located away from the axis of rotation more will be the M.I
- More the mass, more will be the M.I



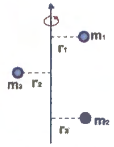


M.I of point mass



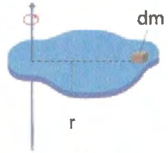
$$I = mr^2$$

M.I of discrete Particle system



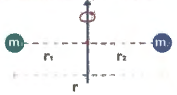
$$I = m_1r_1^2 + m_2r_2^2 + \cdots$$

M.I of body with continuous mass distribution



$$I = \int r^2\,dm$$

M.I of 2 point masses about their COM



$$r_1 + r_2 = r$$

$$m_1r_1 = m_2r_2$$

$$r_1 = \frac{m_2}{m_1+m_2}.r$$

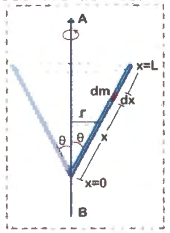
$$r_2 = \frac{m_1}{m_1+m_2}.r$$

$$I = I_1 + I_2 = m_1r_1^2 + m_2r_2^2$$

$$= \frac{m_1m_2}{m_1+m_2}r^2 = \mu r^2$$

$$\mu = \text{Reduced mass} = \frac{m_1m_2}{m_1+m_2}$$

M.I of uniform rod



M.I of element of mass(dm) about AB = dI = $dm.r^2$

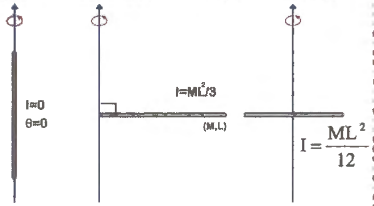
$$dI = \frac{m}{L}.dx.\left[x\sin\theta^2\right]$$

M.I of total rod = $I = \int dI$

$$= \frac{M}{L}.\sin^2\theta.\int_0^L x^2.dx$$

$$I = \frac{ML^2}{3}.\sin^2\theta$$

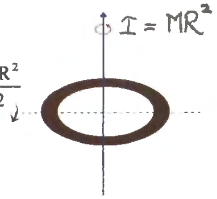
$$AB = \frac{ML^2}{12}.\sin^2\theta$$



$$I = \frac{ML^2}{12}$$

$I = MR^2$

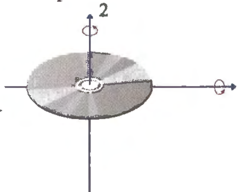
$\frac{MR^2}{2}$



Ring

$I = \frac{MR^2}{2}$

$I = \frac{MR^2}{4}$

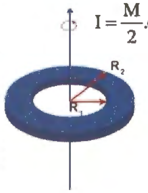


Disc

$I = \frac{MR^2}{2}$

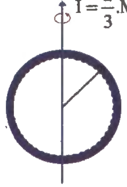
Hollow Cylinder

$I = \frac{M}{2}.(R_1^2 + R_2^2)$



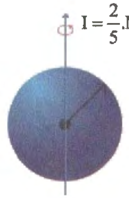
Annular Ring

$I = \frac{2}{3}.MR^2$



Hollow Sphere

$I = \frac{2}{5}.MR^2$

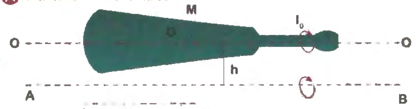


Solid Sphere

$I = \frac{MR^2}{2}$

Solid Cylinder

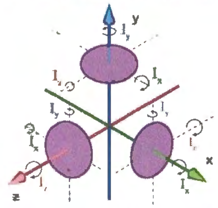
Parallel Axis Theorem:



$$I_{AB} = I_0 + Mh^2$$

Parallel axis theorem is applicable for 2-D or 3-D bodies but, Perpendicular Axis Theorem is applicable for only 2-D bodies

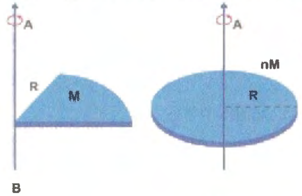
Perpendicular Axis Theorem:



For body in x-z plane:
$I_y = I_x + I_z$

For body in z-y plane:
$I_x = I_y + I_z$

For body in z-y plane:
$I_z = I_x + I_y$



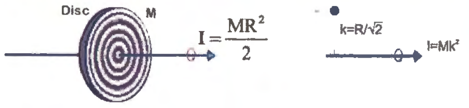
- Formula of M.I of a part of a rigid body (symmetrically cut from the whole mass) is the same as that of whole body about same axis of rotation

Let the sector be nth part of disc of sector of mass = m, then disc mass = n.M

$$I_{disc} = \frac{1}{2}(nM)R^2$$

$$I_{sector} = \frac{1}{n}.I_{disc} = \frac{1}{2}MR^2$$

Radius of gyration of a body about an axis is the perpendicular distance of a point from the same axis, where whole mas of the body were concentrated, the body shall have same M.I as it has with the acual distribution of mass.

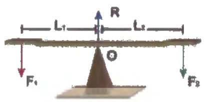


$I = \frac{MR^2}{2}$

$I = Mk^2 = mr_1^2 + mr_2^2 + \cdots$

$(M = n.m)$

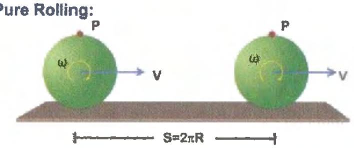
$\boxed{K = \sqrt{\frac{r_1^2 + r_2^2 + \cdots}{n}}}$



Rotational Equilibrium: A body is said to be in rotational equilibrium if net torque on it is zero.
Taking moments about (o)
clock wise moment = anti clock wise moment

$\boxed{F_2 \times L_2 = F_1 \times L_1}$

Pure Rolling:



$S = V.T = Rw.\frac{2\pi}{w};$

$\frac{S}{T} = \frac{2\pi}{T}.R\,;\, V = \omega.R$

$\frac{V}{T} = \frac{\omega}{T}.R\,;$

$S = 2\pi R$
$V = \omega.R$
$a = \alpha.R$

Conditions for pure rolling

If S > 2π R (V > ωR) → Forward slipping
If S < 2π R (V < ωR) → Backward slipping

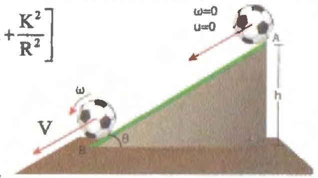
KE of body rolling down an inclined plane

$TE_A = TE_B$

$mgh = \frac{1}{2}mv^2.\left[1 + \frac{K^2}{R^2}\right]$

$V = \frac{\sqrt{2gh}}{\sqrt{1 + \frac{K^2}{R^2}}}$

$V_{Rolling} = \frac{V_{Sliding}}{\sqrt{1 + \frac{K^2}{R^2}}}$



from $v^2 - u^2 = 2as\,;\, \left[u = 0;\, s = \frac{h}{\sin\theta}\right]$

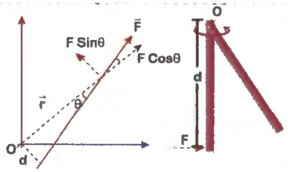
$a_{Rolling} = \frac{g\sin\theta}{1 + \frac{K^2}{R^2}} = \frac{a_{Sliding}}{1 + \frac{K^2}{R^2}}$

$t = \frac{1}{\sin\theta}\sqrt{\frac{2h}{g}\left(1 + \frac{K^2}{R^2}\right)}$

Radius of Gyration

Torque (τ)

Rotational effect of a Force
A body pivoted about a part (o), when rotated under the action of a force, is said to be acted upon by a torque.
If a particle (p) is acted up on by a force $\vec{F}$, whose position vector is $\vec{r}$, then torque,



$\boxed{\vec{\zeta}_0 = \vec{r} \times \vec{F}}$

The radial component of, $\vec{F} = F\cos\theta$
but the transversal component of $\vec{F} = F\sin\theta$ rotates.
In scalar form,

$\boxed{\zeta_0 = r\,F\sin\theta = rd}$

Pure Rolling

Angular Momentum

Rolling Motion

KE of a body rolling on horizontal surface

Translation + Rotation = ROLLING

$KE_{Tr} = \frac{1}{2}mv^2$

$KE_{Rot} = \frac{1}{2}I\omega^2$

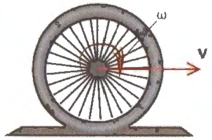
$= \frac{1}{2}mk^2.\frac{v^2}{R^2}$

$KE_{Rolling} = KE_{Tr} + KE_{Rot}$

$= \frac{1}{2}mv^2.\left[1 + \frac{k^2}{R^2}\right]$

$\frac{KE_{Total}}{KE_{Tr}} = 1 + \frac{k^2}{R^2}$

$\therefore \frac{KE_{Rot}}{KE_{Tr}} = \frac{k^2}{R^2}$



Accelerated pure rolling (f = frictional force)

If

$a = R\alpha,\, f = 0$

$a < R\alpha$, f acts forward

$a > R\alpha$, f acts backward

$a = \frac{F + f}{M} - (i)$

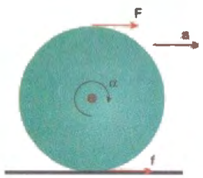
$\alpha = \frac{\zeta}{R} = \frac{(F\text{-}f)R}{I} - (ii)$

$a = R\alpha - (iii)$

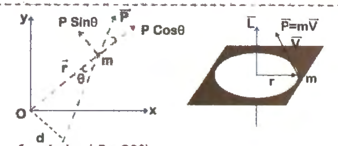
solving $(i),(ii)(iii)$

$f = \left(\frac{MR^2 - I}{MR^2 + I}\right).F$

(If $I = MR^2; f = 0$
$I < MR^2$; f is forward
$I > MR^2$; f is backward)



Angular momentum ($\vec{L}$):
It is the moment of linear momentum of a body w.r.t any axis of notation



$\vec{L} = \vec{r} \times \vec{P}$

$L = mvr\sin\theta$

When a body moves on circumference of a circle, (θ= 90°)

$\therefore L = mvr = mr^2.\frac{V}{r}$

$\boxed{L_{Rot} = I.\omega}$ $\boxed{L_{Translational} = L_{Tr} = mvr}$

$\boxed{K.E_{Tr} = \frac{1}{2}mv^2}$ $\boxed{K.E_{Rot} = \frac{1}{2}mr^2.\frac{v^2}{r^2} = \frac{1}{2}\omega^2 = \frac{L^2}{2I}}$

$\boxed{\text{Power (P)} = \vec{\tau}.\vec{\omega}}$

Work: If the body is initially at rest and angular displacement is 'dθ ' due to torque, then

$\boxed{w = \int \tau.d\theta}$

Law of conservatiion of angular momentum:
External torque

$\tau_{ext} = \frac{d}{dt}[L] = \frac{d}{dt}[I\omega]$

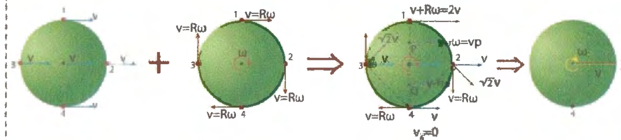
$\tau_{ext} = I.\frac{d\omega}{dt} \Rightarrow \boxed{\tau_{ext} = I\alpha}$

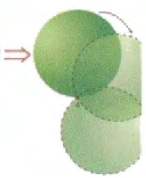
when, $\tau_{ext} = 0$; L = angular moment remains constant

$I\omega$ =constant;

$I\alpha\frac{1}{\omega}$

Translation + Rotation ⇒ Rolling





Point 4 is called instantaneous point of rotation because
a) It's linear velocity is zero
b) Angular velocity (ω) of all other points about point (4) is same.

$\omega_1 = \frac{2V}{2R} = \frac{V}{R};\quad \omega_2 = \frac{\sqrt{2}V}{\sqrt{2}R} = \frac{V}{R} = \omega_3$

Simple Harmonic Motion

Definition

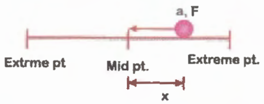
1. SHM is oscillatory motion, where in the time period is independent of amplitude.

2. In SHM, acceleration (Restoring force) is directly proportional and opposite in direction to displacement.

$a = -\omega^2 \cdot y$ $\qquad$ $\alpha = -\omega^2 \cdot \theta$

$F = -ky$ $\qquad$ $\tau = -c \cdot \theta$

(Linear SHM) $\qquad$ (Angular SHM)

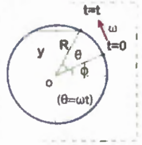


Ⓠ If, F = + ky, then what will happen?

3. In SHM, angular frequency (ω) is constant

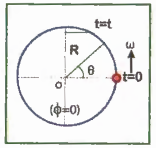
Phase

Fig(a)



Phase constant(ϕ): The constant 'ϕ'is called epoch or phase constant or initial phase. This is to find the position from where time is considered in SHM.

Fig(b)

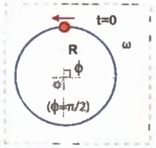


If we consider the motion from equilibrium position ie, e t = 0 ;y=0 , the equation, y=A sin(nωt+ϕ) gives, 0 = A sin(ωx0+ϕ) ⇒ ϕ=0; ∴ y = A sin(ωt).

Y=A sin(ωt+ϕ) (Starting from eq.position)

Y=A sinωt (Starting from M.P)

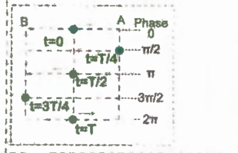
Fig(c)



If we consider the motion from extreme position ie, @ t = 0 , y = a ;the equation, y = A sin(ωx0+ϕ)=A ⇒ sinϕ = 1 ⇒ ϕ = Π/2; y = Acosωt

Y= A cosωt (Starting from ex.p)

Phase: The argument (ωt+ϕ) of sine function is called phase. The phase of an oscillating system at any instant is its state as regards to its position and direction of motion at that instant.



Kinematics in SHM

Relation between y,v, and a

$y = A\sin\omega t$

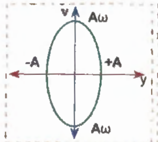
$v = \dfrac{dy}{dt} = A\,\omega\cos\omega t = A\omega\sqrt{1-\sin^2\omega t}$

$v = \omega\sqrt{A^2 - y^2}$

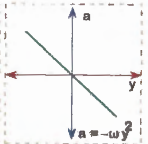
$\dfrac{v^2}{(A\omega)^2} + \dfrac{y^2}{A^2} = 1 \Rightarrow \dfrac{v^2}{v_{max}^2} + \dfrac{y^2}{y_{max}^2} = 1[Ellipse]$

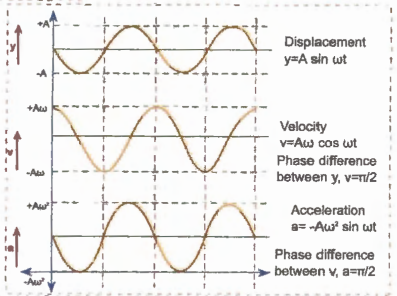
$a = \dfrac{dv}{dt} = -A\omega^2\sin\omega t = -\omega^2 y,$

$a = -\omega^2 y$

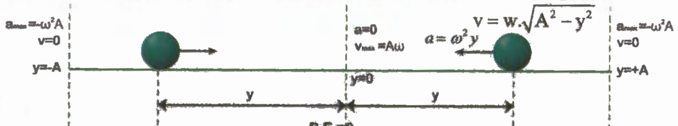


$v^2 = \omega\sqrt{A^2 - y^2}$





Energy consideration in SHM



P.E = $\frac{1}{2}m\omega^2 y^2$

K.E= $\frac{1}{2}m\omega^2(A^2 - y^2)$

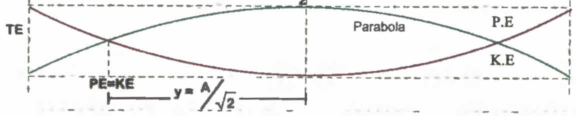
P.E =0

K.E max = $\frac{1}{2}m\omega^2 A^2$

T.E = $\frac{1}{2}m\omega^2 A^2$ = constant

KE =0

$PE_{max} = \frac{1}{2}m\omega^2 A^2$

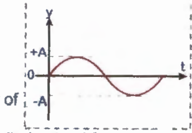


* $KE_{avg} = PE_{avg} = \dfrac{TE}{2} = \dfrac{1}{4}m\omega^2 A^2$

Oscillations

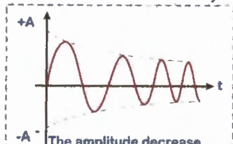
A. Free Oscillations:

1. The oscillations of a particle with fundamental frequency under the influenceof restoring force are free oscillations.
2. The amplitude, frequency and energy of oscillations remain constant
3. The frequency of free oscillations is called natural frequency because it depends upon the nature and structure of the body.



B. Damped Oscillations:

1. In this,amplitude and hence energy of oscillator decrease exponentially due to damping forces like frictional forces, viscous forces etc.
2. If Restoring force F= -KX
Damping force F'= -bv



The amplitude decrease exponentially with time as

$x = x_m \cdot e^{-\left(\frac{b}{2m}\right)t}$

C. Forced Oscillations:

1. The oscillations in which a body oscillates under the influence of an external periodic force is known as forced oscillations.
2. The amplitude of oscillator would have decreased due to damping forces but due to the energy gained from external source, it remains constant.

Superposition of two SHMs

In same direction and of same frequency

$x_1 = A_1\sin\omega t$

$x_2 = A_2\sin(\omega t + \theta)$

Resultant displacement

$x = x_1 + x_2$

$= A_1\sin\omega t + A_2\sin(\omega t + \theta)$

$= A\sin(\omega t + \varphi)$

(which is SHM)

Where , A=amplitude of resultant wave

$A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos\theta}$

$\varphi = \tan^{-1}\left[\dfrac{A_2\sin\theta}{A_1 + A_2\cos\theta}\right]$

In same direction but of different frequencies.

$x_1 = A_1\sin\omega_1 t$

$x_2 = A_2\sin\omega_2 t$

$x = x_1 + x_2$

$= A_1\sin\omega_1 t + A_2\sin\omega_2 t$

The resultant motion is not SHM

In two perpendicular directions

$x_1 = A\sin\omega t$

$x_2 = B\sin(\omega t + \pi/2)$

$= B\cos(\omega t)$

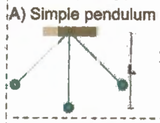
$\dfrac{x_1^2}{A^2} + \dfrac{x_2^2}{B^2} = 1(Ellipse)$

$if\ A = B = R\ (circle)$

$then\ x^2 + y^2 = R^2$

Examples

A) Simple pendulum



$T = 2\pi\sqrt{\dfrac{l}{g}}$

Simple pendulum of very large legth(L) (R=Radius of earth)

$T = 2\pi\sqrt{\dfrac{1}{g\left(\frac{1}{L}+\frac{1}{R}\right)}}$

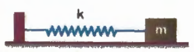
If L >> R,T ≈ 84.6min

If L=R, T = 1Hr

Seconds pendulum: whose T= 2sec

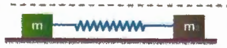
L= 1m

B) Spring pendulum



If the spring is a massive (mass=M). then

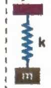
$T = 2\pi\sqrt{\dfrac{m + M/3}{k}}$



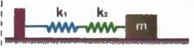
$T = 2\pi\sqrt{\dfrac{\mu}{k}}$

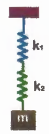
$\mu = reduced\ mass = \dfrac{m_1 m_2}{m_1 + m_2}$

Springs in series



$T = 2\pi\sqrt{\dfrac{m}{k}}$

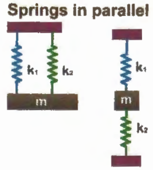




$k_s = \dfrac{k_1 k_2}{k_1 + k_2}$

$T_s = 2\pi\sqrt{\dfrac{m}{k_s}}$

Springs in parallel

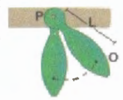


$K_p = K_1 + K_2$

$T_p = 2\pi\sqrt{\dfrac{m}{k_p}}$

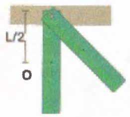
Physical Pendulum:

L = distance from point of projection(p) to COM(o).



$T = 2\pi\sqrt{\dfrac{Ip}{mgL}}$

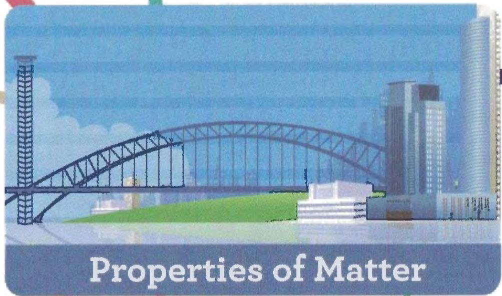
$Ip = M.I\quad about(p)$



$I_p = \dfrac{ML^2}{3}$

eg: $T = 2\pi\sqrt{\dfrac{2L}{3g}}$

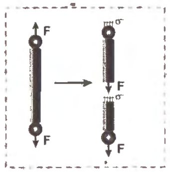
Properties of Matter



Elasticity

The property by virtue of which a body regains its original shape and size when deforming force is applied on it is removed is called Elasticity. If it does not regain so, it is called Plasticity.

Stress is the restoring force (F_R) per unit cross section area and is measured by the magnitude of deforming force (F_d) acting on unit cross sectional area (A) of the body when equilibrium is reached



$$\text{Stress} = \frac{F_d (\text{or } F_R)}{A}$$

It is a TENSOR

Strain

Longitudinal strain = $\frac{\Delta L}{L}$

Volumetric strain = $\frac{\Delta V}{V}$

Hooke's Law

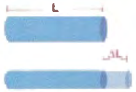
Within elastic limit, stress α strain

Stress = E. Strain

$$E = \frac{\text{Stress}}{\text{Strain}}$$

Types of modulus

Young's modulus (y)



Stress = F / A

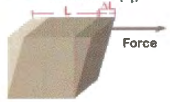
Strain = ΔL / L

$$Y = \frac{F.L}{A.\Delta L}$$

$\frac{F}{\Delta L}$ = Force per unit elongation called as Force Constant(k)

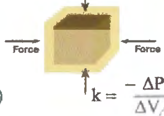
$$k = \frac{A.Y}{L}$$

Shear Modulus (η)



$$\eta = \frac{F}{\Delta L/L}$$

Bulk modulus (k)



$$k = -\frac{\Delta P}{\Delta V/V}$$

Energy stored in a deforming body (strain energy)

$$U = \frac{1}{2} F.\Delta L$$

$$= \frac{1}{2} \text{stress} \times \text{strain} \times \text{volume}$$

Inter atomic force constant (k)

$k = Y.r_0$

r_0 = distance between ends of wire

Poisson's Ratio (s)

$$\sigma = \frac{\text{Lateral strain}}{\text{Longitudial strain}} = -\frac{\Delta r/r}{\Delta L/L}$$

(–) because, when length (L) increases radius (r) decreases.

Theoritical limit for (σ) = -1 to +0.5

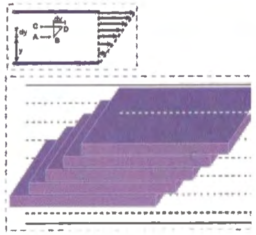
Practical limit for (σ) = 0 to +0.5

Relation between Y, k, η, σ

Y = 3k (1- 2σ); Y = 2η(1+ σ)

Viscosity

Viscosity: It is the property of the liquid by virtue of which it opposes the relative motion between its adjacent layers.



$$\text{Viscous Force (F)} = -\eta A \frac{dV}{dy} \text{ (Newton's law)}$$

(-) direction indicates $\vec{F}$ is opposite to the relative motion of fluid layer.

SI Unit → POISEUILEE (PI)

CGS Unit → dyne-sec / m² = POISE, 1PI = 10 POISE

Points to remember:

1) $\eta_{liquids} \propto \frac{1}{\sqrt{T}}$, $\eta_{gases} \propto \sqrt{T}$

2) Viscosity of water decreases with increase in pressure but of other liquids increases with increase in pressure, for gases **viscosity**, it is independent of pressure.

STOKE'S law: Viscous force (F) on a ball of radius (r) sinking in a liquid with velocity 'v' is given by

$F = 6\pi\eta r v$

POISEULLI'S FORMULA:

Volume of fluid / sec = $\frac{dQ}{dt}$

Length (L), radius (r) of capillary tube, P - pressure difference across the table

$$\frac{dQ}{dt} = \frac{\pi P r^4}{8\eta L}$$

Stoke's law

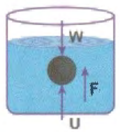
Terminal Velocity:

V = Volume of ball = $\frac{4}{3}\pi r^3$

ρ_L, ρ_b = density of liquid, ball

U = up thrust

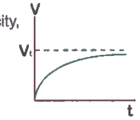
$= \rho_L.V.g$



Net acceleration (a) = $\frac{W-(F+U)}{m}$

When ball gains a constant velocity, called terminal velocity, a = 0

$$V_T = \frac{2}{g} . r^2 \left[\frac{\rho_b - \rho_L}{\eta}\right] g$$



- $V_T \propto r^2$
- Greater the viscosity, density of liquid lesser will be the terminal velocity
- Viscous force acts only when the ball is in motion, but upthrust (ϖ) acts when the body is at rest or motion.
- The maximum velocity up to which fluid motion is steady is called critical velocity (V_c)

$$V_C = R_C . \frac{\eta}{\rho . r}$$

η , ρ = viscosity, density of liquid

r = radius of tube; R_e= Reynold's number

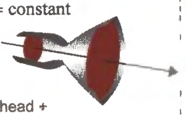
For steady flow $R_e \le 2000$

Equations

Bernoulli's equation: In case of nonviscous, incompressible fluid along a streamline flow,

$$P_{pressure}(P) + \frac{P.E + K.E}{\text{volume}} = \text{constant}$$

$$\frac{P}{\rho g} + \frac{v^2}{2g} + h = \text{constant}$$



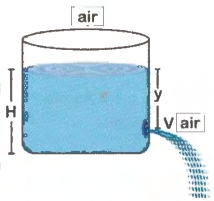
Pressure head + velocity head + gravitational head = constant

TORRICELLI'S theorem:

Velocity of efflux (V) = $\sqrt{2gy}$

1) It is valid if cross sectional area (a) of hole is very very small when compared with surface area of container (A)

2) The medium into which liquid jet is emerging should be same as the medium about liquid surface in container.



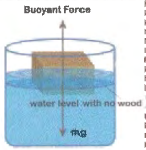
Buoyancy

Buoyancy - Floatation

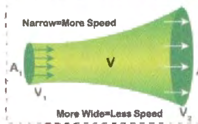
For equilibrium, mg = U

$$\rho_b.V.g = \rho_L.V_{in}.g$$

$$\frac{V_{in}}{V} = \frac{\rho_b}{\rho_L}$$



Continuity

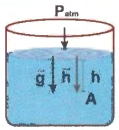


Continuity Equation:

In case of incompressible fluids,

$$\text{Rate of fluid flow} = \frac{\text{volume of fluid}}{\text{time}} = A_1V_1 = A_2V_2$$

Fluid Pressure

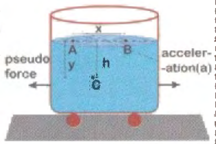


Gauge pressure at (A)

$$= \rho g h = \rho[\vec{g}.\vec{h}]$$

Angle between $\vec{g}, \vec{h} = 0°$)

Absolute Pressure = $P_{atm} + \rho g h$



$$P_B - P_A = \rho\, a.x$$

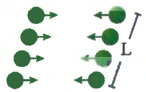
$$P_C - P_B = \rho\, y.g$$

$$P_C - P_A = \rho[ax + gy]$$

Surface Tension

Surface Tension is a property of liquid at rest by virtue of which a liquid surface gets contracted to a minimum area and behaves like a stretched membrane.

It is measured by force per unit length on either side of any imaginary line drawn tangentially over the liquid surface



$$T = \frac{F}{L}$$

Surface energy density: It is the work done per unit increase in surface area.

Capillary rise: $h = \frac{2T\cos\theta}{r\rho g}$

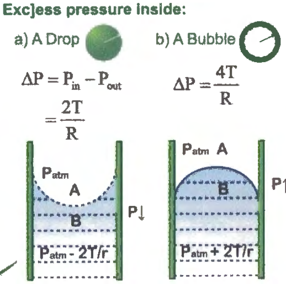
Excess pressure inside:

a) A Drop

$$\Delta P = P_{in} - P_{out} = \frac{2T}{R}$$

b) A Bubble

$$\Delta P = \frac{4T}{R}$$

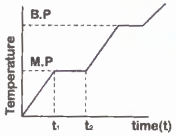


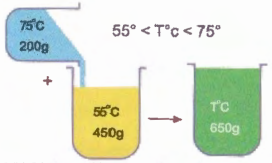
Calorimetry :
When 2 bodies at different temperature are mixed

Heat gained by cold body = Heat lost by hot body

$$\theta_{mix} = \frac{m_1s_1\theta_1 + m_2s_2\theta_2}{m_1s_1 + m_2s_2}$$

(m,s,θ are their respective mass,specific heat,temperature)



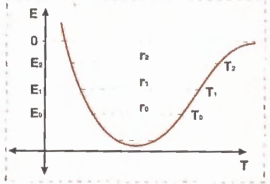


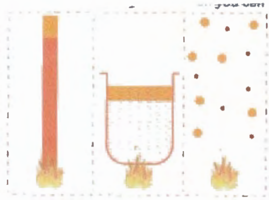
Car radiators work on this principle

Temperature:
It is the degree of hotness or coldness of a body

- Two bodies are said to be in thermal equilibrium, if they are at same temperature.
- Production and measurement of very low temperature (below boiling point of liquid hydrogen ie., 77 k) is called CRYOGENICS while measurement of very high temperature (>1000 k) is called PYROMETRY.
- $\frac{t^\circ c}{100} = \frac{F-32}{180} = \frac{T(k)-273}{100}$

Potential Energy Curve





When matter is heated without change in its state, it expands in general. According to atomic theory of matter, assymetry in potential energy curve is responsible for thermal expansion, as with rise in temperature say from $T_1 \to T_2$, the amplitude of vibration and hence energy of atoms increase from $E_1 \to E_2$, and hence the average distance between them increases from $r_1 \to r_2$,which makes the matter to expand. As the inter molecular forces are maximum for solids and minimum in gases, the expansion of solids is the least, while of gases is the highest.

Calorimetry

Definition

Thermal Expansion

Thermometry

Heat

Types

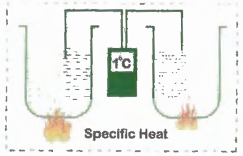
Expansion of Solids

Expansion of Liquids

Specific Heat (s): It is the amount of heat required to raise the temperature of unit mass (m =1) of a substance by 1°C (or 1K)

$$s = \frac{\Delta Q}{m.\Delta t}\left(\frac{cal}{gram\text{-}^\circ C}\right); \Delta Q = m.s.\Delta t$$

- Atomic weight x SP. heat = 6 cal / °C(constant). It means heavier elements will have lesser specfic Heat (except Carbon).
- Specific heat is maximum for hydrogen (3.5 cal / gm - °C) specific heat of water (= 1.0 cal / gm - °C) and less than 1 for all other substances. Specifier of ice 0.5 cal / gm - °C. Specifier of steam 0.47 cal / gm - °C.



Specific Heat

$$C = \frac{\Delta Q}{n.\Delta t}\left(\frac{cal}{mole-^\circ C}\right); \Delta Q = n.C.\Delta t$$

$$\text{Thermal Capacity} = \frac{\Delta Q}{\Delta t} = \text{mass} \times \text{specific heat}$$

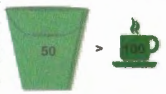
Water equivalent (W): It is the mass of water which when given same amount of heat as to the body, changes the temperature of water through same range as that of the body.

$$W = (m.s)g$$

Latent Heat (L): It is the amount of heat required to change the state of the body at same temperature.

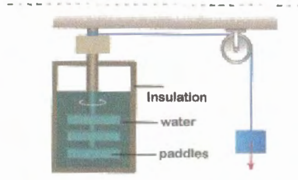
$$L = \frac{\Delta Q}{m}\ ; \Delta Q = mL$$

$$L_{\text{fusion of ice}} = 80\ cal/gm\ ;\ L_{\text{vap. of water}} = 540\ cal/gm$$



A bucket of liquid at 50°C carries more heat than a cup of the same liquid at 100°C

Heat: The energy associated with configuration and random motion of atoms and molecules within a body is called Internal energy. The internal energy which is transferred from one body to another due to temperature difference is called heat.
While temperature depends only on the speed of the molecules, heat depends on **the mass and the speed of molecules.**

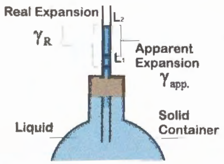


The weight pulled by the person converts the mechanical energy into internal thermal energy

Joules law: The work (w) done and heat (Q) produced, bears a constant ratio called as Joules constant (J) or mechanical equivalent of heat

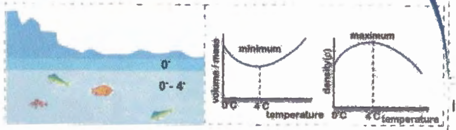
$\frac{W}{Q} = J$ = 4.186 Joule / Calorie ≈ 4.2 Joule/Calorie

ie., to produce 1 calorie of heat 4.2 Joule work is to be done



Expansion of liquids: As liquids can not be heated directly, but to be heated by taking in a solid container, liquids have 2 coefficient of expansions.

$$\gamma_{Real} = \gamma_{apparent} + \gamma_{Container}$$



Anomalous Expansion of Water:
Mater in general expands in heating, but water contracts from 0°C to 4°C , and then above 4°C it expands.
It is due to the fact that water has 3 types of molecules viz., H_2O, $(H_2O)_2$, $(H_2O)_3$
having different volume / unit mass (called sp. volume = 1 / ρ) and at different temperatures their properties in water are different

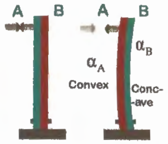
Linear expansion	Area expansion	Volumetric expansion
$\alpha = \frac{L_2 - L_1}{L_1.\Delta t}$	$\beta = \frac{A_2 - A_1}{A_1.\Delta t}$	$\gamma = \frac{V_2 - V_1}{V_1.\Delta t}$
$L_2 = L_1(1+\alpha t)$	$A_2 = A_1(1+\beta.\Delta t)$	$V_2 = V_1(1+\gamma.\Delta t)$
		$\rho_2 = \frac{\rho_1}{1+\gamma.\Delta t}$

$$\alpha : \beta : \gamma = 1:2:3 \qquad \beta = \frac{\alpha+\gamma}{2}$$

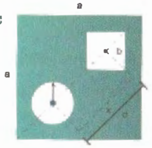
$\frac{L_2 - L_1}{L_1} = \frac{\Delta L}{L}$ = Thermal strain = α. Δt
Thermal stress = Y. α. Δt

Points to note:

Bimetallic strip: A straight metal strip made of two different metals when heated bends with the metal with more 'α' value to be an convex side ? ($\alpha_A > \alpha_B$)

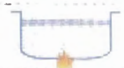
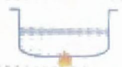


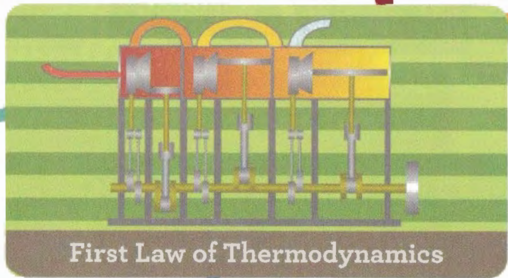
If 2 holes (circle, square) are punched in a square metal plate which dimensions(a,b,r,x,d) increase and which decrease?
Ans: As thermal expansion is just like photographic enlargment, all the dimension increase



Condition for difference in lengths of 2 metal rods to be same at any temperature

$$\alpha \propto \frac{1}{L}\ ;\ \alpha_1 L_1 = \alpha_2 L_2$$

If $\gamma_{Liquid} > \gamma_{Container}$	Level of liquid in container rises on heating	
If $\gamma_{Liquid} < \gamma_{Container}$	Level of liquid in container falls on heating	
If $\gamma_{Liquid} = \gamma_{Container}$	Level of liquid remains same	



Gases and Laws

Macroscopic concept on gases:

1) **Permanant gas:** These are the gases like He, H_2, O_2, which cannot be liquified easily.

2) **Ideal gas:** The gases whose molecules are point masses and due to their very large seperation, do not attract each other and their P.E is negligible. ∴ The total energy of an ideal gas called as Internal energy (U) is only K.E.

$U = KE_{Translational} = \frac{1}{2}m.V_{RMS}^2$

m = mass of gas, $V_{RMS} = \sqrt{\frac{3RT}{m}}$

$U = KE_{Tr} = \frac{3}{2}n.R.T$

$U \propto T$

n = no.of moles = $\frac{\text{mass of gas(m)}}{\text{molecular weight of gas(M)}}$

Ideal gases are non-existent, because gas molecules have definite size and attract each other

Avogadro's law
$V \propto N$ (No of molecules) at constant P, T

Boyle's Law
$V \propto 1/P$
PV=K
P & V Change
N,T constant

Ideal Gas Law
PV=nRT

Charles' Law
$V \propto T$
V/T=K
T & V Change
P,N constant

Combined Gas Law
$V \propto \frac{NT}{P}$
PV=NKT
PV=nRT

k = Boltzman's Constant $= 1.38 \times 10^{-23}$ J/molecule-k

N_A = Avagadro number = $\frac{\text{no.of molecules (N)}}{\text{no.of moles (n)}}$

$PV = kN_A.nT$

$\left[kN_A = R = \text{Ideal gas constant} = 8.31 \frac{\text{Joule}}{\text{mole-k}}\right]$

Microscopic Concept

(Kinetic theory of gases)

1) Pressure of gas = $\frac{2}{3} \times [KE_{Tr}$ per unit volume$]$

2) $V_{sound} = \sqrt{\frac{\gamma RT}{M}}$; $V_{Rms} = \sqrt{\frac{3RT}{M}}$

$\frac{V_{sound}}{V_{Rms}} = \sqrt{\frac{\gamma}{3}}$

Points to remember:

a) When a gas of n_1 moles, C_{v1}, γ_1 values is mixed with n_2 moles C_{v2}, γ_2, then for the gas mixture

$$(C_V)_{mix} = \frac{n_1 C_{V1} + n_2 C_{V2}}{n_1 + n_2}$$

$$\frac{n_1 + n_2}{\gamma_{mix} - 1} = \frac{n_1}{\gamma_1 - 1} + \frac{n_2}{\gamma_2 - 1}$$

Processes

Isothermal Process

1) It is a process which takes place at constant temperature

2) PV = constant(c); log P + log V = log C,

$\frac{\Delta P}{P} + \frac{\Delta V}{V} = 0$

Slope of isothermal curve = $\frac{\Delta P}{\Delta V} = \frac{-P}{V}$

3) Bulk modulus of gas undergoing isothermal process $\Big\} k = \frac{-\Delta P}{\frac{\Delta V}{V}} = P$(Pressure)

4) Sp.heat of gas undergoing isothermal process

$s = \frac{\Delta Q}{m.\Delta T} \rightarrow$ infinity $\quad (as\ \Delta T = 0)$

5) As $\Delta U = n.C_v.\Delta T$, $\Delta U = 0$ is isothermal process, as $\Delta T = 0$

$\therefore \Delta Q = \Delta W$

6) $\Delta W = P.\Delta V = n.R.T.\frac{\Delta V}{V}$

Work done during isothermal process is given by

$$W_{V_1 \to V_2} = nRT \int_{V_1}^{V_2} \frac{\Delta V}{V} = nRT.\log_e\left[\frac{V_2}{V_1}\right]$$

$$W_{P_1 \to P_2} = nRT.\log_e\left[\frac{P_1}{P_2}\right] \quad (\because P_1V_1 = P_2V_2)$$

7) Isothermal process is a Slow Process, and is observed in gases taken in conducting container.

Isochoric Process:

a) It takes place at constant volume

b) as, $\Delta W = P.\ \Delta V = 0$

c) ∴ $\Delta Q = \Delta W + \Delta U$ $\quad \Delta V = 0$

$(\Delta Q) = \Delta U$

Isobaric Process:

a) It takes place at constant pressure

b) $\Delta W = P.\ \Delta V = P[V_2 - V_1] = nR\,[T_2 - T_1]$

Adiabatic Process:

1) It is a process which takes place at constant heat(Q →constant, ΔQ = 0)

2) $P.V^{\gamma} = T.V^{(\gamma-1)} = T^{\gamma}.P^{(1-\gamma)} = \text{Constant(c)}$

3) Slope of adiabatic curve is given by,

$\log P + \gamma.\log V = \log(c), \frac{\Delta P}{P} + \gamma.\frac{\Delta V}{V} = 0$

Slope $= \frac{-\Delta P}{\Delta V} = \gamma.\frac{P}{V}$

∴ Slope of adiabatic curve = γ × slope of isothermal curve

∴ Adiabatic curves are steeper than isothermal curves

4) Bulk modulus of gas under going Adiabatic process = $k = \gamma.P$

5) Specific heat = S = 0 $(\Delta Q = 0)$

6) $\Delta Q = \Delta W + \Delta U; \Delta W = -\Delta U = -nC_V\Delta T$

$\Delta W = \frac{n.R.\Delta T}{\gamma - 1} = \frac{\Delta[nRT]}{\gamma - 1} = \frac{P_2V_2 - P_1V_1}{\gamma - 1}$

7) Adiabatic process is Fast Process, and to keep the heat constant, gas is taken in insulating medium.

Cyclic Process:

In this case, U → constant, ΔU = 0.

∴ ΔQ = ΔW = [Area of P-V graph]

Note:

If cycle is clockwise, ΔW is +ve

If cycle is anti clockwise, ΔW is -ve

Internal Energy

Heat supplied to an ideal gas at constant volume

V → Constant ; ΔV = 0 ; ΔW = ΔV. P = 0

$(\Delta Q)_V = n.C_V.\Delta T = \Delta W + \Delta U = \Delta U = n.C_V.\Delta T$

Heat supplied to an ideal gas at constant pressure

$(\Delta Q)_P = n.C_P.\Delta T = \Delta W + \Delta U = n.C_P.\Delta T$

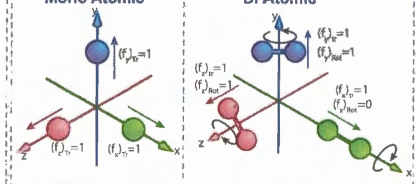
$(\Delta W = \Delta (PV) = nR.\Delta T\ ; \Delta U = n.C_V.\Delta T$

$C_P - C_V = R$ [Mayor's formula]

$\frac{(\Delta Q)_P}{(\Delta Q)_V} = \frac{C_P}{C_V} = \gamma$

$\therefore C_V = \frac{R}{\gamma - 1}; C_P = \frac{\gamma R}{\gamma - 1}$

$(\Delta Q)_P : (\Delta Q)_V : \Delta W = C_P : C_V : R = \gamma : 1 : (\gamma - 1)$



Degree of freedom (f) of a gas molecule:

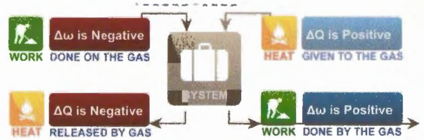
It is the number of independent motion that a gas molecule can posses. The independent motions can be Translational, Rotational & vibrational.

Note: Degree of freedom due vibrational motion is to be considered, if the gas is at high temperature, then add '1' to degrees of freedom due to translational & rotational motion

Sign convention for I Law of thermodynamics

Heat supplied to a System = Work Done + Change in Internal Energy

$\Delta Q = \Delta W + \Delta U$



Law of Equi-Partition of energy:

The internal energy (U) of an ideal gas per molecule per degree of degree is equal to $(\frac{1}{2}kT)$.

If there are 'n' moles of gas with degree of freedom as 'n',

$U = \frac{1}{2}f.n.R.T$

∴ For mono atomic gas $\quad U = \frac{3}{2}n.R.T(f = 3)$

For di atomic gas $\quad U = \frac{5}{2}n.R.T(f = 5)$

For tri atomic gas $\quad U = 3n.R.T(f = 6)$

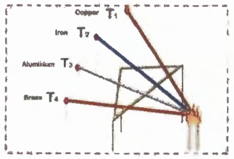
$\Delta U = \frac{1}{2}f.n.R.\Delta T = n.C_V.\Delta T$

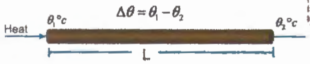
$C_V = \frac{fR}{2};$

$C_P = C_V + R = R\left[1 + \frac{f}{2}\right]$

$\gamma = \frac{C_P}{C_V} = 1 + \frac{2}{f}$

Heat transfer through a metal rod: Metals are better conductors of heat compared to liquids and gases because metals have free electrons which have larger thermal speeds. $(V \propto \sqrt{T})$





$Q \propto \frac{At\Delta\theta}{L}$; $Q = K \cdot \frac{At\Delta\theta}{L}$

K = Coefficient of thermal conductivity

$\frac{Q}{t}$ = Rate of heat flow = Thermal current (H)

$H = \frac{\Delta\theta}{\frac{L}{KA}}$; $\frac{L}{KA} \to$ Thermal resistance (R)

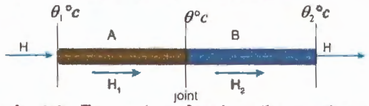
$\therefore H = \frac{\Delta\theta}{R} \Rightarrow similar\ to\ i = \frac{V}{R}$

Without motion of molecules of the body heat transfers from higher temperature to lower temperature in the body.
eg: heat flow in solids

Laws of Conduction

Conduction

RODS connected in series:



Steady state: Temperature of each section remains constant through temparature difference exists between them. No part of body absorbs heat and heat entering the rod is equal to heat leaving the rod.

In steady state: Temparature of first of 2 rods, (interface)

$(R_{eq})_s = R_1 + R_2$

$(K_{eq})_s = \frac{2K_1K_2}{K_1+K_2} [L_1 = L_2]$

$\theta = \frac{\theta_1 R_2 + \theta_2 R_1}{R_1 + R_2}$

$\theta = \frac{\theta_1 K_1 L_2 + \theta_2 K_2 L_1}{K_1 L_2 + K_2 L_1}$

$\theta = \frac{\theta_1 K_1 + \theta_2 K_2}{K_1 + K_2} [if\ L_1 = L_2]$

RODS connected in parallel

$(R_{eq})_P = \frac{R_1 R_2}{R_1 + R_2}$

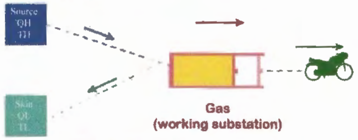
$(K_{eq})_P = \frac{K_1 + K_2}{2}$

Convection

With motion of molecules of the body heat transfer in the body. eg: heat flow in fluids.

Heat Engine

It is a device which converts heat into work through a cyclic process.



$Q_H \to$ heat available at source $\quad \Delta U = 0$

$Q_L \to$ heat wasted to sink (surroundings) $\Delta Q = \Delta W$

$\Delta W = Q_H - Q_L$

Efficiency of heat engine$(\eta) = \frac{\text{Workdone}}{\text{heatgiven}} = \frac{W}{Q_H}$

$\eta = \frac{Q_H - Q_L}{Q_H} = 1 - \frac{Q_L}{Q_H} = 1 - \frac{T_L}{T_H}$

Kelvin's Statement

Whole of work $\xrightarrow{possible}$ heat

Whole of heat $\xrightarrow{not\ possible}$ work

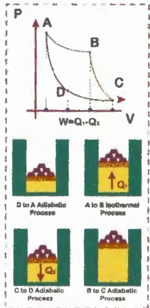
Carnot Cycle

(Working of heat engine)

The working subject (gas) in the cylinder is taken through a reversible cyclic process. Containing following 4 steps.

1) A→B : Isothermal expansion
2) B→C : Adiabatic expansion
3) C→D : Isothermal compression
4) D→A : Adiabatic compression

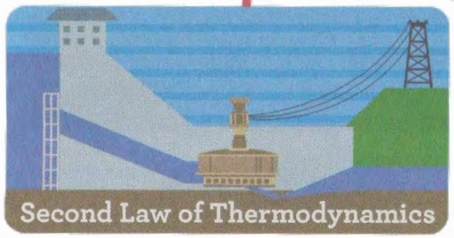
To get more efficiency, temperature of sink (T_L) should be minimum and temp of source (T_H) should be maximum.



Classius Statement

$t_1{}^0 c \xrightarrow[(t_1>t_2)]{heatflow} t_2{}^0 c$

Definition



a) The heat radiated by the hot body is called as Radiant energy and is in the forms of EM waves, which travel with speed of light and do not effect the medium through which they travel.

b) The EM wave λ ranging from 0.0001mm to 1mm are called IR radiations or heat radiation or heat radiation

c) Thermal radiation travel in straight line sand intensity of radiation follow inverse square law

$\left(I \propto \frac{1}{r^2}\right)$

d) Perfect Black body : It absorbs all the radiations incident on it.

e) Abosorptive power (a) : (No units, dimensions) = $\frac{\text{Radiant energy absorbed}}{\text{Radiant energy incident}}$, in a given time

f) Emissive power (e) (Watt/m²) = Radiant energy emitted per second per unit surface area

WEIN'S Displacement Law: The product of wavelength (λ) corresponding to maximum intensity of radiation and temparature (T) of body is constant

$\lambda_m \cdot T = b = a\ \text{constant} = 2.89 \times 10^{-3} mt - k$

Kirchoff's Law:

$\left(\frac{e}{a}\right)_{body} = \left(\frac{e}{a}\right)_{blackbody} = E_{(constant)}$

(e,a are considered to be at the same wavelength (λ), temparature)

$\frac{e_\lambda}{a_\lambda} = E_\lambda$ = constant

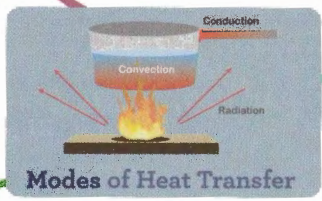
ie, $e_\lambda \propto a_\lambda$ [ie, good emitters are good obsorbers]

Radiation

Without role of molecules, heat flows. eg: heat flows from sun to earth.

Newton's law of cooling: Rate of cooling of a body is directly proportional to the temparature difference between the body and its surroundings ($\theta_0{}^\circ c$) of the temparature of body drops from $\theta_1{}^\circ c$ to $\theta_2{}^\circ c$ in time (dt) then,

$\frac{d\theta}{dt} = K \cdot \left[\frac{\theta_1 + \theta_2}{2} - \theta_0\right]$



Stefan's Law: Radiant energy (Q) of a hot body per unit second (t) per unit surface area (A) called as emissive power (P) or radiancy (R) or intensity (I)is directly proportional to the fourth power of absolute temperature of hot body.

$R\left(=\frac{Q}{t-A}\right) \propto T^4$ {when hot body is in vaccum}

$\frac{Q}{t} = \sigma e A T^4$ (e = emissibility or relative emittance. e = 1 for perfect black bodies)

σ = stefain constant $= 5.67 \times 10^{-8} W/m^2\text{-}K^4$

The term $\frac{Q}{t} = \frac{dQ}{dt}$, indicates the rate of heat loss

$\frac{dQ}{dt} \propto A \propto r^2 \quad (r = radius\ of\ body)$

$As\ \frac{dQ}{dt} = ms\frac{d\theta}{dt}$ (dθ/dt = rate at which temparature is lost called Rate of Cooling)

$\frac{d\theta}{dt} = \frac{\sigma e A T^4}{ms}$ $[as\ m = density \times volume\ of\ sphere = \frac{4}{3}\Pi R_3$

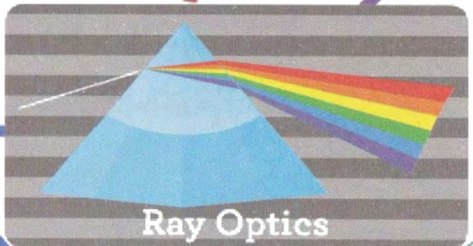
$A = surface\ area\ of\ sphere = 4\Pi R^2$

$\frac{d\theta}{dt} \propto \frac{1}{r}$

if the hot body kept in surrounds at temperature T_0 Kelvin

$\frac{dQ}{dt} = \sigma \cdot e \cdot A \cdot \left[T^4 - T_0^4\right]$

Ray Optics



Definition

It is the branch of optics, deals with the propagation of light in terms of rays, and explains the phenomenon like reflection, refraction, dispertion & deviation of light

Sign convention New Cartesian Convention

Reflection at Curved Surface

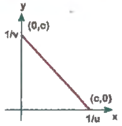
Reflection at curved surface

(Convex and concave mirrors),1/u,1/v graph

If u,v are the object and image distances and f is the focal length, then the mirror formula is,

$\frac{1}{u}+\frac{1}{v}=\frac{1}{f}$ and it resembles y=x+c

$y=\frac{1}{v}; x=\frac{1}{u}; C=\frac{1}{f}$ It is a straight line



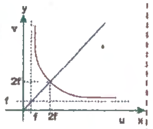
u, v graph

The u-v graph will be a **hyperbola** as

$v\to\infty$; v=f

$u\to\infty$; v=u

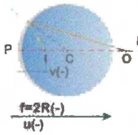
The straight line will cut the hyperbola at (2f, 2f)



1) All the distances are measured from pole (p) of mirror.

2) The distances measured in the direction of incident ray are taken +ve, otherwise -ve.

3) Heights measured upward and perpendicular to the \ principal axis are +ve and otherwise -ve.



differentiating b/s w.r.t t

$\frac{-1}{u^2}\cdot\frac{du}{dt}-\frac{1}{v^2}\cdot\frac{dv}{dt}=0$

$\frac{du}{dt},\frac{dv}{dt}$ are Velocities of object and image (v_o, v_i)

$\frac{v}{u}=$ Magnification (m)

$$\frac{dv/dt}{du/dt}=\frac{v_i}{v_o}=\left(\frac{v}{u}\right)^2=m^2 \qquad m=\frac{f}{f-u}$$

Reflection at Plane Surface

1) All the angles are measured with the normal drawn to the surface.

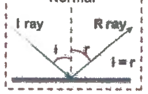
2) Incident ray, reflected ray and normal are to be in the same plane.

3) The angle of incidence [i] = angle of reflection (r)

4) If the incident ray falls normally (i = 0), then, reflected ray retraces its path (r =0)

5) **On reflection:** frequency , λ , speed - do not change intensity, amplitude - change (decrease)

6) There is a phase change of 'π' , if reflection takes place from denser medium.



For the same incident ray, if the mirror is rotated by an angle ' θ ', the reflected ray rotates by an angle ' 2θ '

To see his fall image in a plane mirror, a person requires a mirror of at least half of his height.

Refraction at Plane Surface

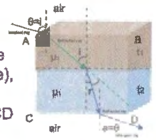
Multiple refractions of light ray passing through a composite slab

Incident and emergent rays will be parallel.

ie., angle of incidence (i) will be equal to angle of emergence (e), if

a) Interfaces AB is parallel to CD

b) Incident ray & emergent ray should be in same medium

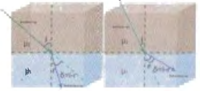


Refraction at plane surface: The bending of light ray as it passes obliquely from one transparent medium to another is called refraction. (If ray incidents normally on the interface there won't be any refraction)

From snell's law,

$\mu_1 \sin i=\mu_2 \sin r$

$\frac{\mu_2}{\mu_1}=\frac{\sin i}{\sin r}={}_1\mu_2$

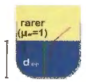


On refraction velocity, λ of light change but frequency remain same.

Refractive index (μ) of medium = $\frac{\text{velocity of light in vaccum}}{\text{velocity of light in medium}}$

' μ ' in term of real and apparent depth:

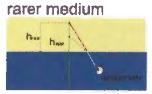
Observer (o) in rarer medium object in denser medium



$d_{real}=\mu . d_{app}$

$V_{object}=\mu . V_{image}$

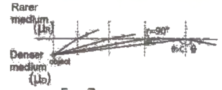
Observer (o) in denser medium object in rarer medium



$h_{app}=\mu . h_{real}$

$V_{image}=\mu . V_{object}$

Total internal reflection and critical angle (C)



$C=\sin^{-1}\left[\frac{\mu_R}{\mu_D}\right]$ $\sin c.\mu_D=\sin 90°.\mu_R$

If rarer medium in air, μ = 1 then $C=\sin^{-1}\left(\frac{1}{\mu}\right)$

Principle of reversibility of light ${}_2\mu_1=\frac{1}{{}_1\mu_2}$

μ of 1 wrt 2 = 1/μ of 2 wrt 1 medium

Refraction at Curved Surface

$\frac{\mu_2}{v}-\frac{\mu_1}{u}=\frac{\mu_2-\mu_1}{R}$ Comparing this with standard mirror formula

$\frac{1}{\frac{v}{\mu_2}}-\frac{1}{\frac{u}{\mu_1}}=\frac{1}{\frac{R}{\mu_2-\mu_1}}$ $\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$

$v\to\frac{v}{\mu_2}; u\to\frac{-u}{\mu_1}$

Transverse magnification (m) = $\frac{I}{O}$

$=\frac{v/\mu_2}{-u/\mu_1}=\frac{\mu_1}{\mu_2}\cdot\frac{v}{u}$

Magnification (m) = $\frac{\mu_1}{\mu_2}\cdot\frac{v}{u}$

Focal length (f) = $\frac{R}{\mu_2-\mu_1}$

If the refracting surface is plane (R→∞)

$\frac{v}{u}=\frac{\mu_1}{\mu_2}=\frac{d_{real}}{d_{apparent}}$

Len's maker's formula

$\frac{1}{f}=\frac{1}{v}-\frac{1}{u}=\left(\frac{\mu_L}{\mu_m}-1\right)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$

(without sign convention)

μ_L - R.Index of lens

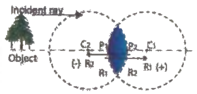
μ_m - R.Index of medium in which lens is kept

$\frac{1}{f}=\frac{1}{v}-\frac{1}{u}$

Len's formula (without sign convention)

(Sign convention is same as for refraction at curved surface)

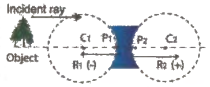
Equi convex lens (R1 = R2)



$\frac{1}{f}=(\mu-1)\left(\frac{2}{R}\right)$

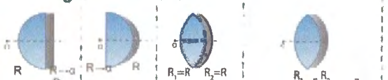
$f=\frac{R}{2(\mu-1)}$

Equi concave lens



$f=\frac{-R}{2(\mu-1)}$

Silvering a lens surface



a) $f=\frac{R}{2(\mu-1)}$ b) $f=\frac{R}{2\mu}$ c) $f=\frac{R}{2(2\mu-1)}$ d) $\frac{1}{f}=2(\mu-1)\left[\frac{1}{R_1}+\frac{1}{R_2}\right]+\frac{2}{R}$

Sign convention - New cartesian convention

1, 2,3 are covered in " Reflection at curved surface "

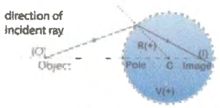
4) μ1 → Ref. Index of medium in which ray pertains to object exist.

5) μ2 → Ref. Index of medium in which ray pertains to image exist.

Refraction at curved surfaces

The bending of light ray as it passes obliquely from one transparent medium to another is called refraction. (If ray incidents normally on the interface there won't be any refraction)

$\frac{\mu_2}{v}-\frac{\mu_1}{u}=\frac{\mu_2-\mu_1}{R}$ without sign convention

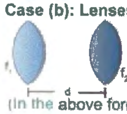


Combination of lenses:

Case (a): Lenses in contact

$f'=f_{combined}=\frac{f_1 f_2}{f_1+f_2}$

Case (b): Lenses with seperation



$\frac{1}{f'}=\frac{1}{f_1}+\frac{1}{f_2}-\frac{d}{f_1 f_2}$

(In the above formulae, if the lenses are concave take (-) sign for focal length)

Prism

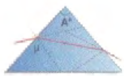
a) Condition for no emergence:

$\mu>\operatorname{cosec}\left(\frac{A}{2}\right)$

b) Condition for Grazing emergence:

$(i_1)_{min}=\sin^{-1}\left[\sqrt{\mu^2-1}.\sin A-\cos A\right]$

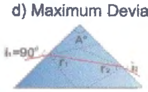
c) Deviation (δ) by a prism



$\delta=i_1+i_2-(r_1+r_2)$

$r_1+r_2=A$

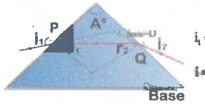
d) Maximum Deviation (δ max)



$i_2=\sin^{-1}[\mu\sin[A-C]]$

Critical angle (C) = $\sin^{-1}\left[\frac{1}{\mu}\right]$

e) Minimum Deviation (δ min)



$i_1=i_2; r_1=r_2=\frac{A}{2}$

$i=\sin^{-1}\left[\mu\sin\left(\frac{A}{2}\right)\right]$

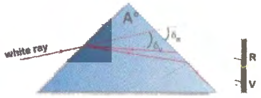
$i=\left(\frac{A+D}{2}\right)$ $\mu=\frac{\sin\left(\frac{A+D}{2}\right)}{\sin\left(\frac{A}{2}\right)}$

Conditions:

a) Refracted ray PQ is parallel to Base

for thin prism D = A(μ-1)

f) **Dispersion:** Splitting of a white ray into constituent colours, as it passes through a prism is called dispersion.



Angular Dispersion (θ) = δ_V - δ_R

Dispersive power (ω) $=\frac{\theta}{\delta_{mean}}=\frac{\delta_V-\delta_R}{\frac{\delta_V+\delta_R}{2}}$

$\delta_{mean}\approx\delta_{yellow}$

δ= A(μ−1)

$\therefore\omega=\frac{\mu_V-\mu_R}{\mu_y-1}$

Case (a): Deviation without dispersion

Achromatic prism

Deviation without Dispersion (A)

$\theta=0,ie., A'=-\frac{(\mu_V-\mu_R)}{(\mu'_V-\mu'_R)}A$

Case (b): Dispersion without deviation

Direct vision prism

Deviation without Dispersion (B)

$\delta=0,ie., A'=-\frac{(\mu-1)}{(\mu'-1)}A$

Factors Affecting Velocity of Sound

$$v_{medium} = \sqrt{\frac{E}{\rho}}$$ E, S are modulus of elasticity and density of medium

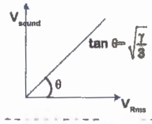
Newton's formula
$$v = \sqrt{\frac{P}{\rho}}$$
(based on assumption that sound travels under isothermal condition) which is experimentally wrong

Laplace correction
(based on fact that sound travels under adiabatic condition)

$$v = \sqrt{\frac{\gamma P}{\rho}}$$

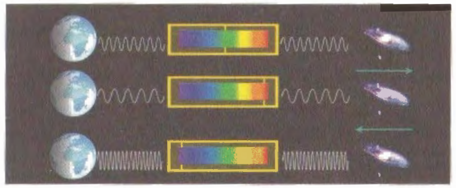
$$v = \sqrt{\frac{\gamma RT}{M}}$$

$$\therefore \frac{v}{v_{Rms}} = \sqrt{\frac{\gamma}{3}}$$



$$v \propto \sqrt{T}$$ Pressure has no effect on velocity of sound as long as there is no change in temperature

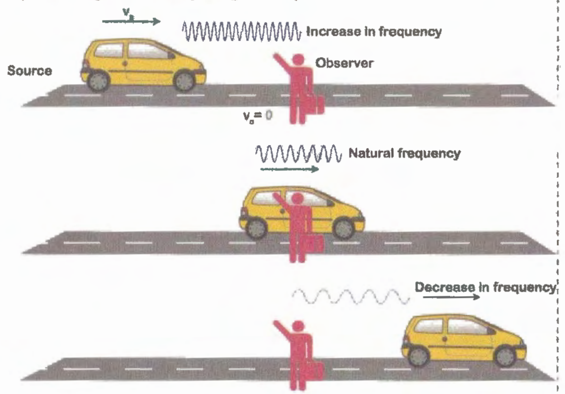
Doppler Effect



Doppler Effect is the principle by which it is suggested that the universe is ever expanding

The apparent change in frequency or pitch due to the motion of source (s) and observer (o) relative to the medium along the line of sight is called Doppler effect.

(Line of sight - The line joining S → 0)



$$n_{app} = n\left[\frac{v - v_O}{v - v_S}\right]$$

If w = speed of wind
If wind blows from s → 0 ; v = v + w
If wind blows from 0 → s ; v = v - w

Point to note:
Doppler effect is not valid if v_o, w are comparable to velocity of sound (v)

Pressure wave

A Pressure wave is a Sound Wave that is caused by excess pressure due to compressions and rarefactions.

$$y = A\sin(\omega t - kx)$$

$$\frac{dy}{dx} = -Ak\cos(\omega t - kx)$$

$$P = B\left[\frac{-\Delta v}{v}\right] = -B\left[\frac{dy}{dx}\right]$$

$$P = AKB\cos(\omega t - kx)$$

$$P_0 = \text{maximum pressure} = ABK$$

(B = Bulk modulus of the medium in which sound travels)

$$\text{Intensity}(I) = \frac{1}{2}\rho v\omega^2 A^2 = \frac{P_0^2}{2\rho v}$$

$$I \propto A^2$$

Intensity at point (P) from source,

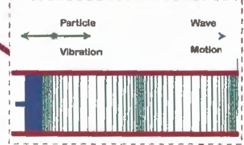
$$I = \frac{P}{4\pi r^2}, \quad I \propto \frac{1}{r^2} \quad (P = Power)$$

Intensity of sound is expressed in $\frac{watt}{m^2}$

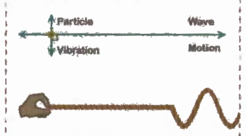
Sound level (SL) $= 10\log\left(\frac{I}{I_0}\right)$

I_0 = Threshold of human ear = 10^{-12} w / m²

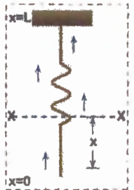
Longitudinal



Transverse



Transverse wave in a hanging rope



at section X-X

$$\text{Tension}(T) = \left[\frac{M}{L}x\right]g$$

$$v = \sqrt{\frac{T}{m}} = \sqrt{gx}$$

$$a = \frac{dv}{dt} = \frac{g}{2}$$

Time for pulse to reach the top

$$L = ut + \frac{1}{2}at^2 \quad (u = 0, a = g/2)$$

$$t = 2\sqrt{\frac{L}{g}}$$

Velocity if transverse wave in stretched string

$$(v) = \sqrt{\frac{T}{m}} = n\lambda$$

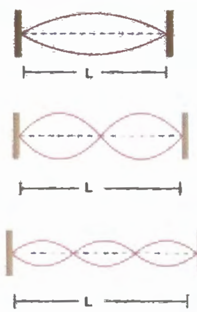
T = Tension in string
m = Linear density
m = mass / length = ρ A
ρ = density of wire material
A = c/s area of wire

$$\text{wave speed} = n\lambda = \sqrt{\frac{T}{m}}$$

Standing waves in stretched strings

Case (a): Fixed at both the ends

$$V = n.\lambda = n.2L$$



$$n_0 = \frac{V}{2L} = \frac{1}{2L}\sqrt{\frac{T}{m}}$$ [Fundamental frequency or I Harmonic (IH) or 0th OT]

$$n_1 = \frac{V}{L} = 2.\frac{V}{2L}$$ [II H or I OT]

[III H or II OT]

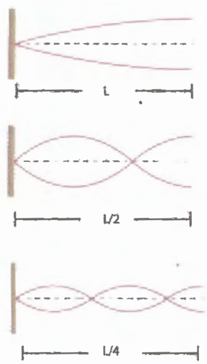
$$n_2 = 3.\frac{V}{2L}$$

$$\therefore N^{th}\ \text{Harmonic} = N.\frac{V}{2L}$$

$$n_0 : n_1 : n_2 = 1:2:3\cdots\left(\frac{V}{2L}\right)$$

[N =Harmonic number = no.of loops]

Case (b): Fixed at one end, free at other end



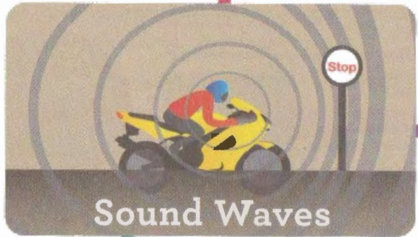
$$n_0 = \frac{V}{4L}$$ [Fundamental frequency or I H or oth OT]

$$n_1 = 3.\frac{V}{4L}$$ [III H or 1st OT]

$$n_2 = 5.\frac{V}{4L}$$ [V H or 2nd OT]

$$n_0 : n_1 : n_2 = 1:3:5\cdots\left(\frac{V}{4L}\right)$$

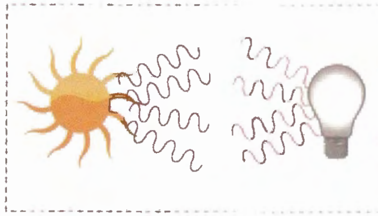
Sound Waves



Waves

Definition

A wave is a disturbance which propagates energy (& momentum) from one place to another without transport of matter.



Mechanical

Waves which require medium for propogation. eg., seismic, sound waves

Non-Mechanical

Waves which donot require any medium for propogation. eg., light, radio waves

Types based on energy propagation

Progressive wave

A wave which travels in a medium with constant velocity (v) amplitude (A) and wave length (λ) and transfers energy from one point to another in a medium is called Progressive wave.

Different forms of Progressive wave

$$y = A\sin[\omega t - kx] \quad \text{—— } 1$$

$$y = A\sin 2\pi\left[\frac{t}{T} - \frac{x}{\lambda}\right] \quad \text{—— } 2$$

$$y = A\sin k[vt - x] \quad \text{—— } 3$$

$$y = A\sin\omega\left[t - \frac{x}{v}\right] \quad \text{—— } 4$$

where $k = \frac{2\pi}{\lambda}$ = propoagation constant

$$\frac{\omega}{k} = v = n\lambda$$

from eq 1 , velocity of particle = v_p

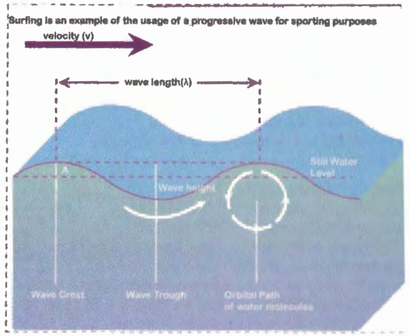
$$v_p = \frac{dy}{dt} = A\omega\cos(wt\text{-}kx)$$

$$(v_p)_{max} = A\omega \quad \text{—— } 5$$

Slope of wave = $\frac{dy}{dx} = -Ak\cos(\omega t\text{-}kx)$ —— 6

from eqns 5 & 6

$$\boxed{v_p = -v \times \text{slope}}$$



Superposition of waves

The phenomenon of intermixing of two or more waves to produce a new wave is called as superposition of waves

Stationary wave

The harmony produced by musical instruments are examples of stationary / standing wave patterns

When 2 waves of same frequency and amplitude travel in opposite direction at same speed, their superposition gives rise to stationary waves, in which energy is confined between nodes.

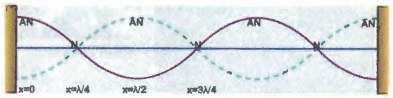
$$y_1 = A\sin(\omega t - kx),\ y_2 = A\sin(\omega t + kx)$$

$$y = y_1 + y_2 = 2A\cos kx.\sin\omega t$$

Amplitude of stationary wave = 2 A Coskx

If cos kx = 0 ; $x = \frac{\lambda}{4}, \frac{3\lambda}{4}, \frac{5\lambda}{4}, \ldots$ nodes form

If cos kx = ±1 ; $x = 0, \frac{\lambda}{2}, \frac{2\lambda}{2}, \frac{3\lambda}{2}, \ldots$ ati nodes form



Beats

The distinct sound that the blades of a helicopter produce are the example of beats

When 2 sound waves of nearly equal (but not exactly equal) frequencies travel in same direction, at a given point due to their superposition, frequ-ency alternately increases (waving) and decreases (waning) periodicaly.

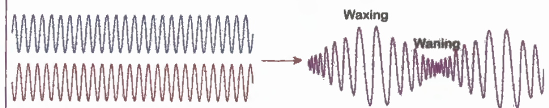
$$y_1 = A\sin(2\pi n_1 t),\ y_2 = A\sin(2\pi n_2 t)$$

$$y = y_1 + y_2 = 2A\cos 2\pi\left(\frac{n_1 - n_2}{2}\right)t.\sin 2\pi\left(\frac{n_1 + n_2}{2}\right)t$$

frequency of resultant wave = $\left(\frac{n_1 + n_2}{2}\right)$

frequency of max. to max. = $\left(\frac{n_1 - n_2}{2}\right)$

frequency of max. to min = $n_1 - n_2$ = Beat frequency

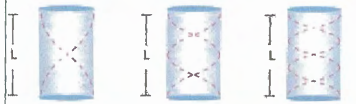


Resonance

When a body is maintained in a state of osiclations by a periodic force having the same frequency as the natural frequency of the body, the oscillations are called Resonant Oscillations, and the phenomenon is called Resonance.

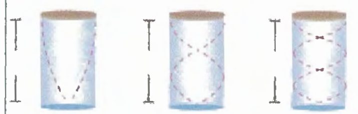
Standing waves in air columns

Case (a): Open at both the ends



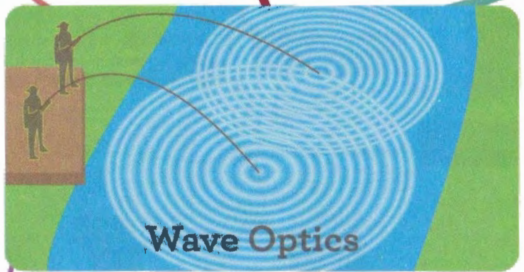
$$n_0 = \frac{V}{2L} \qquad n_1 = \frac{V}{L} = 2\frac{V}{2L} \qquad n_2 = 3\frac{V}{2L}$$

Case (b): Open at one end and closed at other



$$n_0 = \frac{V}{4L} \qquad \eta_1 = 3.\frac{V}{4L} \qquad \eta_2 = 5.\frac{V}{4L}$$

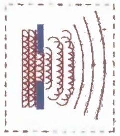
Wave Optics



Nature of Light

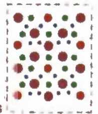
The branch of optics which deals with the nature of light waves exhibiting the properties Interference, diffraction and polarisation

Huygen's Wave Principle



a) Each point of light is a centre of disturbance from which waves spread in all directions.
The locus of all the particles of medium vibrating in the same phase at a given instant is called a wave front.
b) The new position of the wavefront at any instant is a line drawn tangent to the edges of the wavelets at that instant.

Newton's Corpuscular Theory



According to this, body emits light in the form of particles called CORPUSCLES.
This theory **could not** explain phenomenon like interference, diffraction and polarisation.
According to this theory, difference in colours is due to the different sizes of particles.
Speed of light in denser medium is more than that in rarer medium, which is against experimental observation of **Focault.**

Maxwell's EM Theory

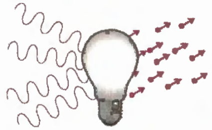
According to this, light is not a mechanical wave but EM wave, travelling with speed,
$C = 3 \times 10^8$ m/sec $C = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$

In it electric vector $\vec{E}$ and magnetic vector are relates as,

$\frac{E}{B} = C, E \perp B$

Einstein's Quantum Theory

According to this, EM radiations also propagate as PHOTONS. It would explain photoelectric effect but could not explain interference, diffraction & polarisation.

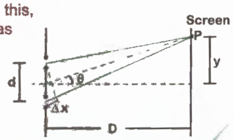


Dual Nature theory

At present, it has been universally accepted that, Light has both wave and particle nature. Due to this, Eddington described light as consisting of WAVICLES (= wave + particles)

Interference

The modification in the distribution of light intensity in the region of superposition due to waves is called INTERFERENCE.



At point (P) on the screen, let 2 waves 1,2 superpose

$y_1 = A_1 \sin[\omega t - kx]$; ——— 1 $y_2 = A_2 \sin[\omega t - k(x + \Delta x)]$ ——— 2

But, Φ (phase difference) = 2π/λ = k.path difference = $\frac{2D}{\lambda}.\Delta x$

By superposition, $y = y_1 + y_2$ $\Delta x = d \sin\theta$; $\tan\theta = \frac{y}{D}$

$y = [A_1 + A_2 \cos\phi]\sin(\omega t - kx) - [A_2 \sin\phi]\cos(\omega t - kx)$

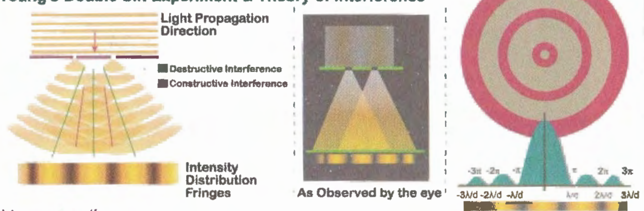
A = Amplitude of resultant wave = $\sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos\phi}$

If I_1, I_2 are the intensities of 2 waves,
I = Intensity of resultant wave = $I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$

Condition for constructive interference or formation of bright fringes $\cos\phi = \text{maximum} = 1$;
$\phi = \pm 2\pi n$, $(n = 0, 1, 2, ...)$
$x = \pm n\lambda$
$\therefore A_{max} = (A_1 + A_2)^2$; $I_{max} = (\sqrt{I_1} + \sqrt{I_2})^2$

Condition for destructive interference or formation of dark fringes $\cos\phi = \text{minimum} = -1$;
$\phi = \pm(2n-1)\pi$, $x = \pm(2n-1)\frac{\lambda}{2}$; $(n = 1, 2, 3, ...)$
$\therefore A_{min} = (A_1 - A_2)^2$; $I_{min} = (\sqrt{I_1} - \sqrt{I_2})^2$

Young's Double Slit Experiment & Theory of Interference



Fringe Width (β)

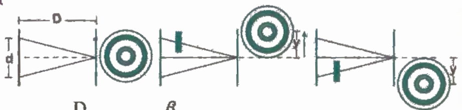
Width of bright fringe (BF) = Width of dark fringe (DF) = $\frac{\lambda D}{d}$

Position of nth BF = $\frac{D}{d}(n\lambda)$; $(n = 0, 1, 2, ...)$

Position of nth DF = $\frac{D}{d}(2n-1)\frac{\lambda}{2}$; $(n = 1, 2, 3, ...)$

a) If interference expt. is conducted with bichromatic light, the fringer of 2 wavelengths will coincide for first time, if
$y = n\lambda_{longer} = (n+1)\lambda_{shorter}$
b) When a transparent sheet of thickness 't', refractive index t is introduced, path diff (x) = t (μ -1)

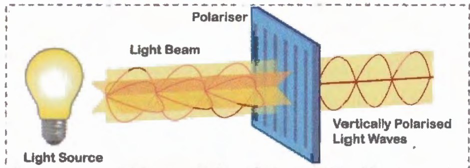
Fringe pattern shifts by y, but fringe width does not change.



$y = \frac{D}{d}(\mu - 1)t = \frac{\beta}{\lambda}(\mu - 1)t$

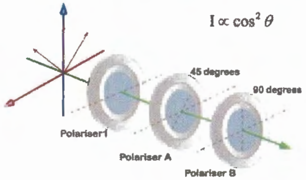
Polarisation

The phenomenon of restricting the vibrations of a transverse wave in a particular direction is called Polarisation.



When all the vibrations of a light wave are in a single plane which contains the direction of propagation of wave, the wave is said to be plane-polarised (or linearly polarised). The variation of E only in the xy plane. Therefore, the light wave is plane polarised in xy plane, and vibrations of light means vibrations of electric vector of light, and electric component of light is mainly responsible for optical effects.

Malus Law



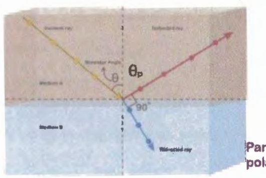
$I \propto \cos^2\theta$

I = Intesity of light transmitted by the analyser.
θ = The angle between transmission axies of analyser, polariser

Brewster's Law

At an angle of incidence θ_p = (Polarising angle) reflected ray and refracted ray are perpendicular

$\frac{\mu_2}{\mu_1} = \tan\theta_p$



Diffraction

The phenomenon of bending of light waves as they pass through a narrow opening or pass an obstacle is called **diffraction**. The diffraction is more prominant if size of opening or obstacles is comparable to the wavelength of the wave.

Diffraction at a single slit

For n^{th} secondary minimum

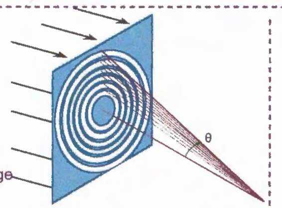
Path difference = nλ = $d\sin\theta_n$

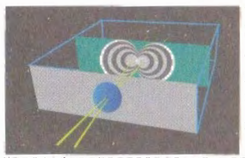
$\sin\theta_n = \frac{n\lambda}{d}$

sinθ = 0 , n = 0 ; Central bright fringe

$\sin\theta = \frac{\lambda}{d}$; n = 1 ; I secondary minimum

$\sin\theta = \frac{\lambda}{2d}$; n = 2 ; II secondary minimum



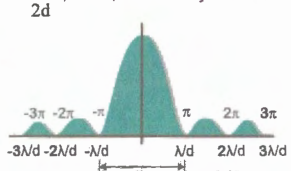


For n^{th} secondary maximum

Path difference = $\sin\theta_n = (2n+1)\frac{\lambda}{2}$

$\frac{3\lambda}{2d}$, n = 1 ; I secondary maxima

$\frac{5\lambda}{2d}$; n = 2 ; II secondary maxima

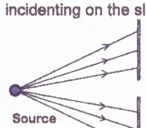


Width of central maximum(x)= $\frac{2\lambda D}{d}$

Types of diffraction

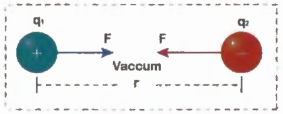
FRAUNHOFER
Parallel rays incidenting on the slit

FRESNEL
Non Parallel rays incidenting on the slit



Electrostatics-I

Columb's Law



The electro static force between 2 stationary changes

$$= F_0 = \frac{1}{4\pi\varepsilon_0}\cdot\frac{q_1 q_2}{r^2}$$

$$\frac{1}{4\pi\varepsilon_0} = k = 8.98\times10^9 \approx 9\times10^9 \frac{\text{N-m}^2}{\text{C}^2}$$

$$\varepsilon_0 = \text{Permitivity of free space} = 8.85\times10^{-12}\frac{\text{C}^2}{\text{N-m}^2}$$

Columb's law is valid for point changes only.
(If not, please refer problem 1)

If a dielectric medium (insulator) is introduced between the 2 charges completely.

$$F_m = \frac{1}{4\pi\varepsilon_m}\cdot\frac{q_1q_2}{r^2} = \frac{1}{4\pi\varepsilon_0 k}\cdot\frac{q_1q_2}{r^2} = \frac{F_0}{k}(F_m < F_0)$$

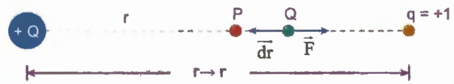
ε_m = Permittivity of medium = $\varepsilon_0 k = \varepsilon_0\varepsilon_r$

Where, $k(=\varepsilon_r)$ is called as relative permittivity or dielectric constant or sp.inductive capacity.

$k_{metal} = \infty$; $k_{vaccum} = 1$; $k_{air} = 1.006$; $k_{water} = 81$

Columb's law is applicable if there is continuity of medium. ie., same medium present between the 2 charges.
(If not, please refer problem 2)

Electric potential (V) due to a point charge (Q)



Work done in shifting unit charge from Q → P "steadily"

$$dW = \vec{F}.\vec{dr} = \frac{1}{4\pi\varepsilon_0}\cdot\frac{Q}{r^2}.dr\cos180^\circ = -\frac{1}{4\pi\varepsilon_0}\cdot\frac{Q}{r^2}.dr$$

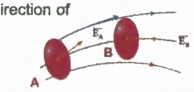
Work done in shifting 'q' from infinite distance to the point (P)

$$W = \int_\infty^r dW = \frac{1}{4\pi\varepsilon}\cdot\frac{Q}{r} = +\frac{k.Q}{r} = \text{Potential at P }(V_p)$$

$$V_p = -\frac{k.Q}{r}$$

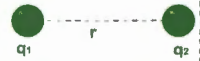
Electric lines of force

It is an imaginary curve, the tangent to which at a point gives the direction of $\vec{E}$ at that point.



Electrostatic P.E between 2 point charges

$$U = \frac{k.q_1.q_2}{r}$$



$$U = \frac{k.q_1}{r}.q_2 = V_1\times q_2 \qquad U = \frac{k.q_2}{r}.q_1 = V_2\times q$$

Potential energy(U)= Potential(v) X change(q)

Electric Field (E)

Due to a point charge(Q)

Electric field (E) due to a point change (Q):

Electric firled at a point near a point charge is the electrostatic force experienced by a unit positive charge kept at that point.

$$\vec{E} = \frac{kQ}{r^3}\vec{r} \Rightarrow \vec{E} = \frac{kQ}{r^2}$$

Relation between E, V

As electrostatic field is coneservative,

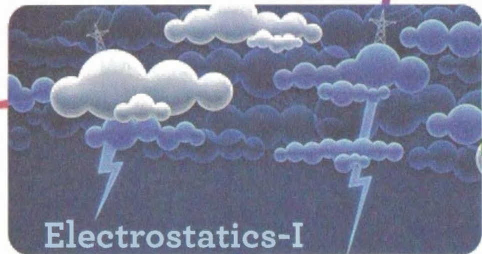
$$E = \frac{-dV}{dr}\ ; \quad \vec{E} = E_x\hat{i} + E_y\hat{j} + E_z\hat{k}$$

Note 1: In case of uniform electric field,
$F = E.q$

Note 2: $E = \frac{V_A - V_B}{d} = \frac{V}{d}$
$\Rightarrow V = E.d$

V - It is the potential difference between two points seperated by distance (d).

Note 3: A (+q) charge moves along $\vec{E}$
A (- q) charge moves opposite to $\vec{E}$



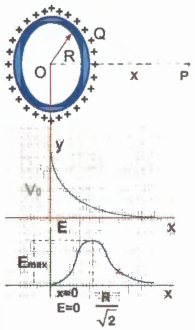
Important Cases

Electric potential (V) and field (E) due to different shaped charged bodies

A Charged Rod

At point (P), $V = \frac{kQ}{L}.\log_e\left[\frac{a+L}{a}\right]^a, E = \frac{kQ}{a(a+L)}$

A Charged Ring



$$V_p = \frac{kQ}{\sqrt{R^2+x^2}}, V_o = \frac{kQ}{R}$$

if $x \gg R, V = \frac{kQ}{x}$

(Behaves as point charge)

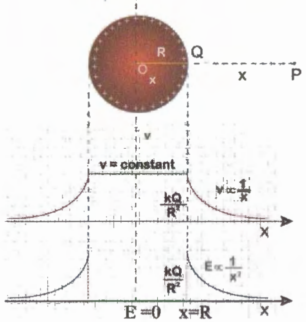
as $x\to\infty, V\to 0$

$$E_p = \frac{kQx}{(R^2+x^2)^{\frac{3}{2}}}$$

$$E_{max} = \frac{2kQ}{3\sqrt{3}R^2}\ ; \quad E_O = 0$$

at $x = \frac{R}{\sqrt{2}}$

Charged Conducting Sphere



at (P) external points

$$V_p = \frac{kQ}{x}, E_p = \frac{kQ}{x^2}$$

at Q (on surface)

$$V_Q = \frac{kQ}{R}, E_Q = \frac{kQ}{R^2}$$

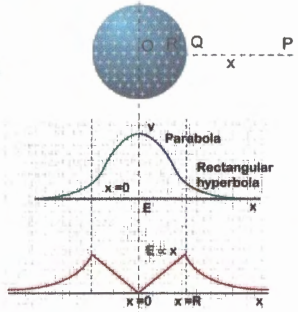
at R (an internal point)

$$V_R = \frac{kQ}{R}, E_R = 0$$

at O (Centre of sphere)

$$V_O = \frac{kQ}{R}, E_O = 0$$

Charged Non Conducting Sphere



at (P) external points

$$V_p = \frac{kQ}{x}, E_p = \frac{kQ}{x^2}$$

at Q (an surface)

$$V_Q = \frac{kQ}{R}, E_Q = \frac{kQ}{R^2}$$

at R (an internal points)

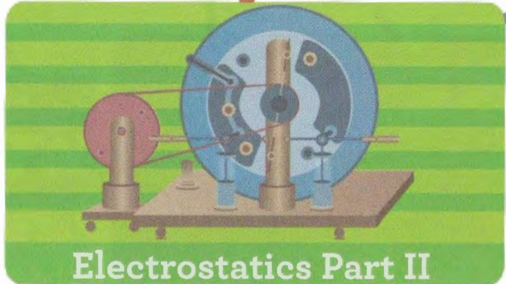
$$V_R = \frac{kQ}{2R^3}[3R^2 - x^2], E_R = \frac{kQ}{R^3}x \Rightarrow (E_R \propto x)$$

at O (Centre of sphere)

$$V_O = \frac{3}{2}\frac{kQ}{R} = \frac{3}{2}V_Q, E_O = 0$$

$$V_{center} = \frac{3}{2}.V_{surface}$$

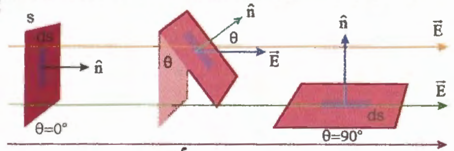
Electrostatics Part II



Electric Flux (Φ)

Gauss's Law:
According to this, the total flux (Φ) linked with a closed surface is $\frac{1}{\varepsilon_0}$ times the charge enclosed by the closed surface

$$\phi = E\int ds.\cos\theta = \frac{q_{enclosed}}{\varepsilon_0}$$



$\phi_{max} = S.E$ $\quad \phi = E\int ds\cos\theta \quad \phi = 0$

Electric Flux (Φ) is the no.of electric lines of force passing per unit surface area perpendicular to the area.

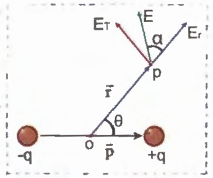
Electric Dipole

Dipole moment $= \vec{P} = q.\vec{r}$

$(\hat{r}, \vec{P}$ are directed from -q to +q)

d=distance between +q, -q.

Electric potential (V) and field (E) due to Dipole



At Point (P) (r >>> d)

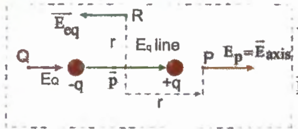
$V = \frac{k.p.\cos\theta}{r^2}$

(θ The angle between $\vec{r}, \vec{p}$)

E_r = Radial comp.of $\vec{E} = \frac{2k.p.\cos\theta}{r^3}$

E_T = Transversal comp.of $\vec{E} = \frac{k.p.\sin\theta}{r^3}$

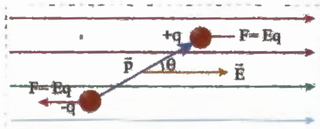
$$E = \sqrt{E_r^2 + E_T^2} = \frac{k.p}{r^3}\left[1+3\cos^2\theta\right]^{\frac{1}{2}}$$



$V_P = \frac{k.p}{r^2}; V_Q = -\frac{k.p}{r^2}, V_R = 0$

$\vec{E}_{axis} = -2\vec{E}_{eq} = \frac{2k.\vec{p}}{r^3}$

Torque (τ) on dipole kept in $\vec{E}$



Work done in rotating the dipole from angle $\theta_1 \to \theta_2$

$W = PE(\cos\theta_1 - \cos\theta_2)$

Potential Energy$= -\vec{P}.\vec{E} = -PE.\cos\theta$

Time period of dipole(T)

$= 2\pi.\sqrt{\frac{I}{P.E}}$

Capacitors

Spherical capacitors

$Q = C.V\ ; C = \frac{Q}{V} = \frac{R}{k} = 4\pi\varepsilon_0.R$

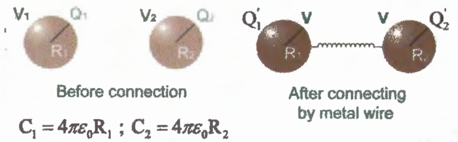
Energy stored in capacitor $= \frac{Q^2}{2C} = \frac{CV^2}{2} = \frac{QV}{2}$

Work done in storing charge = q.v

ie., Only 50% of work done is stored as energy in the capacitor.

Self energy of capacitor $= \frac{Q^2}{2C} = \frac{Q^2}{8\pi\varepsilon_0.R}$

Redistribution of changes on connected spheres.,



$C_1 = 4\pi\varepsilon_0 R_1\ ;\ C_2 = 4\pi\varepsilon_0 R_2$

Total charge $= Q = Q_1 + Q_2 = Q_1' + Q_2'$

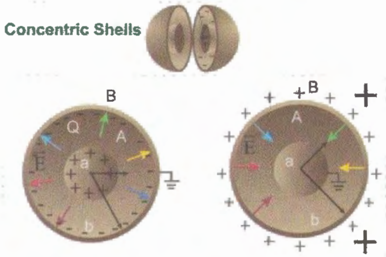
After connecting by metal wise, the common potential (V), they acquire $= V = \frac{C_1V_1 + C_2V_2}{C_1 + C_2}$

Loss in energy $= \Delta U = -\frac{1}{2}\left[\frac{C_1C_2}{C_1+C_2}.(V_1-V_2)^2\right]$

Charges appearing on spheres after connection,

$$Q_1' = Q\left[\frac{R_1}{R_1+R_2}\right];\ Q_2' = Q\left[\frac{R_2}{R_1+R_2}\right]$$

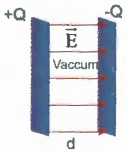
Concentric Shells



$V = V_A - V_B$	$V = V_B - V_A$
$V = k.Q\left[\frac{b-a}{ba}\right]$	$V = k.Q\left[\frac{b-a}{b^2}\right]$
$C = \frac{ab}{k(b-a)}$	$C = \frac{b^2}{k(b-a)}$
$E = \frac{V}{b-a} = \frac{k.Q}{b.a}$	$E = \frac{k.Q}{b^2}$

Parallel plate capacitors (PPC)

$C_0 = \frac{\varepsilon_0 A}{d}$



d = seperation between plates

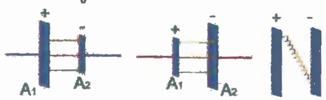
A = Area within which electric field is confined

$E = \frac{\sigma}{\varepsilon_0} = \frac{Q}{A\varepsilon_0}$

V = Potential difference between plates

$V = E.d = \frac{Q.d}{A\varepsilon_0}\ ;\ C_0 = \frac{Q}{V} = \frac{\varepsilon_0 A}{d}$

Note:



$C = \frac{\varepsilon_0 A_2}{d} \qquad C = \frac{\varepsilon_0 A_1}{d} \qquad C \to \infty$

Capacitors in series

$\frac{1}{C_S} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$

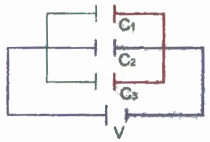
If $C_1 = C_2 = C_3 = C$

$C_S = \frac{C}{n}$

If 'n' plates are arranged in series, they constitute (n-1) capacitors

$C_S = \frac{\varepsilon_0 A}{d(n-1)}$

Capacitors in parallel



$C_P = C_1 + C_2 + C_3$

$C_1 = C_2 = C_3 = C$

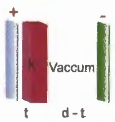
$C_P = n.C$

If 'n' plates are arranged in parallel, they constitute (n-1) capacitors.

$C_P = (n-1)\frac{\varepsilon_0 A}{d}$

Dielectric slab kept between the plates

$C_K = \frac{\varepsilon_0 A}{d - t\left(1 - \frac{1}{k}\right)}$



If (t = d);

$C_K = \frac{\varepsilon_0 A}{d}.k = C_0.k$

Force of attraction between the plates: $F = \frac{Q^2}{2\varepsilon_0 A}$

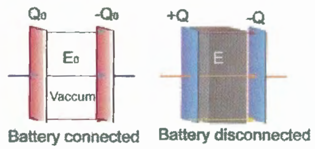
Effect of dielectric slap kept between the plates

Case A: Battery disconnected (change will be constant)

$Q = Q_0\ ;\ C = C_0.k$

$V = \frac{V_0}{k}\ ;\ E = \frac{E_0}{k}$

$U = \frac{U_0}{K}\ ;\ F = \frac{F_0}{K}$



Case B: With battery still in connection (Pot.diff.between plates will be constant)

$Q = Q_0.k\ ;\ C = C_0.k$

$V = V_0;\ E = E_0$

$U = U_0.k\ ;\ F = F_0.k$

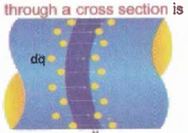
Energy stored in a capacitor

$$\overline{u_E} = \text{Energy density} = \frac{\text{Energy}}{\text{Volume}} = \frac{\frac{1}{2}CV^2}{A.d} = \frac{1}{2}\varepsilon_0 E^2 = \frac{\sigma^2}{2\varepsilon_0}$$

Current Electricity

Current

The time rate of flow of change through a cross section is current.



Instantaneous current

$\Rightarrow I = \frac{dq}{dt}$

1: Unit: SI - Ampere (A)
Biot (Bi)

$1A = \frac{1C}{sec} = \frac{\left(\frac{1}{10}\right)\text{emu of charge}}{sec} = \frac{1}{10}Bi$

2. q motion → current (i)
motion ← -q → current (i)

3. Current is not a vector, though it has magnitide direction, because it does not follow law of vector addition.

vector addition

$i = i_1 + i_2$

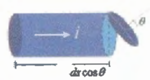
$i \neq \sqrt{i_1^2 + i_2^2 + 2i_1 i_2 \cos\theta}$

4. Current due to translatory motion of charge
$i = n.q.v_d.s$
(n = no.of free electrons per unit volume) $S = c/s$ area
q= change of free electron ; V=drift speed of free electron

5. Current due to rotational motion of charge

$i = \frac{q}{T} = \frac{qV_0}{2\pi r} = \frac{q\omega}{2\pi}$

Current density ($\vec{J}$)



$i = \int \vec{J}.\vec{ds}$

$i = J.ds.\cos\theta$

Current flowing in a conductor = $i = n.e.s.V_d$

$\frac{\text{Pot.diff}}{\text{Current}} = \frac{V}{I} = R = \rho.\frac{L}{S}$ = constant

$\therefore \rho = \text{Sp.resistance} = \frac{2m}{n.e^2.\tau}$

$V = iR$ Ohm's law

m = mass of electron
τ = Relaxation time
R = Resistance of wire

$\rho = \frac{V}{I}.\frac{S}{L} = \frac{V}{L}.\frac{1}{\left(\frac{I}{S}\right)} = \frac{E}{J}$;

$R \propto \frac{1}{\tau}$; but $\tau \propto \frac{\lambda}{V_{rms}}$

$\therefore E = \rho.J$

λ = mean free path of electrons.
as temperature increases,
V_{rms} increases, λ decreases,
$\therefore \tau$ decreases, hence 'R' increases.

Variation of resistance of a streching wire

$R = \rho.\frac{L}{A} = \rho.\frac{L^2}{A.L} = \rho.\frac{L}{V} = \rho.\frac{L}{A^2}$ Area=A

On stretching ρ,V remain same (r = radius of wire)

$R \propto L^2 \propto \frac{1}{A^2} \propto \frac{1}{r^4}$

Resistance in series & parallel

$R_S = R_1 + R_2$

$R_P = \frac{R_1 R_2}{R_1 + R_2}$

In series
1) Same current flows through R_1, R_2
$V = V_1 + V_2 = iR_1 + iR_2$

In parallel
1) Current splits, $i = i_1 + i_2$
2) Same potential diff a/c R_1, R_2

R-C Circuits

Case (A): Charging a capacitor and growth of current through a capacitor

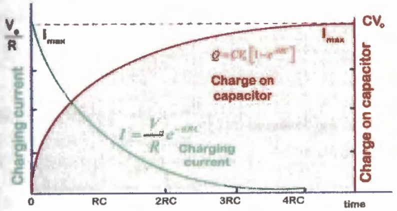
Charge on a capacitor plates at time (= t) = q

$q = q_0\left[1 - e^{-t/RC}\right]$

RC = Time constant (τ)
q_0 = maximum change = CE

$i = \frac{dq}{dt} = i_0\left[e^{-t/RC}\right]$

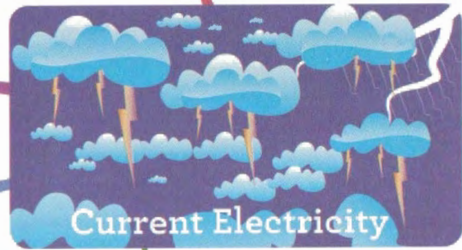
$i_0 = \frac{\varepsilon}{R}$



Case (B): Dischanging a capacitor and decay of current through a capacitor

$q = q_0\left[e^{-t/RC}\right]$

$i = -i_0\left[e^{-t/RC}\right]$



Measuring Instruments

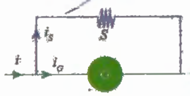
a) **Galvanometer:** It is to measure current up to 10^{-8} A and is very sensitive instrument. Deflection (θ) is directly proportional to current (i)

$i \propto \theta; i = k\theta$

k = Galvanometer constant
1/k = Current sensitivity (Cs)

b) **Ammeter:**

$S = \frac{ig}{i - ig}.G$

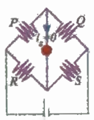


c) **Voltmeter:**

$R = \frac{V}{i_g} - G$

d) **Wheatston's Bridge (Balance condition)**

$\frac{P}{Q} = \frac{R}{S}$



Kirchoff's Laws

First law: The algebraic sum of currents meeting at a junction is zero.

$\sum i = 0$

$i_1 + i_2 = i_3 + i_4$

Second law: The sum of potential difference across each element is equal to net emf's of the cells in closed loop

Sign Convention

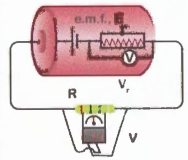
E(+ve) — Observer's motion
E(−ve) — Observer's motion

Observer's motion $\Delta V = -iR$
Observer's motion $\Delta V = +iR$

Cells

Electromotive force (emf): The emf of a cell is defined as the work done by the cell in moving unit positive charge in the whole circuit once including the cell.

emf = w/q



Potential difference (V): It is work done in moving unit positive charge from one point to another in a specified part of circuit.

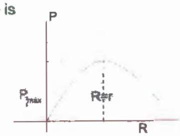
Internal resistance (r): It is the opposition to the flow of current by the cell it self.

Cell charging	Cell discharging	Cell in open circuit
V = E+iR	V = E-iR	V = E

Maximum power transfer theorem: Power transfer to the load (R) is

$P = i^2R = \frac{\varepsilon^2 R}{(R+r)^2}$ for P → maximum $\frac{dp}{dR} = 0$

R = r $\quad P_{max} = \frac{\varepsilon^2}{4r} = \frac{\varepsilon^2}{4R}$



Grouping of Cells

Series

$i = \frac{nE}{R + nr}$

Parallel

$i = \frac{nE}{nR + r}$

Mixed Grouping

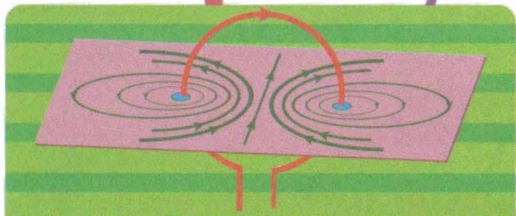
(n) Columns
(m) Rows

$i = \frac{mnE}{mR + nr}$

In mixed grouping; for maximum current $R = \frac{nr}{m}$
(External resistentance = Total internalresistentance)

$i_{max} = \frac{m\varepsilon}{2r} = \frac{n\varepsilon}{2R}$ $\quad P_{max} = \frac{m^2\varepsilon^2}{4r^2}.R$

Thermal and Magnetic Effects of Current Part I



Ampere's Laws

Ampere's Circuital Law

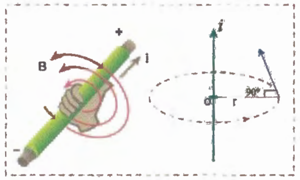
The line integral of magnetic field around any closed path in vaccum/air $\vec{B}$ is equal to 'μ₀' times the total current/enclose by that path.

$$\oint \vec{B}.\vec{dl} = \mu_0 i$$

Note: This law is valid for steady curreny only

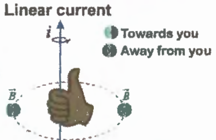
Proof: $\oint \vec{B}.\vec{dl} = \frac{\mu_0 i}{2\pi r}.\oint dL = \frac{\mu_0 i}{2\pi r}.2\pi r$

$$\oint \vec{B}.\vec{dl} = \mu_0 i$$

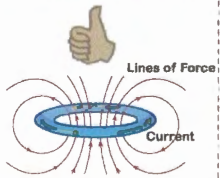


Ampere's Right hand Rule

Linear current

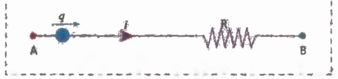


(keep right hand thumb along current, direction of rotation of other fingers gives direction of $\vec{B}$)



(Rotate right hand fingers along current direction of thumb gives direction of $\vec{B}$)

Thermal Effects



V = Pot. diff. between A, B

The work done in driving current (+q charge)

$$\Rightarrow W = q.v = v.i.t \quad (q = i.t)$$

$$= \frac{v^2}{R}.t = i^2.R.t \quad (v = i.R)$$

Thermal power $\Rightarrow P = \frac{W}{t} = i^2.R = \frac{v^2}{R} = v.i$

Heat produced = $Q = \frac{W}{J} = \frac{i^2.Rt}{J} = \frac{v^2t}{R.J} = \frac{v.i.t}{R.J}$

Force on Current carrying conductor

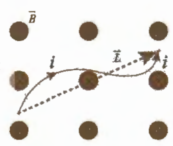
Force on a current - carrying conductor in magnetic field($\vec{B}$)

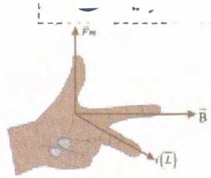
$$\vec{F}_m = i\left[\vec{L} \times \vec{B}\right]$$

$\vec{L}$ Displacement vector of current

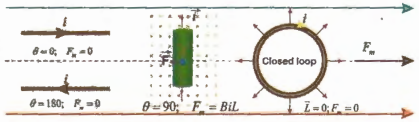
$$F_m = i.LB.\sin\theta$$

θ → The angle between $\vec{L}, \vec{B}$





Flemmings left hand rule



Biot-Savart's Law

Biot- Savart's law

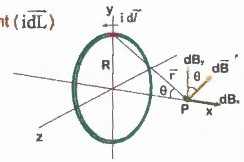
at 'P' magnetic field due to element ($i\vec{dL}$)

$$= dB = \frac{\mu_0}{4\pi}.\frac{i.dL.\sin\theta}{r^2}$$

$$\vec{dB} = \frac{\mu_0}{4\pi}.\frac{i\left(\vec{dL} \times \vec{r}\right)}{r^3}$$

Magnetic field due to entire wire

$$= B = \int dB = \frac{\mu_0}{4\pi}.\frac{i\left(\vec{dL} \times \vec{r}\right)}{r^3}$$

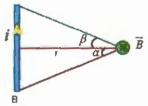


Cases of Interest

Straight wire

a)

b)



$$B = \frac{\mu_0}{4\pi}.\frac{i}{r}\left[\sin\alpha + \sin\beta\right]$$

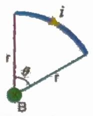
c) In case of infinitely long wire

$$(\alpha, \beta \to 90°)$$

$$B = \frac{\mu_0 i}{2\pi r}$$

d) Circular arc carrying current

$$\vec{B} = \frac{\mu_0 i}{4\pi r}.(\theta) \quad (\theta \to \text{In radians})$$



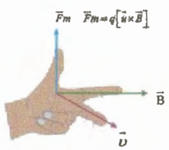
Magnetic Effects

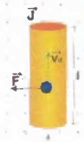
Magnetic force ($\vec{F}_m$) on a charge in uniform magnetic field ($\vec{B}$)

$\vec{F}_m$ is perpendicular to $\vec{u}, \vec{B}$

$\vec{u}$ may or may not be perpendicular to $\vec{B}$

$$\vec{F}_m = q.u.B\sin\theta$$



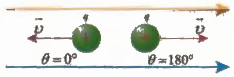


Flemmings left hand rule

(i): The above expression is for (+q) charge. If the charge is (-q), then

(ii): A charge at rest (u =0) does not experience force ($F_m = 0$)

(iii): If the charge is moving along or opposite to $\vec{B}$ (θ= 0° or 180°) also does not experience force



(iv): If θ = 90° $\left(F_m\right)_{max} = q.u.B$

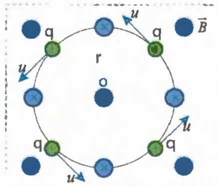
The charge performs uniform circular motion

$$\frac{mu^2}{r} = B.q.u;$$

$$r = \frac{mu}{Bq} = \frac{\sqrt{2m.KE}}{Bq}$$

$$T = \frac{2\pi r}{u} = 2\pi.\frac{m}{Bq}$$

$$\omega = \frac{2\pi}{T} = \frac{Bq}{m}$$



(v): If (θ ≠ 0°,90°,180°)

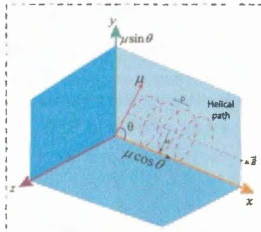
u cosθ → Pushes the charge along the axis of helix.

u sinθ → makes the charge to perform perpendicular motion.

$$r = \frac{mu.\sin\theta}{Bq};$$

$$T = \frac{2\pi r}{u\sin\theta} = 2\pi.\frac{m}{Bq}$$

Pitch (P) = $T \times u.\cos\theta = 2\pi r.\cot\theta$



Note (vi): Motion of charged particle in combined electric and magnetic fields ($\vec{E}, \vec{B}$)

$\vec{F} = q\left[\vec{E} + (\vec{u} \times \vec{B})\right]$ (Lorentz - Force equation)

Case (a): $\vec{u}, \vec{E}, \vec{B}$ collinear

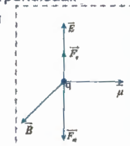
$$F_m = 0; F_e = E_q$$

Case (b): $\vec{u}, \vec{E}, \vec{B}$ mutually perpendicualr

If $F_m = F_e$, charge moving with same velocity undeviated,

Bqu = Eq

$$u = \frac{E}{B}$$



Consider a circular coil of radius 'R', having 'n' turns and carrying current 'i' as shown. At a point 'P' on the axis of the coil is given by, (OP = x)

$$B = \frac{\mu_0 . n.i.R^2}{2(R^2+x^2)^{\frac{3}{2}}}$$

at centre (O) of ring (x =0)

$$B_0 = B_{max} = \frac{\mu_0 . n.i}{2R}$$

at very large distance (x>>R)

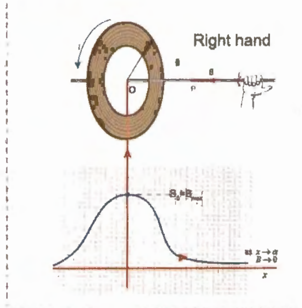
$$B = \frac{\mu_0 . n.i.R^2}{2x^3} \;;\; (R^2+x^2 \approx x^2), (B \propto \frac{1}{x^3})$$

at infinite distance (x →α)

$$B = 0$$

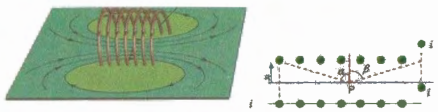
at very short distance (x<<R), $(R^2+x^2 \approx R^2)$

$$B = \frac{\mu_0 . n.i}{2R}$$



Magnetic field ($\vec{B}$) on the axial point of a Circular Coil

If many turns of an insulated current carrying wire is wound around a cylinder, then that coil is a SOLENOID



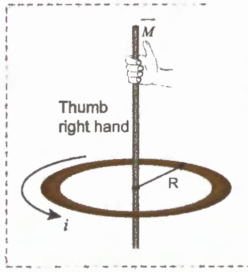
n= no.of turns per unit length

$$\text{at (P)}, B = \frac{\mu_0 . n.i}{2} . [\sin\alpha + \sin\beta]$$

For infinitely long solenoid, α,β→ π/2

$B = \mu_0 ni$

Magnetic field ($\vec{B}$) on the axial point of a solenoid



$M = i.A$

$M = i.\pi r^2$

$M = \left(\frac{q}{2m}\right).L$

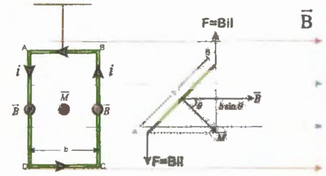
L = angular momentum

L = mvr

Magnetic Moment ($\vec{M}$) of current loop

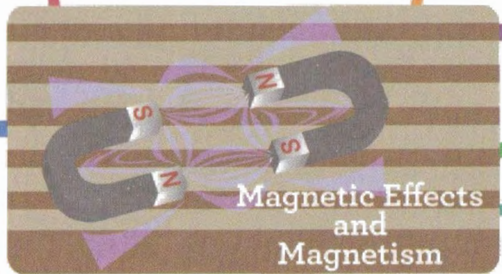
Torque (τ) = F x perpendicular distance
= B.i. (L.b)sinθ
= B(iA)sinθ

$\vec{\tau} = \vec{M} \times \vec{B} \;;\; M = iA$



$F_{AB} = F_{DC} = 0$

Torque (τ) on current loop in magnetic field (B)



Magnetic Effects and Magnetism

Properties of Magnetic Materials

Intensity of magnetic field ($\vec{H}$)

It is the strength of the magnetic field used for magnetising the magnetic material

$$\vec{H} = \frac{\vec{B_0}}{\mu_0}$$

Intensity of magnetisation(I)

$$\vec{I} = \frac{\text{magnetic moment (M)}}{\text{Volume of magnet (V)}} = \frac{m \times 2L}{A \times 2L}$$

$$= \frac{\text{Pole strength (m)}}{\text{c/s area (A)}}$$

Magnetic suscpetibility (χ):
It represents the ease with which a magnetic material can be magnetised, and is given by,

$$\chi = \frac{I}{H}$$

Relative permlability (μ_r):

$$\mu_r = 1 + \chi$$

Curie's law

$$\chi \propto \frac{1}{T\ (\text{kelwin})}$$

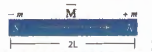
Magnet and its Characteristics

Length of magnet:
It is the shortest distance between the poles of a magnet.

Pole strength (m): It is the strength of a magnetic pol depends on cross sectional area, intensity of magnetisation, nature of the material of magnet.
S.I Unit for (m): Amp-mt

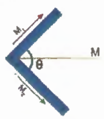
Magnetic moment ($\vec{M}$)

$M = 2L \times m$

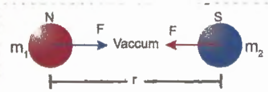


$$\text{SI units for 'M'}: \frac{\text{Joule}}{\text{Tesla}} ; \text{wb - mt, Amp - mt}^2$$

$$M = \sqrt{M_1^2 + M_2^2 + 2M_1M_2\cos\theta}$$



Inverse Square Law

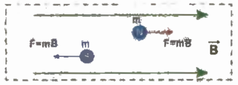


$$F = \frac{\mu_0}{4\pi} . \frac{m_1 m_2}{r^2}$$

μo - Permlability of free space.
S.I unit for 'μ' - Henry/mt

Force experienced by magnetic pole (m) in magnetic field

The behaviour of a short magnet is similar to electric dipole in electric field.



Magnetic field ($\vec{B}$)

It is the force experienced by unit pole (m =1) kept in magnetic field

$$B = \frac{\mu_0}{4\pi} . \frac{m}{d^2}$$

It is also defined as the no.of magnetic lines(Φ) passing per unit cross sectional area (A) perpendicularly

$$B = \frac{\phi}{A} = \frac{\text{weber}}{\text{mt}^2} = 1\ \text{Tesla}$$

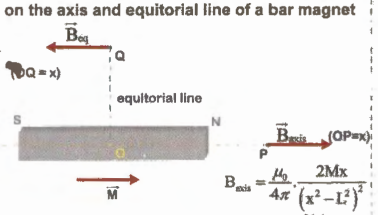
Magnetic field ($\vec{B}$) on the axis and equitorial line of a bar magnet

$$B_{eq} = \frac{\mu_0}{4\pi} . \frac{M}{(x^2+L^2)^{\frac{3}{2}}} \quad (OQ = x)$$

$$B_{eq} = \frac{\mu_0}{4\pi} . \frac{M}{x^3}$$

(for short bar magnet)

equitorial line

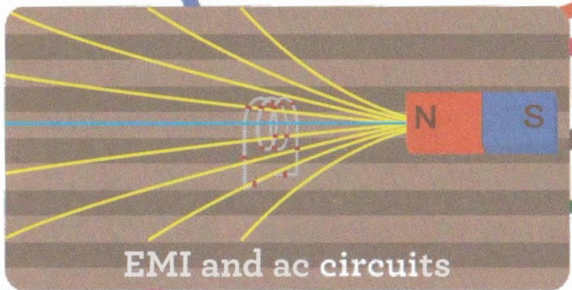


$$B_{axis} = \frac{\mu_0}{4\pi} . \frac{2Mx}{(x^2-L^2)^2}$$

$$B_{axis} = \frac{\mu_0}{4\pi} . \frac{2M}{x^3}$$

(for short bar magnet)

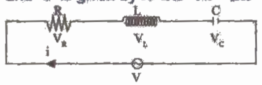
EMI and ac circuits



A-C Circuits | L-C Oscillations | L-R Circuits | Types of EMI | Cases | Faraday's Laws of EMI | Transformer

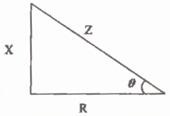
Analysis of L-C-R circuits

1) **Reactance (X):** It is the net resistance offered by 'L' and 'C' is given by $X = X_L - X_C = L\omega - 1/\omega C$



2) **Impedence (z):** It is the net resistance offered by R,L,C. $Z = \sqrt{(X^2+R^2)}$

Φ = phase difference between v & i $\Rightarrow \Phi = \tan^{-1}(X/R)$



3) **Power Factor:** $\cos\Phi = R/Z$

4) **Powder consumed by LCR circuit:**

$P_{avg} = V_{rms}.\ I_{rms}.\ \cos\Phi = 1/2.\ V_{max}\ I_{max}.\ \cos\Phi$

5) **Admittance (Y):** $Y = 1/Z$

6) If $\Phi = 0$; If v,i are in same phase, P→ Maximum

If $\Phi = \pi/2$; Power is zero, current is called as **Wattless current.**

7) **Relation between potential drops across R,L,C (ie,. V_R, V_L, V_C):**

$V^2 = V_{ii}^2 + V_{\perp}^2 = V_R^2 + (V_L - V_C)^2$

$V_{\perp} = iX = i(X_L - X_C) = (V_L - V_C)$

8) L,C,R circuit is in **Resonance**. when, X=0; $X_L = X_C$

$\omega = 1/\sqrt{LC}$

In Resonance:

a) Z = R

b) $\cos\Phi = 1$, $\Phi = 0$; v,i are in same phase

c) Power consumption is maximum

d) X=0 ; $X_L = X_C$; $iX_L = iX_C$; $V_L = V_C$

e) Quality factor (Q) = $V_L/V = iX_L/iR = X_L/R = L\omega/R = \sqrt{L/C}.1/R$

A-C Circuits

Alternating current: Current is said to be alternating if current (i) and voltage (v) vary sinusoidally with time and angular frequency (ω) is constant.

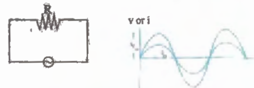
$i = i_0 \sin \omega t = i_{max} \sin \omega t$

$$i_{avg} = \frac{\int_0^{T/2} i.dt}{\int_0^{T/2} dt} = \frac{2}{\pi} I_0\ ,\ i_{rms} = \left[\frac{\int_0^T i^2.dt}{\int_0^T dt}\right]^{\frac{1}{2}} ; i_{rms} = \frac{I_0}{\sqrt{2}}$$

Form factor (F-factor)

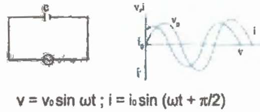
$i_{rms}/i_{avg} = \pi/2\sqrt{2}$

a) A Resistor (R) in an 'ac' circuit



$v = v_0 \sin \omega t$ v, i are in same phase

b) A Capacitor (C) in an 'ac' circuit



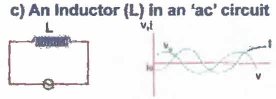
$v = v_0 \sin \omega t$; $i = i_0 \sin(\omega t + \pi/2)$

$= v_0 c\omega \sin(\omega t + \pi/2)$

Current leads voltage by $\pi/2$

The term ($1/c\omega$) indicates the resistance offered by capacitor called as "Capacitive reactance" (X_C)

c) An Inductor (L) in an 'ac' circuit



$v = v_0 \sin \omega t$; $i = i_0 \sin(\omega t - \pi/2)$

$= v_0/\omega L$. $\sin(\omega t - \pi/2)$

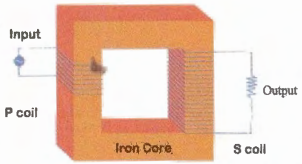
Voltage leads current by $\pi/2$.

The term ($L\omega$) indicates the resistance offered by inductor called as "Inductive reactance" (X_L)

Transformer

It is a device which is used to increase or decrease the voltage in ac-circuits, through mutual induction

$$\frac{N_p}{N_s} = \frac{\phi_p}{\phi_s} = \frac{emf_p}{emf_s} = \frac{i_s}{i_p}$$



Faraday's Laws of EMI

1) When ever there is a change of flux (Φ) (no.of magnetic lines) linked with a circuit an emf is induced in it. This is called as Electro Magnetic Induction (EMI),and emf is called Induced emf (e) and the current is called induced current.

2) The magnitude of induced emf is equal to the rate of change of flux ie., $e = d\Phi/dt$

3) The direction of induced emf is such as to oppose the change in flux (Lenz's law);$e = -d\Phi/dt$ (-ve) sign shows that if flux increases 'e' is -ve, and vice versa)

Induced charge:

$$e = iR = \frac{d\phi}{dt} \quad \left[i = \frac{dq}{dt}\right]$$

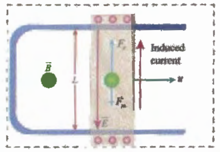
∴Induced charge

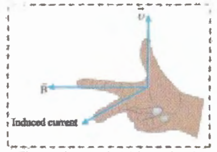
$$(dq) = \frac{d\phi}{R} = \frac{\text{change in flux}}{\text{Resistance of loop}}$$

Cases

Motion of a conducting rod in uniform magnetic field (B): When the conductor moves as shown, the free electron experiences forces F_m, F_e due to external magnetic field (B) and internally developed electric field (E). (When F_m, F_e are equal the free electron will be at rest)

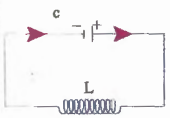
$F_m = F_e$; $B.q.u = E.q = \frac{e}{L}.q$ **e = BuL** (e = induced or motional emf)

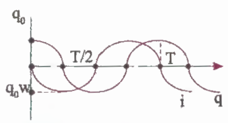




Flemming's right hand rule

L-C Oscillations



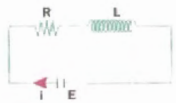


A charged capacitor (C) having an initial charge 'q_0' is discharged through an inductor 'L' charge oscillates in the circuit with natural angular frequency $\omega = \sqrt{LC}$. Change (q) at time (t) = $q_0 \sin(\omega t + \Phi)$; $\Phi = \pi/2$

Current (i) at time (t) = $i = -q_0\ \omega \sin \omega t$; $i_0 = i_{max} = q_0\ \omega$

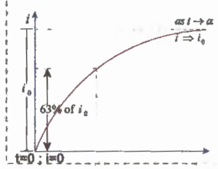
L-R Circuits

Growth of current



$i = i_0(1 - e^{-t/\tau})$; $i_0 = i_{max} = E/R$

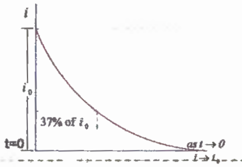
τ = Time constant = L/R



Decay of current

$i = i_0 e^{-t/\tau}$

(Graphs similar to R-C circuits)



Types of EMI

Self-Induction: Whenever the electric current (i) passing through a coil changes, flux (Φ) linked with it also changes,which induces emf 'e' (back emf) in the coil. The phenomenon is called **self Induction.**

$\Phi \propto i$, $\Phi = Li$, $d\Phi/dt = e = L.\ di/dt$

L = coeff of self induction

Note: The energy stored in a coil $U = 1/2\ Li^2$

Self-Induction of a coil

$B = \mu_0 ni/2R$; S=Area

$= \pi R^2 n$; $\Phi = B.S = Li$;

$L = \mu_0 \pi n^2 R/2$

Self-Induction of a solenoid

$L = \mu_0\ n^2 S.L$

length of solenoid = L

c/s area = $S = \pi R^2$

Mutual Induction:

Whenever the current flowing in a coil (primary-p) changes, the magnetic flux linked with a neighbouring coil (secondary-s) also changes, and hence an emf will be induced in the secondary coil. This phenomenon is called as mutual induction.

$\phi_s \propto I_p$; $\phi_s = M.I_p$; M = coeff of mutual induction

$$e_s = -M.\frac{dI_p}{dt}$$

SI unit for L or M: henry

Dimensional formula for L or M: $ML^2T^{-2}A^{-2}$

Note: If L_1, L_2 are self inductances of 2 coils, then mutual inductance (M) of 2 coils

$$M = k.\sqrt{L_1 L_2}$$

k = coefficient of coupling (0<k<1)

If the winding is tight, k = 1

Atomic Physics

Photons

Properties of Photons

1) They travel with the speed of light in straight lines.
2) Energy of photons depend on frequency but not on medium. With change in medium, v, λ of photon change, frequency(f) remains same.
3) Photons are electrically neutral and are not deflected by electric and magnetic fields.
4) Under suitable conditions, they show diffraction.
5) Mass of a photon, $m = m_0/\sqrt{1-(v^2/c^2)}$

As a photon travels with speed of light, v=c; $m = m_0/0$
If m_0 were not zero, the photon should have infinite mass. Since $E = mc^2$,
Infinite mass means infinite photon energy.
This is not possible.
∴ We conclude, photon is never at rest and its rest mass (m_0) is zero.

Equivalent mass(m) of photon, $E = mc^2 = hf$
$m = hf/c^2 = h/c\lambda$

6) Momentum of a photon, $P = mc = h/\lambda$
7) No. of photons emitted by a source =
n = Power of source (P) / Energy of photon (E)

8) Intensity light beam= Power(P)/ Cr.Sec. Area(A)

Photon Flux(Φ)= I/E=Iλ/nc

9) The pressure experienced by surface exposed to radiation is known as Radiation Pressure (P).
P = I/C (If radiation is absorbed by surface)
P = 2I/C (If radiation is reflected by surface)

Einstein's Qunatum Theory of Light

Light is not emitted continuously as waves, instead it is emitted as packets of energy called PHOTONS
Energy of photons (E) = hf = hc/λ ≈ [12400/ λ(Å)] eV
where **h** = plank's constant = 6.63×10^{-34} Joule-Sec
f = frequency of radiation; **λ** = wave length of photon
c = speed of light (= photon) = 3×10^8 m/sec

de Broglie waves

According to de Broglie: Energy moving particle is associated with a wave which controls the particle in every respect. These waves are called as **matter waves** or de Broglie waves and are also called as **Pilot waves,** as their function is to pilot or guide the matter particles.

The wavelength (λ) of de Broglie waves = h/mu = h/p
$m = m_0/\sqrt{1-(v^2/c^2)}$; If v = 0 ; λ→α ; if v→α , λ = 0
It means de Broglie waves are associated with material particles if they are in motion.

de Broglie waves travel with speed of EM waves. ie., that of light. de Broglie waves are not EM waves because their velocity is not constant (depending on velocity of particle) but the velocity of EM waves is constant. EM waves are associated with accelerated charged particles, but de Broglie waves are associated with moving particles whether charged or not.

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2.m.KE}} = \frac{h}{\sqrt{2.m.q.V}} \quad (KE = W = qV)$$

(for a change (q) accelerated through potential difference 'V')

$$\lambda_{electron} = \frac{12.27}{\sqrt{v}} A^\circ \; ; \; \lambda_{proton} = \frac{0.286}{\sqrt{v}} A^\circ;$$

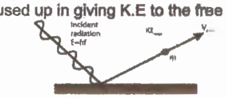
$$\lambda_{deuteron} = \frac{0.202}{\sqrt{v}} A^\circ \; ; \; \lambda_{\alpha-particle} = \frac{0.101}{\sqrt{v}} A^\circ;$$

$$\lambda_{neutron} = \frac{0.286}{\sqrt{E(eV)}} A^\circ \quad \lambda_{gas\ molecule} = \frac{h}{\sqrt{3mkT}}$$

As mvr= n . h/2π; 2πr= n . h/mv
Circumference of nth orbit = n x de Broglie wavelength
(for an electron rotating around nucleus in n^{th} orbit)

Einstein's Photo Electric Effect

When a radiation of energy (E = hf) incident on a metal surface, part of energy called as Work function (W_0) is utilised in liberating the free electron with zero-velocity for metal surface and the balance energy is used up in giving K.E to the free electron, ranging from 0 to KE_{max}

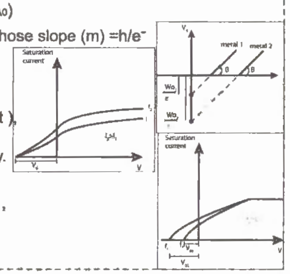


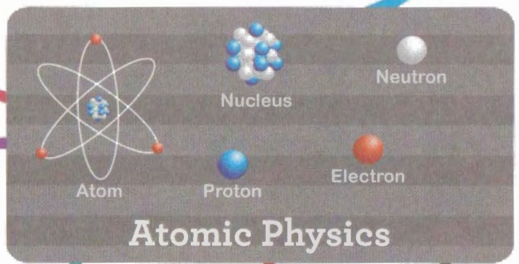
If the incident energy is just sufficient to remove the free electron, then KE = 0
$W_0 = hf_0 = hc/\lambda_0$; ($f_0$, λ_0 are threshold frequency and wave length)

If $f < f_0$ - No photoelectric effect, for what ever be the intensity of incident radiation is.

Stopping Potential (V_0):
It is the minimum retarding potential at which photoelectric current becomes zero.
$e.V_0 = KE_{max} = 1/2.mV^2_{max} = hf - hf_0 = hf\,(1/\lambda - 1/\lambda_0)$
$V_0 = (h/e^-)\,f - (h/e^-)\,f_0$. It is in y = mx-c format whose slope (m) =h/e⁻
intercept = $hf_0/e = W_0/e$

Note 1 :
If Intensity (I) is increased (keeping 'f' constant), then saturation (maximum) current increases. ie., KE_{max} is independent of intensity.
Note 2 :
If light of different frequencies (keeping intensity constant) is used, then as 'f' increases, V_0 also increases.
ie., KE_{max} also increases.





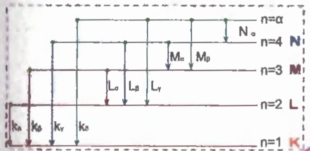
X-Rays

These are EM radiations of short wavelength (0.1 Å→100 Å), and high energy, which are emitted when fast moving electron or cathode rays strike a target of high atomic mass, which travel with a speed equal to that of light.

Types of x-ray spectra :

Characteristic x-rays

They are called so, because they are characteristic of element used as target anode



Continuous x-rays

This is of varying intensities and are independent of material. They end abruptly at a certain lower wavelength for a given voltage.

Moseley's Law

The frequency (f) of characteristic x-rays depends on atomic number (z) of an element as $\sqrt{f} = a(z-b)$
a is a constant, **b** is called as screening constant depends on series K, L, M,N (b = 1 for k_α)

Spectrum

Emission spectrum:
When an electron is excited state makes transition to lower energy state, a photon is emitted. Collection of these photon frequencies is emission spectrum.

a) **Continuous emission:** It covers wide range of wavelength without gap. Generally solids, liquids emit this. eg , solar spectrum.
b) **Line emission:** It contains bright lines (wave length) seperated from each other by dark spaces. eg., It is emited by atom of an element
c) **Band emission:** It is one which consists of bright bands in (group of wavelength) of light, sharp at one end and fading off at other end.

Spectral lines of hydrogen atom:

Series	n1	n2 in spectrum	Position (in nm)	Range of 'λ'
LYMAN	1	2,3,4,....∞	UV	120 - 90
BALMER	2	3,4,5,....∞	Visible	650 - 350
PASCHEN	3	4,5,6,....∞	IR	1900 - 820
BRACKETT	4	5,6,7,....∞	IR	4053 - 1500
PFUND	5	6,7,8,....∞	IR	7500 - 2300

Emission spectrum: When white light composed of all visible photon frequencies is passed through atomic hydrogen gas, certain frequencies are absent, being absorbed by gas. The resulting specturm consists of bright background with some dark lines and is called **absorption spectrum**

Kirchoff's law: If a substance emits light of a certain wavelength at a given temperature, then it will absorb the same wavelength, when light is passed through it is vapour state.

Bohr's Atomic Model of Hydrogen Atom :
I Postulate: Energy electron revolves around the nucleus in stationary orbit. $mV^2_0 = (k.ze^2)/r^2$
II Postulate: The angular momentum of an electron is integral multiple of h/2π. mvr = n. h/2π
As long as an electron revolves in stationary orbit. It neither emits nor absorbs energy.
If the electron jumps to lower orbit, it emits energy and if it jumps to higher orbit, it absorbs energy.

The frequency (f) and wavelength (λ) of such emitted radiations given by, $hf = \Delta E = E_1 - E_2$, therefore.
$f = (E_1-E_2)/h$; $\lambda = hc / (E_1-E_2)$
$1/\lambda = \upsilon$ (wave number) $= z^2.R\,[(1/n_1^2) - (1/n_2^2)]$
R= Rydberg constant = $1.097373 \times 10^7 m^{-1}$

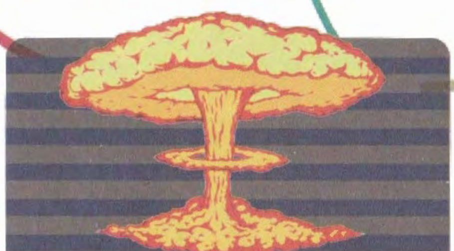
Note 2 : When an electron jumps from $n_1 \to n_2$ orbit, the total no.of emission lines = $(n_1-n_2)(n_1-n_2+1)/2$

Points to remember:

1) For an electron in n^{th} orbit of hydrogen like atom

Radius of orbit	=	$r_n = 0.53 \times \frac{n^2}{z}\ (A^\circ)$
Orbit velocity of electron	=	$V_n = 2.16 \times 10^6 \left(\frac{z}{n}\right)$ (m/sec)
Energy of electron	=	$E_n = -13.6 \times \left(\frac{z}{n}\right)^2$ eV
Time period	=	$T_n \propto (r_n)^{\frac{3}{2}} \Rightarrow T_n \propto \left(\frac{n^2}{z}\right)^{\frac{3}{2}}$
Angular momentum	=	$L_n = m.V_n.r_n \Rightarrow L_n \propto n$
P.E : K.E : TE	=	-2 : +1 : -1

Nuclear Physics



Nuclear Composition

Isobars

The nuclei with same number(A), but different atomic number (z).

$_{6}C^{16}$ $_{7}N^{16}$ $_{8}O^{16}$ $_{9}F^{16}$

Isotopes

The nuclei having same atomic number (z). but different number of neutrons

$_{6}C^{11}$ $_{6}C^{12}$ $_{6}C^{13}$ $_{6}C^{14}$

Isotones

The nuclei with same number of neutrons, But different 'z' values.

$_{6}C^{14}$ $_{7}N^{15}$ $_{8}O^{16}$ $_{9}F^{17}$

Shape of Nucleus

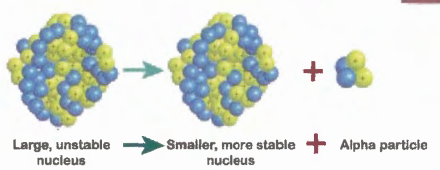
A nucleus can be modelled as a cluster of tighly packed spheres, each of which is a NUCLEON.

Size of Nucleus

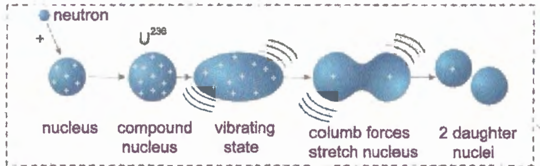
Radius (R) of nucleus = $R_0.A^{\frac{1}{3}}$

R_0 = 1.2 fermi (1 fermi = 10^{-15} mt)

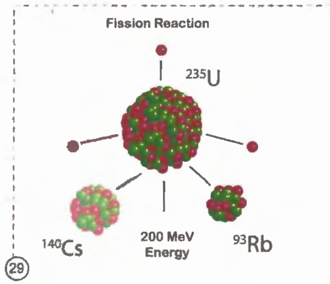
A = mass number.



Nuclear Fission: The splitting of a heavy nucleus (A > 230) (getting bombarded by a neutron) into 2 daughter nuclei releasing huge amount of energy neutron(n).

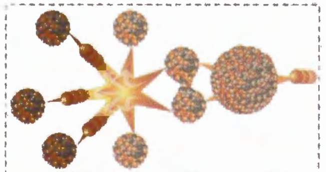


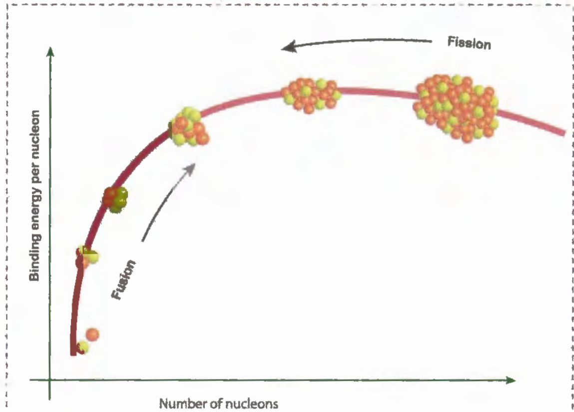
- Total energy released per fission = 200 MeV
- On an average 2.5 neutrons are emitted in each fission
- Mass lost per reaction = 0.2 amu



Nuclear Waves

Nuclear Fusion: The process of combining two light nuclei to form a heavy nucleus with the release of huge amount of energy is known as Fusion. For making the nuclei to come closer for fusion, they are imported with very high K.E (about 0.1 MeV), which requires a very high temperature of about 10^8 k and high pressure about 10^6 atm.





Natural Radioactivity

It is the process of spontaneous disintegration of the nuclei of heavy elements (Z > 82) with the emission of radiations like α,β particles, γ- rays.

Law of decay

Let @ t = 0, no.of atoms in the sample = N_0

@ t = t, no.of atoms Undecayed = N

no.of atoms decayed = N_0 - N

Rate of disintegration α No.of atoms undecayed

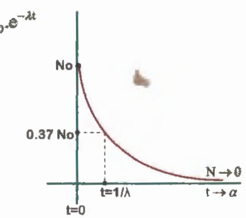
$$\frac{dN}{dt}=-\lambda.N\ ;\ N=N_0.e^{-\lambda t}$$

1) @ $t=\frac{1}{\lambda}$(mean life)

$$N=\frac{N_0}{e}=0.37N_0$$

$$N=37\%\text{ of }N_0$$

2) $T_{1/2}=\frac{1}{\lambda}[\log_e 2]=\frac{0.693}{\lambda}$



3) Let, @ t = 0 ; activity = A_0, mass = m_0

@ t = a activity = A, mass = m

$A=A_0.e^{-\lambda t}$

4) n = no.of half lives

t = total time of disintegration

$t=n.T_{1/2}$

$$\frac{N}{N_0}=\frac{A}{A_0}=\frac{m}{m_0}=\left(\frac{1}{2}\right)^n$$

5) SI uint of activity = 1 becquerel (bq)

1 bq = 1 disintegration per second (dps)

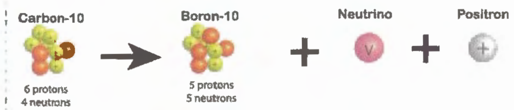
1 curie = 1Ci = 3.7 x 10^{10} dps

$$_ZX^A\xrightarrow{\alpha\text{-decay}}{}_{Z-2}Y^{A-4}+{}_2He^4\quad(\alpha-\text{particle})$$

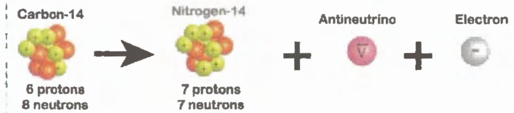
$$_ZX^A\xrightarrow{\beta\text{-decay}}{}_{Z+1}Y^A+{}_{-1}e^0\quad(\beta-\text{particle})$$

$$_ZX^A(\text{excited state})\xrightarrow{\gamma\text{-decay}}{}_ZX^A(\text{ground state})+\gamma$$

Beta-puls Decay

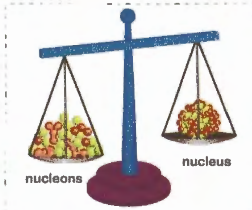


Beta-minus Decay



Gamma Decay

Mass Defect (Δm)



Weight of collective nucleons is greater than the nucleus

Nuclear Binding Energy (B.E): If M = mass of nucleus $_ZX^A$ in kg A = mass number, Z = Atomic number, M_N,M_P = mass of neutron, proton. Then,

B.E of nucleus = $[Z.M_P+(A-Z)M_N-M]\,a.m.u$

$$1\text{ a.m.u}=\frac{1}{12}[\text{mass of }{}_6C^{12}\text{atom at rest}]$$

$$=1.6603\times10^{-27}\text{ kg}=931.5\text{ MeV}$$

Except for a few light nuclei, the average B.E per nucleus is about 8 MeV for all nuclei.

Packing fraction = $\frac{\Delta m}{A}$

A particle of mass (m) possesses an equivalent energy (E), $E=mc^2$

Δm = mass of nucleus - sum of masses of NUCLEONS in free state

Δm = M - A

a) They travel with speed (C) of light. $\frac{E}{B} = C = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$ do not require medium for their propagation

b) They show all wave properties interference, diffraction, polarisation etc., and wave eqn, c = nλ is applicable.

c) The components of $\vec{E}, \vec{B}$ are in same phase.

d) Average energy stored in electric & magnetic fields per unit volume are equal and given by,

$$\mu_E = \frac{1}{2}\varepsilon_0 E_0^2 = \mu_B = \frac{B^2}{2\mu_0}$$

e) Intensity (I) = $\frac{1}{2}\varepsilon_0 C E_0^2 = \frac{C B_0^2}{2\mu_0} = \frac{E_0 B_0}{2\mu_0}$

f) Momentum of EM waves

$P = \frac{\upsilon}{C}$ (When falling on absorbing surface)

$P = \frac{2\upsilon}{C}$ (When falling on reflecting surface)

g) The electric vector is an EM wave is responsible for then optical effects. Therefore electric vector is also called as light vector.

Properties

EM waves are generated by accelerating electric changes. These consist of oscillating mutually perpendicularelectric and magnetic fields $(\vec{E}, \vec{B})$, which are perpendicular to the directions of wave propogation. These $(\vec{E}, \vec{B})$ are transverse to direction of wave propagation.

Sources

The orderly classification of EM waves according to the increasing order of wavelengths or decreasing order of frequencies is called EM spectrum, which consists of Gamma rays, X rays, UV rays, IR rays, heat radiation, micro waves, radio waves, etc.,

EM spectrum

Basic Elements

Source of information eg., speech, text, photo etc.,

Message Signal

Transmitter converts message into electrical vibrations called signals and modulates it.

Transmitted Signal

Channel

Receiver demodulates the signal and feeds to Loudspeaker TV picture take etc

User of information

Modulation
It is the process of change of some characteristics. eg., amplitude, frequency or phase of a carrier wave in accordance with the intensity of signal

Demodulation
It is the process of recovering the radio signal from the modulated wave

Why modulation?

By increasing the frequency, energy is increased, there by height of the antenna can be reduced, range for transmission can be increased and the communication can be made without wires.

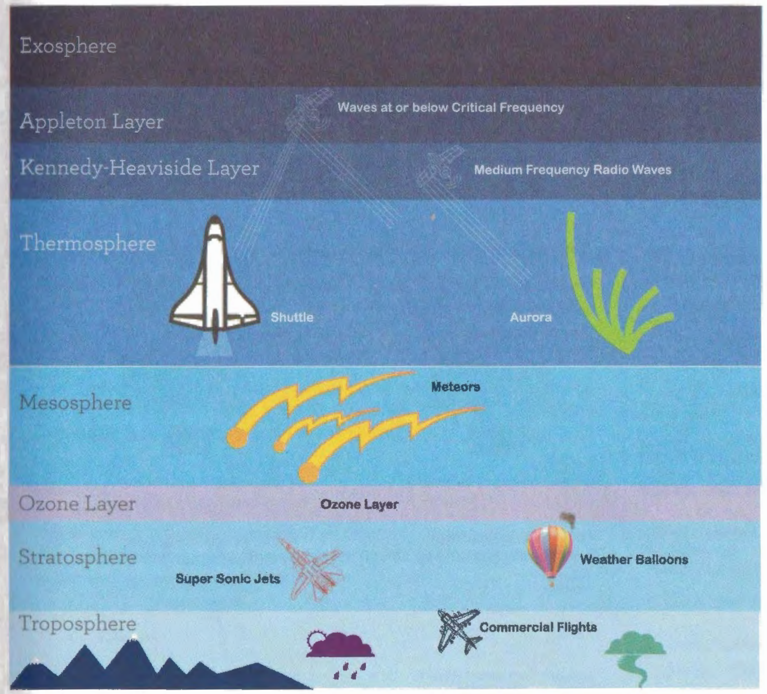
Types of modulation

Amplitude Modulation (AM): Amplitude of carrier wave is changed

Frequency Modulation (FM): Frequency of carrier wave is changed

Phase Modulation (PM): Phase angle (Φ) of carrier wave is changed

Atmosphere



10,000 km

690 km

85 km

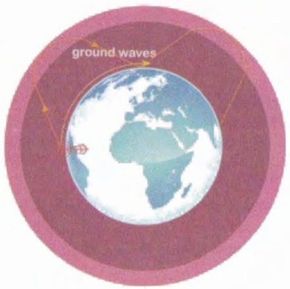
50 km

6-20 km

Wave Propagation

Radio communication use EM waves propagated through the earths atmosphere or space to carry information over long distances without use of wires. The EM waves having frequency ranging from 3 KHz to 300 GHz are called radio waves.

Types of radio wave propagation



Ground waves:
Radio waves travel very close to earth

Sky waves:
Radio waves are radiated away from earth surface up to ionosphere, from where they are reflected back to earth.

Direct or Surface waves: The frequencies above 30 MHz cannot be transmitted by ground or sky waves.
Such waves are transmitted by direct waves by means of antenna.

Range of transmission = d
Height of antenna = h

R = Radius of earth
$d = \sqrt{2Rh}$

Semi Conductor Devices

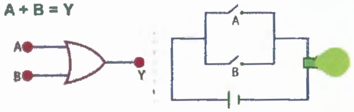
Basic Logic Gates

Truth Table

Input A	Output B	Y= A+B	Y= A.B	Y= $\overline{A.B}$	Y= $\overline{A+B}$
0	0	0	0	1	1
0	1	1	0	1	0
1	0	1	0	1	0
1	1	1	1	0	0

OR Gate

Boolean expression for OR gate is
A + B = Y



a. Clearly, if either or both of A and B are ON, the bulb is ON

AND Gate

Boolean expression for AND gate is
A . B = Y

The bulb lights when both A and B are ON. Similarly the voltage across output will be batter voltage only when bot A and B are connected to 1.

NOT Gate

Boolean expression is for NOT Gate is
$\overline{A}$ Y

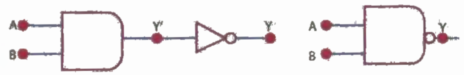
The truth table:

Input A	Output Y
1	1
0	1

Combination of Gates

NAND gate

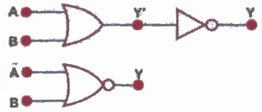
It is the combination of 'AND' and 'NOT' gates.
The symbol of a NAND gate is



The Boolean expression for 'NAND' gate is
$Y = \overline{A.B}$

NOR gate

The symbol of NOR gate is



The Boolean expression for NOR gate is
$Y = \overline{A+B}$

The NAND and NOR gates are universal gates because the use of one or more of these gates can yield any of the OR, AND and NOT gates.

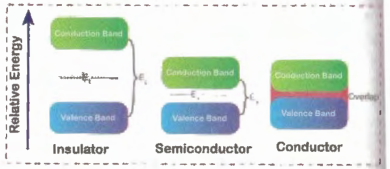
Energy Bands in Solids

The electrons revolving in a particular orbit possess the energy of that orbit. The electron, say in first orbit have different energies as they will be in different change environment due to the atoms closely packed. The range of the energies possessed by an electron in a solid is called **Energy Band.**

Types

Valence band (VB)
It is the range of energies of valence electron

Conduction band (CB)
It is the range of energies of conduction electron
a) For insulators Eg ≈ 15 eV
b) For conductors Eg ≈ 0
c) For semi conductor Eg ≈ 1 eV



Types of Semi Conductors

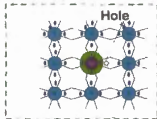
They have almost filled VB, nearly empty CB

A semi conductor in extremely pure form is **Intrinsic**

An intrinsic semi conductor, added with impurities (dopants) to increase conducting properties is **Extrinsic**

n - type: Pure semi conductor added with pentavalent (Arsenic, Antimony) impurities. The no. of free electrons for exceeds no.of holes

p - type: Pure semi conductor added with trivalent (Gallium, Indium) impurities.The no.of holes for exceeds no.of electrons

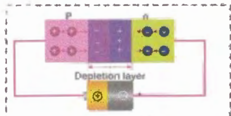


S.C Diode OR Crystal OR pn

It is a semi conductor crystal. So prepared that one half is p-type and other n-type.

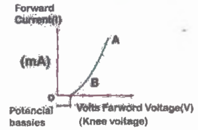
Biasing of p-n junction diode: It is the arrangement of applying an external potential difference across a p-n junction.

Forward Biasing

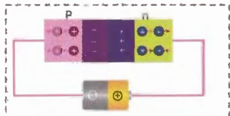


p connected to (+) terminal }of battery
n connected to (-) terminal

Width of depletion layer and potential barrier reduced

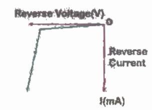


Reverse Biasing



p connected to (-) terminal }of battery
n connected to (+) terminal

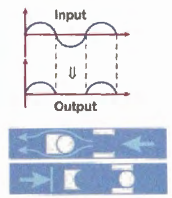
Width of depletion layer and potential barrier increases



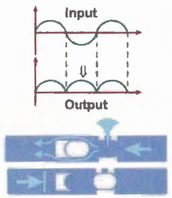
Junction Diode as Rectifier

Rectification is the process of converting ac → dc, and the device for this is a rectifier

Diode as a half wave rectifier



Diode as a full wave rectifier



Types of Junction Diodes

Solar cell: It converts solar energy into electrical energy

Light Emitting Diode: It emits visible light when forward biased.

Zener diode: It works in break down region of reverse biasing

Junction Transistor

Transistor is a device having 2 pn junctions and 3 terminals.
A transistors works when its Emitter Base (EB) junction is forward biased and Collector Base (CB) junction is reverse biased.

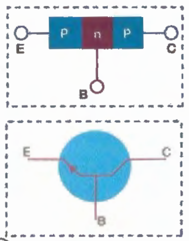
If I_c, I_e, I_b are collector, emitter and base currents

$$I_e = I_b + I_c$$

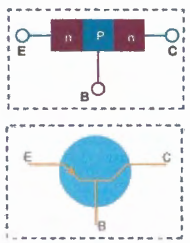
$$\alpha = \frac{I_c}{I_e}; \beta = \frac{I_c}{I_b};$$

$$\alpha = \frac{\beta}{1+\beta}$$

p - n - p transistor



n - p - n transistor



Transistor as an amplifier

It amplifier or increases the amplitude of alternating current, voltage or pomer

Transistor as an oscillator

It is a device which produces an output of a desired frequency and amplitude without any external input voltage
The frequency of such oscillator

$$f = \frac{1}{2\pi\sqrt{LC}}$$